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Disquisitiones generales circa curvaturam et situm varietatum latentium

quas machinae discentes efformant, ope theorematis Chernii-Gaussii-Bonnetii

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Information Geometry and Latent Topology of Machine Learning

Investigating curvature and topology of statistical latent manifolds via Chern–Gauss–Bonnet, with application to variational autoencoders

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Abstract

Real-world high-dimensional data typically concentrates on a low-dimensional manifold. Generative modelling learns a representation of this manifold in a low-dimensional latent space. Such a manifold can carry non-trivial topology, for instance when the data generating process has a compact symmetry group, which a contractible Euclidean latent space cannot encode continuously without necessarily increasing in dimension. This thesis investigates the differential geometric connection between the topology and the information geometry of latent spaces, by constructing a latent space formalism to which the Chern–Gauss–Bonnet theorem applies. We develop an intrinsic information geometric framework for latent spaces from the Fisher information metric and construct two variational autoencoders whose latent spaces carry the prescribed topologies of the two-dimensional sphere and the torus, together with the methodology needed to evaluate the latent geometry. On datasets with enforced topology, the numerically integrated curvature reproduces the correct Euler characteristic of the latent space and matching latent topology to data topology improves reconstruction fidelity. Visualising the learned geometry shows that curvature concentrates where a Euclidean latent space representation fails. The result is numerical access to intrinsic geometric and topological quantities of learned representations.

Zusammenfassung

Hochdimensionale Daten lassen sich häufig durch eine niedrigdimensionale Mannigfaltigkeit darstellen. Generative Modelle lernen eine approximative Darstellung dieser Mannigfaltigkeit in einem niedrigdimensionalen latenten Raum. Eine solche Mannigfaltigkeit kann topologisch nichttrivial sein, beispielsweise wenn der Prozess der Datenerzeugung auf einer kompakten Symmetriegruppe basiert. Ein kontrahierbarer euklidischer latenter Raum kann eine solche Topologie nicht stetig darstellen, ohne höhere Dimensionen zu benötigen. Diese Bachelorarbeit untersucht den differentialgeometrischen Zusammenhang zwischen der Topologie und der Informationsgeometrie latenter Räume durch Konstruktion eines Formalismus für latente Räume, auf den der Satz von Chern–Gauß–Bonnet anwendbar ist. Ausgehend von der Fisher-Informationsmetrik entwickeln wir eine intrinsische informationsgeometrische Beschreibung latenter Räume und modifizieren einen Variational Autoencoder, sodass dessen latenter Raum die Topologie der zweidimensionalen Sphäre beziehungsweise des Torus trägt. Zusammen mit der notwendigen Methodik zur Auswertung der latenten Geometrie erzeugen wir Datensätze mit vorgegebener Topologie und zeigen, dass die numerisch integrierte Krümmung die korrekte latente Euler-Charakteristik reproduziert. Weiterhin können wir zeigen, dass eine Übereinstimmung zwischen der Topologie des latenten Raums und der Daten die Rekonstruktionsgüte merklich verbessert. Die Visualisierung der gelernten Geometrie macht deutlich, dass sich Krümmung dort konzentriert, wo die euklidische Darstellung nicht ausreicht. Somit schaffen wir einen numerischen Zugang zu intrinsischen geometrischen und topologischen Größen gelernter Darstellungen.

Contents

1 Introduction **5**

1.1 Contributions 6

2 Theory **7**

2.1 Measures, Probability and Information 7

2.2 Geometry of Information 9

2.3 Machine Learning as Statistical Inference 11

3 Relating Topology and Geometry of Latent Spaces **13**

3.1 Chern–Gauss–Bonnet 13

3.1.1 Theorema Egregium to Global Gauss–Bonnet 13

3.1.2 Meaning and Motivation of the Theorem 14

3.1.3 Euler Class, Pfaffian and Chern's General Theorem 15

3.1.4 The Theorem in 2 Dimensions 15

3.1.5 The Theorem in 4 Dimensions 16

3.2 Limits of the Euler Characteristic 19

3.3 Building a Suitable Latent Space Formalism 20

3.3.1 Enforcing Topology 20

3.3.2 Dimensionality of Latent Spaces 22

3.4 Gauss–Bonnet and Gravity 22

4 Methodology of Statistical Latent Spaces **25**

4.1 Motivating Variational Autoencoders 25

4.2 Topology-Inducing Architecture 27

4.2.1 The Toroidal Latent Space 28

4.2.2 The Spherical Latent Space 28

4.3 Preserving Topology 29

4.4 Latent Metric 30

4.5 Determinant of Latent Metric 33

4.6 Latent Curvature 33

4.7 Gaussian Decoder 34

4.7.1 Homoscedasticity as Numerical Necessity 35

4.7.2 Curvature as Measure of Non-Linearity 36

4.7.3 Metric Determinant 37

4.8 Amari–Chentsov Duality and Non-Metricity . . . 38
4.9 Dealing with the Curse of Dimensionality . . . 38
4.9.1 Generalization the Calculation of the Pfaffian . . . 38
4.9.2 Numerical Advantages of Extrinsic Curvature . . . 40

5 Experiment and Result 41
5.1 Setup . . . 41
5.1.1 Orbits of Group Action as Generator for Periodic Datasets . . . 41
5.2 Technical Implementation . . . 45
5.2.1 Integration . . . 46
5.3 Results . . . 49
5.3.1 Canonical Latent Geometry and Topology Mismatch . . . 49
5.3.2 Validating Topology . . . 61
5.3.3 Verifying Chern–Gauss–Bonnet . . . 71
5.3.4 Allowing Higher Dimensionality . . . 73

6 Discussion and Conclusion 77

7 Outlook and Generalization 81
7.1 On Identifying Symmetries . . . 81
7.2 Generalization of Chern–Gauss–Bonnet . . . 82
7.2.1 Amari Curvature . . . 82
7.2.2 Operators on Statistical Manifolds . . . 82
7.3 Diffusion Models . . . 85

8 Appendix 87
8.1 Explicit Proof of Chern–Gauss–Bonnet for Low Dimensionality . . . 87
8.1.1 $\mathbb{S}^2$ . . . 87
8.1.2 $\mathbb{T}^2$ . . . 88
8.2 Edge Crossing Flat Data Modelling . . . 91
8.3 Ressources and References . . . 94
8.4 Acknowledgements . . . 101

Chapter 1

Introduction

Most real-world high dimensional data can be described by a low dimensional representation. This lower dimensional manifold, or rather the union of several disjoint manifolds with varying dimension, is embedded into the ambient space of data. Generative artificial intelligence implicitly learns an approximate parametrization of this manifold and a density on it. Generating new data involves sampling from this density and decoding back into data space. This encoded representation is not just a compression of the high-dimensional data, but describes how the model organizes information and features. It furthermore shows learned importance of direction of variance, category boundaries or similarities.

If the process generating the observed data has a symmetry group, for example rotations, phase periodicity or gauge symmetries in physics, an approximatively correct representation should reflect the topology of the group's orbit. Standard generative machine learning uses the Euclidean $\mathbb{R}^n$ to build this structure, which can force tearing or discontinuities, while trying to represent non-trivial topologies. Although increasing the dimension is possible, a global parametrization of the data is lost, as latent coordinates then describe the ambient Euclidean space. This motivates topologically non-trivial models, enabling correct, continuous representation or investigations regarding symmetry breaking.

As coordinates of manifolds are arbitrary, curvature, geodesic distance and topological invariants are the intrinsic learned quantities on which investigations should focus. The theorem of Chern–Gauss–Bonnet ties local geometry together with global topology, relating *how infinitesimal changes in the low-dimensional representation behave* to *how the representation can be characterized globally*. Therefore, it creates a computable diagnostic that we can evaluate numerically to determine which topology the model has learned and how closely the representation approaches the targeted geometry. As data generation includes sampling from a distribution, the only canonical and meaningful choice of distance is one defined between distributions, therefore making an information geometric approach necessary.

As generative modelling is vastly growing not only in deployment but also in our dependence on it, we need principled, quantitative methods to interrogate and understand relevant mechanisms, not only by analysing output, but also by mathematically describing

internal representation. This creates the necessity of a mathematical approach to AI interpretability and safety, especially regarding "analysis of feature manifolds"[1].

Furthermore, these methods find use in particle physics, as the observation of both instrumental symmetries and fundamental physical symmetries benefits from methodical diagnostics of representation learning. Geometric and topological analysis of generative artificial intelligence models could help detect, verify or falsify symmetry structure in high dimensional detector data.

Some of the arising questions have already been answered. Latent spaces with prescribed topology have already been constructed [2] and investigated with respect to persistent homology [3], latent spaces have been endowed with Riemannian geometry [4] and curvature has been calculated for them [5]. Information geometry in machine learning itself is not new, as it has been applied to generative neural networks [6] for over 30 years and has come to fruition in broadly implemented methods [7]. The notion of investigating learned *latent* structures using information geometry is newer and of current interest (e.g. [8]). Contributions in this field, done by the CosmoStatistics group at the Department of Physics and Astronomy in Heidelberg, substantially motivated and guided the following work (especially important was a paper on information geometry of categorizing gluon jets [9]).

This thesis sits right at the intersection of these areas. It treats the latent space as a statistical Riemannian manifold by introducing the Fisher information metric, asks whether a global theorem of differential geometry can be verified on it and used to identify the symmetry structure of the data.

1.1 Contributions

We will address this, by deducing topological and geometric concepts related to the Chern–Gauss–Bonnet theorem applied to latent spaces and elaborate the methods needed to investigate latent topology and information geometry of variational autoencoders. Finally, we will verify that the learned latent geometry satisfies the Chern–Gauss–Bonnet relation numerically, while also comparing the curvature and performance of different topological latent spaces using datasets with different underlying topologies.

[1] This precise formulation closely follows that of Jacob Tsimerman in [1].

Chapter 2

Theory

2.1 Measures, Probability and Information

The acquisition and processing of information is a stochastic process, as repeated measurements of the same state yield varying outcomes. Its rigorous treatment therefore begins with a measure on a sample space X, taken to be finite-dimensional throughout[1]. Comparability across subsets of X requires a conservational constraint. This is done by introducing the Kolmogorov axioms [11], thereby restricting the space of all measures to $\mathcal{P}(X)$, where $P \in \mathcal{P}$ is positive and normalized.

From here on, we choose to simplify notation and calculations to densities p of measures P. Wherever it is necessary to return back to measures, it is stated so. This implies the assumption that a density exists and is well defined. Where not stated differently, the Lebesgue measure is assumed to be sufficient for integrating.

Beyond distributions over a single variable, we need distributions that relate the quantity x entering a measurement and the data y it produces. Therefore, we introduce the *joint* distribution, which factorizes in two equivalent ways

$$p(x, y) = p(x)p(y \mid x) = p(y)p(x \mid y) \tag{2.1}$$

and whose marginals follow by integrating out the other variable

$$p(x) = \int_Y p(x, y)\mathrm{d}y, \qquad p(y) = \int_X p(x, y)\mathrm{d}x. \tag{2.2}$$

Equating it with the second one results in

Theorem 2.1 (Bayes' theorem)**.**

$$p(x \mid y) = \frac{p(x)p(y \mid x)}{p(y)}, \qquad p(y) = \int_X p(x)p(y \mid x)\,\mathrm{d}x. \tag{2.3}$$

Here x is the unknown quantity of interest and y the measured data, so that $p(x)$ is the

[1]When X is infinite, $\mathcal{P}(X)$ "is infinite-dimensional and therefore functional analytical problems arise" [10]. We will assume these problems to be irrelevant for our purposes.

prior, $p(y \mid x)$, a function of x at the fixed, observed value of y, is the likelihood, $p(x \mid y)$ is the posterior and $p(y)$ is the evidence acting as the normalization constant. We further define the expectation of a function f under the posterior as

$$\mathbb{E}_{x|y}[f] = \mathbb{E}_{x\sim p(\,\cdot\,|y)}[f(x,y)] \tag{2.4}$$

$$= \int_X f(x,y)\, p(x \mid y)\, \mathrm{d}x, \tag{2.5}$$

which is itself a function of the observed data y.

Measuring an information state $x_0 \in X$ does not return x_0 itself, but an observation y with a probability distribution $p(\,\cdot \mid y)$ on X, which localizes x_0 only up to the measure of a subset $U \ni x_0$, where $P(U \mid y)$ quantifies how confidently the measurement resolves the state at the scale of U.

This presupposes a measurable structure on X. If X carries a topology τ (see [12]), the canonical choice is the Borel σ-algebra $\mathcal{B}(\tau)$ generated by the open sets, so that every open set is assigned a probability. If one assumes further that X is a topological manifold, which is guaranteed in the cases, we will be dealing with, global properties of X are characterised by topological invariants. One such invariant is the **Euler Characteristic** $\chi(X)$.

Let X to be a compact surface. Choosing a decomposition into finitely many curvilinear triangles, only meeting at shared edges and vertices, we can define the Euler characteristic [13] as follows

Definition 2.2 (Euler characteristic through triangulation)**.** *Let X be a compact surface with a regular triangulation (V, E, F), then*

$$\chi(X) = V - E + F, \tag{2.6}$$

where V is the number of vertices, E the number of edges and F the number of triangles. This definition can be generalized to simplicial complexes [14]. $\chi(X)$ is a topological invariant of X, meaning it does not depend on the triangulation. For a closed, orientable surface of genus g (genus characterizes the number of handles or holes a topological sphere admits), this is equivalent to

Definition 2.3 (Euler characteristic through genus)**.** *Let X be a closed, orientable surface with genus g, then*

$$\chi(X) = 2 - 2g. \tag{2.7}$$

This can be generalized to compact surfaces and orientable manifolds [15], with $\chi = 0$ for closed odd-dimensional manifolds.

As it is necessary, to not only compare densities evaluated at single data points, but also to define differences and proximity between them, we introduce the **Kullback–Leibler divergence** [16]. Originally, a measure of the information available to discriminate between statistical hypotheses, the KL-divergence, as we will reference it from now on, is defined as

Definition 2.4 (Kullback–Leibler divergence)**.**

$$D_{\mathrm{KL}}(p \,\|\, q) = \int_Y p(y) \log \frac{p(y)}{q(y)} \,\mathrm{d}y = \mathbb{E}_{y\sim p}\left[\log \frac{p(y)}{q(y)}\right]. \tag{2.8}$$

As we can see, this functional is not symmetric in its arguments. Given some continuously differentiable parametrization $\theta = (\theta^1, \ldots, \theta^n) \in \Theta \subseteq \mathbb{R}^n$ of the inferred information x, resulting in $p(\cdot \mid \theta)$, we can expand the KL-divergence to second order. This results in

$$D_{\mathrm{KL}}\left[p(\cdot \mid \theta) \| p(\cdot \mid \theta + \delta\theta)\right] \tag{2.9}$$

$$= \tfrac{1}{2}\mathbb{E}_{y\sim p(\cdot|\theta)}\left[\partial_i \log p(y \mid \theta)\, \partial_j \log p(y \mid \theta)\right] \delta\theta^i \delta\theta^j + \mathcal{O}(\|\delta\theta\|^3). \tag{2.10}$$

$\partial_i = \frac{\partial}{\partial \theta^i}$ denotes the partial derivative with respect to the parameters of p.

2.2 Geometry of Information

We define

$$F_{ij}(\theta) = \mathbb{E}_{y\sim p(\cdot|\theta)}[\partial_i \log p(y|\theta) \partial_j \log p(y|\theta)] \tag{2.11}$$

as the **Fisher information metric** and identify it as Riemannian. Furthermore, it is the, up to constant factorization and sufficient statistic [17] unique Riemannian metric on continuously parametrized distributions and precisely the Hessian of relative entropy at coincidence. This establishes the basic framework of information geometry (most of what follows can be referenced by [10, 18–20]).

Let Θ be an open subset of Euclidean space[2] and assume that the map $\Theta \ni \theta \mapsto p(\cdot \mid \theta)$ is injective and smooth. Furthermore, assume that $F_{ij}(\theta)$ is finite and of full rank throughout Θ. Then a family of distributions with the Fisher information metric defines a smooth n-dimensional Riemannian manifold, which we will call a **statistical manifold**

$$\mathcal{S}(\Theta) = \{p(\cdot \mid \theta) : \theta \in \Theta\}. \tag{2.12}$$

For our purposes, we will exclusively use families of conditional distributions as statistical manifolds. Generally, all probability measures over an orientable manifold can be considered as infinite dimensional statistical manifold, with the smooth continuously parametrized case as finite dimensional submanifold.

The following will assume fundamental understanding of differential geometry (foundations can be found in, e.g., [21]), as only necessary definitions are included.

The **Levi-Civita connection** is the unique affine connection on a Riemannian manifold that is torsion-free and metric-compatible. In coordinates it is given by the Christoffel symbols

$$\Gamma^k{}_{ij} = \frac{1}{2} g^{kl}(\partial_i g_{jl} + \partial_j g_{il} - \partial_l g_{ij}), \tag{2.13}$$

[2] As this thesis focuses on the case, where Θ has non-trivial topology this has to be altered to a local Euclidean parametrization of Θ, see Section 4.4.

where g_{ij} denotes the Riemannian metric, in our case the Fisher information metric and g^{ij} its inverse. The **Riemann curvature tensor** is a $(1,3)$ tensor measuring the failure of second covariant derivatives to commute. In coordinates it can be defined using the Christoffel symbols

$$R^i{}_{jkl} = \partial_k \Gamma^i{}_{jl} - \partial_l \Gamma^i{}_{jk} + \Gamma^i{}_{ka}\Gamma^a{}_{lj} - \Gamma^i{}_{la}\Gamma^a{}_{kj}. \tag{2.14}$$

If R vanishes identically, the manifold is locally isometric to Euclidean space.

An **orthonormal frame** is a local basis $\{e_i\}$ of the tangent bundle with $g(e_i, e_j) = \delta_{ij}$. The dual coframe $\{\epsilon^i\}$ satisfies $\epsilon^i(e_j) = \delta^i{}_j$. The connection form ω is defined as $\nabla e_j = \omega^i{}_j \otimes e_i$.

The **Cartan structure equations** express torsion and curvature as forms,

$$\mathrm{d}\epsilon^i + \omega^i{}_j \wedge \epsilon^j = T^i, \tag{2.15}$$

$$\mathrm{d}\omega^i{}_j + \omega^i{}_k \wedge \omega^k{}_j = \Omega^i{}_j, \tag{2.16}$$

where T^i is the torsion form and $\Omega^i{}_j$ the curvature two-form, related to the Riemann tensor by $\Omega^i{}_j = \frac{1}{2} R^i{}_{jkl}\, \epsilon^k \wedge \epsilon^l$. For the Levi-Civita connection the first equation reduces to $\mathrm{d}\epsilon^i = -\omega^i{}_j \wedge \epsilon^j$, since T^i vanishes and together with the antisymmetry of ω_{ij} this determines the connection uniquely.

Integration on manifolds with coordinates $\{x^i\}$ requires a volume form rather than a coordinate measure, since coordinate volume $\mathrm{d}^n x$ alone is not reparametrization invariant. On an oriented Riemannian manifold the metric supplies a choice of covolume,

$$\mathrm{d}V = \sqrt{\det g}\; \mathrm{d}x^1 \wedge \ldots \wedge \mathrm{d}x^n = \epsilon^1 \wedge \ldots \wedge \epsilon^n. \tag{2.17}$$

The factor $\sqrt{\det g}$ exactly compensates the Jacobian arising under a change of coordinates.

Information geometry contains Riemannian geometry, but provides more structure than it. Where a Riemannian manifold is equipped with a metric alone, from which the Levi-Civita connection follows uniquely, a statistical manifold carries a second invariant object. The **Amari-Chentsov tensor**

$$C_{ijk}(\theta) = \mathbb{E}_{y\sim p(\cdot|\theta)}[\partial_i \log p\; \partial_j \log p\; \partial_k \log p]\,, \tag{2.18}$$

is a totally symmetric 3-tensor, the third moment of the score of the distribution[3]. It basically describes the skewness of distributions, while the metric describes their variance. Since C is a 3-tensor it cannot correct the metric, but it can modify the connection, deforming it into a one-parameter family of **α-connections**

$$^{(\alpha)}\Gamma_{ijk} =^{\mathrm{LC}} \Gamma_{ijk} - \tfrac{\alpha}{2} C_{ijk}, \qquad \alpha \in \mathbb{R}, \tag{2.19}$$

[3]Higher moments of the score remain invariant but introduce no new geometric structure.

which reduces to the Levi-Civita connection at $\alpha = 0$. For $\alpha \neq 0$ the connections are dual with respect to the metric and are only metric-compatible in pairs.

$$X\langle Y, Z\rangle = \langle \nabla_X^{(\alpha)} Y, Z\rangle + \langle Y, \nabla_X^{(-\alpha)} Z\rangle, \tag{2.20}$$

with X, Y, Z vector fields. Following, C exactly measures metric incompatibility, as

$$\nabla_i^{\alpha} F_{jk} = \alpha C_{ijk}. \tag{2.21}$$

Curvature is a property of the connection, therefore a statistical manifold possesses a family of curvatures, which collapses to be Riemannian if C vanishes. In the following we will focus on the Levi-Civita connection of the Fisher information metric, as Chern–Gauss–Bonnet is a theorem of Riemannian geometry.

2.3 Machine Learning as Statistical Inference

In the simplest terms, a machine learning model is a function approximator [22]. A function f_ϕ from a hypothesis class $\mathcal{H} = \{f_\phi : \phi \in \Phi\}$ infers x from data y, $f : Y \to X$. One now observes a finite sample $((y_1, x_1), \ldots, (y_m, x_m)) \overset{\text{iid}}{\sim} P$, where P is a fixed but unknown joint probability measure on $Y \times X$. Learning is the act of picking a member of $\mathcal{H}$ that minimizes the expected loss incurred on unseen data and allows us to approximate P.

In general, the object we called a *function* above, does not need to be a function in the mathematical sense. This becomes apparent as we deconstruct P into marginal P_Y and conditional $P(\cdot|y)$

$$P = P_Y \otimes P(\cdot|y) \tag{2.22}$$

A function $f : Y \to X$ induces a distribution of $(y, f(y))$ with $y \sim P_Y$. Therefore, f represents P iff

$$P(\cdot|y) = \delta_{f(y)} \text{ almost everywhere with respect to the probability measure of } Y. \tag{2.23}$$

This Dirac measure can be interpreted as deterministic network output, i.e. normalized distributions with vanishing variance[4], which is an assumption true for supervised hard labeling without noise (such as an idealization of a categorization task). Generative models seek a model distribution Q_ϕ approximating P_Y, which until now was just a nuisance factor. This is done based on a finite observation $y_1, \ldots, y_m \overset{\text{iid}}{\sim} P_Y$.

This motivates defining the objective of machine learning more generally, as fitting a conditional family $P_\phi(\cdot|y)$, where the conditioning variable is observed and a marginal model $Q_\phi \approx P_Y$ where it is not. From now on, we will return to using a density based notation. It is important to notice that a conditional model is parametrized by two sets of

[4]We will later see that a homoscedastic Gaussian decoder's Fisher information metric is $F \propto \sigma^{-2}$, which identifies these point predictions as metrically divergent.

parameters. Keeping one fixed, we can distinguish

$$\phi \mapsto p_\phi = p_\phi(\cdot|y)|_{y \text{ fixed}} \tag{2.24}$$

and

$$y \mapsto p(\cdot|y) = p_\phi(\cdot|y)|_{\phi \text{ fixed}}. \tag{2.25}$$

$\phi \in \Phi$ represents the network weights, meaning the tunable parameters with respect to which the loss function of the model gets minimized. Inducing the Fisher metric onto Φ is exactly what led Amari to define a *natural* gradient descent [7]. Heuristically one could identify this parametrized family p_ϕ and the operations on Φ with answering *how the model was found*. A large class of generative models is constructed using the following architecture. An encoder model $\mathcal{E}$ maps the input data into a low-dimensional space. This so called *feature* or *latent*[5] space Θ is a manifold, which represents the features or structure embedded in the input data. Then, a decoder model $\mathcal{D}$ maps this latent space back into data space, creating newly generated samples. Therefore, we actually need *two* models, which both approximate a conditional distribution

$$\mathcal{E}_\psi : Y \to \mathcal{P}(\Theta), \qquad y \mapsto q_\psi(\,\cdot \mid y), \tag{2.26}$$

$$\mathcal{D}_\phi : \Theta \to \mathcal{P}(Y), \qquad \theta \mapsto q_\phi(\,\cdot \mid \theta), \tag{2.27}$$

together with a fixed prior $\pi(\theta)$ (see Section 4.1). The data is assumed to be drawn from an unknown p_{data}.

The goal is to fully approximate the true distribution of samples in data space by calculating the marginal

$$q(y) = \int q(y|\theta)\pi(\theta)\mathrm{d}\theta \approx p_{\text{data}}(y). \tag{2.28}$$

This is exactly where a deterministic decoder (as described in Eq. (2.23)) would collapse, as the marginal admits no density. This formulation of machine learning models necessitates including an additional noise distribution. Θ is the latent space, whose points θ parametrize the decoder $\mathcal{D}(\theta) = q(\cdot|\theta)$, therefore we will often use the term *parameter space* when describing the latent space Θ. Observing Θ with the Fisher information metric at fixed ϕ lets us ask the question *what the model represents*. Understanding Θ and its geometry, with the aim of explaining of *how* generative machine learning works through *what* the model represents in its latent space, is exactly what this thesis sets out to do.

[5] Lat. *latēns*, *latēre*, meaning "to lie hidden, to lurk, to be concealed".

Chapter 3

Relating Topology and Geometry of Latent Spaces

3.1 Chern–Gauss–Bonnet

3.1.1 Theorema Egregium to Global Gauss–Bonnet

The classical theory of surfaces in $\mathbb{R}^3$ defines the curvature of two-dimensional surfaces extrinsically, meaning in terms of how the surface is embedded into the ambient Euclidean space. In *Disquisitiones generales circa superficies curvas* [23], C. F. Gauss showed that this quantity is fully determined by the metric, that is, by lengths and angles measurable within the surface itself, independent of how it is embedded into ambient space.

$$K = \frac{R_{1212}}{\det g} \tag{3.1}$$

where R is the Riemann curvature tensor. This *Theorema Egregium* [24] can be formulated as such

Theorem 3.1 (Theorema Egregium)**.** *Let $S, S' \subset \mathbb{R}^3$ be regular surfaces and $\varphi\colon S \to S'$ a local isometry. Then $K' \circ \varphi = K$.*

This enables defining abstract Riemannian manifolds with intrinsic geometry, as the curvature is an invariant of the metric, not of an embedding into $\mathbb{R}^n$. In the same publication, Gauss also proposed and proved a relation between the metrically integrated curvature of a geodesic triangle and the angular differences to the Euclidean value. This is now known as the local Gauss–Bonnet theorem.

Theorem 3.2 (Local Gauss–Bonnet)**.** *Let $\Delta \subset S$ be a geodesic triangle with interior angles $\alpha_1, \alpha_2, \alpha_3$. Then*

$$\int_\Delta K \mathrm{d}A = \sum_{i=1}^{3} \alpha_i - \pi. \tag{3.2}$$

O. Bonnet then extended this [25] to regions D homeomorphic to a closed disc and

bounded by piecewise smooth curves.

$$\int_D K \, \mathrm{d}A + \int_{\partial D} \kappa_g \, \mathrm{d}s + \sum_j \varepsilon_j \;=\; 2\pi, \tag{3.3}$$

where κ_g denotes the geodesic curvature and ε_j the exterior angles. Summing over the triangulation of a closed, oriented Riemannian manifold, we find that all inner contributions cancel pairwise and the combinatorial count reassembles exactly the a global statement.

Theorem 3.3 (Global Gauss–Bonnet)**.** *Let $(\mathcal{M}, g)$ be a closed, oriented Riemannian 2-manifold. Then*

$$\frac{1}{2\pi} \int_{\mathcal{M}} K \, \mathrm{d}A \;=\; \chi(\mathcal{M}). \tag{3.4}$$

3.1.2 Meaning and Motivation of the Theorem

The most natural route to visualize curvature might be holonomy. By parallel transporting a vector around a closed loop, which bounds a simply connected region U with curvature K, the vector returns rotated

$$\Delta\varphi = \int_U K \mathrm{d}A, \tag{3.5}$$

where φ is an angle. The Gauss–Bonnet theorem now states that if a closed surface is covered by regions U, the summed angular difference is a multiple of 2π, determined by the topology.

Following Gauss' intuitive way of thinking, one might be able to imagine that for closed, oriented manifold local changes in the curvature have to cancel in the integral, as the total is defined by the topology alone. This is a genuine constraint on which Riemannian metrics a manifold can admit. The sign of the Euler characteristic forces the curvature to be negative or positive somewhere on the manifold, thus determining whether the manifold is spherical, flat or hyperbolic. Furthermore, it becomes apparent how meaningful a constraint on total curvature can be, once one compares it with the theorem of Poincaré–Hopf [26], which asigns precise requirements to isolated zeros p_i of arbitrary continuous vector fields X on an oriented, closed manifold, only dependent on the topologically invariant Euler characteristic.

This motivates the following equality

$$\int_{\mathcal{M}} K \mathrm{d}A = 2\pi \sum_i \mathrm{ind}_{p_i}(X) = 2\pi\chi(\mathcal{M}) \tag{3.6}$$

A direct application of this theorem is the *Hairy-Ball theorem* as introduced in [27], since $\chi(\mathbb{S}^2) = 2 \neq 0$, every continuous tangent vector field on the two-sphere must vanish somewhere, whereas on the torus, with $\chi(\mathbb{T}^2) = 0$, nowhere-vanishing fields can exist.

3.1.3 Euler Class, Pfaffian and Chern's General Theorem

The non-trivial generalization of the classic Gauss–Bonnet theorem to higher-dimensional Riemannian manifolds with even dimension was done by Shiing-Shen Chern, thereby providing the first proof [15]. It is often quoted as

Theorem 3.4 (Chern–Gauss–Bonnet)**.** *Let $\mathcal{M}$ be a closed, oriented Riemannian manifold of even dimension $\dim \mathcal{M} = 2n$ and let Ω denote its curvature two-forms with respect to a local orthonormal frame. Then*

$$\chi(\mathcal{M}) = \int_{\mathcal{M}} \mathrm{eu}(\Omega), \tag{3.7}$$

where $\chi(\mathcal{M})$ is the Euler characteristic and $\mathrm{eu}(\Omega)$ the Euler class (defined below) of the manifold.

Definition 3.5 (Euler class)**.** The Euler class is defined as

$$\mathrm{eu}(\Omega) = \frac{1}{(2\pi)^n}\mathrm{Pf}(\Omega) \tag{3.8}$$

where $\mathrm{Pf}(\Omega)$ denotes the Pfaffian (general definitions can be found in [28], definitions of Euler class and Pfaffian in the forms we are using can be found in, e.g., [29]).

The Pfaffian is the canonical square root of the determinant of a real-valued skew-symmetric $2n \times 2n$ matrix A.

$$\mathrm{Pf}(A)^2 = \det(A) \tag{3.9}$$

$$\mathrm{Pf}(A) \coloneqq \frac{1}{2^n n!}\, \epsilon^{i_1 i_2 \ldots i_{2n}} \prod_{k=1}^{n} a_{i_{2k-1} i_{2k}}, \tag{3.10}$$

where $\epsilon^{i_1 i_2 \ldots i_{2n}}$ is the Levi-Civita symbol and $a_{i_{2k-1} i_{2k}}$ the entries of A. In contrast to the determinant, the Pfaffian is invariant under $SO(2n)$, but admits a sign flip under $O(2n) \backslash SO(2n)$, which is why it depends on a choice of orientability. One might wonder how this polynomial embodies the total information of curvature of the manifold and why it, rather than the determinant of the curvature form, appears in the formula. The answer is that one needs to construct a differential form from Ω which is invariant under change of frame. As $\Omega \mapsto B^\top \Omega B$ under a change of orthonormal frame, $\mathrm{Pf}(\Omega)$ needs to be globally invariant under this transformation. The Pfaffian can be proven to be unique with respect to this requirement.

3.1.4 The Theorem in 2 Dimensions

Calculating the Pfaffian for $d = 2 = 2 \cdot 1$:

Following from Eq. (3.10), we have

$$\begin{aligned}\mathrm{Pf}(\Omega) &= \frac{1}{2^1 \cdot 1!}\epsilon^{ij}\Omega_{ij} \\ &= \frac{1}{2}\epsilon^{ij}\Omega_{ij} \\ &= \frac{1}{2}(\epsilon^{12}\Omega_{12} + \epsilon^{21}\Omega_{21}) \\ &= \frac{1}{2}(\Omega_{12} - \Omega_{21}) \end{aligned} \tag{3.11}$$

using $\Omega_{ij} = -\Omega_{ji}$, we conclude

$$\mathrm{Pf}(\Omega) = \Omega_{12}, \tag{3.12}$$

for the case where $d = 2$. This results in

$$\int_{\mathcal{M}} eu(\Omega) = \frac{1}{2\pi}\int_{\mathcal{M}} \Omega_{12} = \chi(\mathcal{M}). \tag{3.13}$$

This is equivalent to Theorem 3.3, as

$$\Omega^1{}_2 = R^1{}_{212}\epsilon^1 \wedge \epsilon^2 = K\mathrm{d}V. \tag{3.14}$$

A short remark on a different view onto the meaning of Gauss–Bonnet, motivated by this formulation of the theorem:

$$\Omega^1{}_2 = \mathrm{d}\omega^1{}_2 \tag{3.15}$$

is equivalent to saying that the curvature form is locally exact.

$$\int_{\mathcal{M}} K\mathrm{d}A = \int_{\mathcal{M}} \mathrm{d}\omega^1{}_2 \tag{3.16}$$

should now vanish by Stokes' theorem, but $\omega^1{}_2$ requires a global orthonormal frame, which exists exactly for $\chi(\mathcal{M}) = 0$, where the integral actually vanishes. Therefore Section 3.1.2 can be modified by stating that Gauss–Bonnet and the Euler characteristic measure how globally non-exact the locally exact curvature form is. Explicit proofs for standard $d = 2$ manifolds ($\mathbb{S}^2$ and $\mathbb{T}^2$) can be seen in Chapter 8.

3.1.5 The Theorem in 4 Dimensions

Analogously to $d = 2$, we will now derive the explicit Chern–Gauss–Bonnet integrand for $d = 2n = 4$, assuming a general Riemannian metric $g = (g_{ij})$. Also, we will assume metric compatibility of the connection and no torsion. As we recall, the Pfaffian form of a real valued $2n \times 2n$ skew-symmetric matrix $A = (a_{ij})$ is defined as

$$\mathrm{Pf}(A) := \frac{1}{2^n n!}\sum_{i_1,\ldots,i_{2n}=1}^{2n} \epsilon^{i_1 i_2 \ldots i_{2n}} a_{i_1 i_2} a_{i_3 i_4} \ldots a_{i_{2n-1} i_{2n}}, \tag{3.17}$$

where $\epsilon^{i_1 i_2 \dots i_{2n}}$ denotes the Levi-Civita symbol. We identify A with the curvature form $\Omega = (\Omega_{ij})$, where each Ω_{ij} is a 2-form and $\Omega_{ij} = -\Omega_{ji}$. By definition, Eq. (3.10) becomes

$$\mathrm{Pf}(\Omega) := \frac{1}{2^n n!} \sum_{i_1,\dots,i_{2n}=1}^{2n} \epsilon^{i_1 i_2 \dots i_{2n}} \Omega_{i_1 i_2} \wedge \Omega_{i_3 i_4} \wedge \dots \wedge \Omega_{i_{2n-1} i_{2n}} \tag{3.18}$$

as the wedge product is the unique associative product on the exterior algebra of forms. In general just replacing the regular product by the wedge product would be ill defined, as for a p-form ω and a q-form ρ

$$\omega \wedge \rho = (-1)^{pq} \rho \wedge \omega. \tag{3.19}$$

But as all forms Ω_{ij} here are 2-forms, meaning $p \cdot q = 4$, the wedge product pairwise commutes. So applying this formula to $d = 4 = 2n$, $n = 2$, we get

$$\mathrm{Pf}(\Omega) = \frac{1}{4 \cdot 2} \epsilon^{ijkl} \Omega_{ij} \wedge \Omega_{kl} \tag{3.20}$$

Furthermore, we will use the definition of Ω using the Riemann curvature tensor R^i_{jkl}

$$\Omega^i{}_j = \frac{1}{2} R^i{}_{jkl} \, \mathrm{d}x^k \wedge \mathrm{d}x^l. \tag{3.21}$$

Although this is standard in the literature dealing with curvature of Riemannian manifolds, we will briefly derive it. It has to be noted here, that neither Greek nor Latin indices follow the conventions of general relativity, they are simply used to make distinguishment of different parts of tensors easier. We start by using the second Cartan structure equation for a torsionless geometry:

$$\begin{aligned}
\Omega^i{}_j &= \mathrm{d}(\Gamma^i{}_{jk} \, \mathrm{d}x^k) + \Gamma^i{}_{kl} \, \mathrm{d}x^l \wedge \Gamma^k{}_{mj} \, \mathrm{d}x^m \\
&= \partial_l \Gamma^i{}_{jk} \, \mathrm{d}x^l \wedge \mathrm{d}x^k + \Gamma^i{}_{kl} \, \Gamma^k{}_{mj} \, \mathrm{d}x^l \wedge \mathrm{d}x^m \\
&= \Big(\Gamma^i{}_{jm,l} + \Gamma^i{}_{kl} \, \Gamma^k{}_{mj} \Big) \mathrm{d}x^l \wedge \mathrm{d}x^m \\
&= \tfrac{1}{2} \Big[\Big(\Gamma^i{}_{jm,l} + \Gamma^i{}_{kl} \, \Gamma^k{}_{mj} \Big) - \Big(\Gamma^i{}_{jl,m} + \Gamma^i{}_{km} \, \Gamma^k{}_{lj} \Big) \Big] \mathrm{d}x^l \wedge \mathrm{d}x^m \\
&= \tfrac{1}{2} \Big(\Gamma^i{}_{jm,l} - \Gamma^i{}_{jl,m} + \Gamma^i{}_{kl} \, \Gamma^k{}_{mj} - \Gamma^i{}_{km} \, \Gamma^k{}_{lj} \Big) \mathrm{d}x^l \wedge \mathrm{d}x^m \\
&= \tfrac{1}{2} \, R^i{}_{jlm} \, \mathrm{d}x^l \wedge \mathrm{d}x^m
\end{aligned} \tag{3.22}$$

Now, explicitly computing Eq. (3.18), while using Eq. (3.22)

$$\begin{aligned}
\epsilon^{ijkl} \, \Omega_{ij} \wedge \Omega_{kl} &= \epsilon^{ijkl} \, g_{i\alpha} \Omega^\alpha{}_j \wedge g_{k\beta} \Omega^\beta{}_l \\
&= \frac{1}{4} \, \epsilon^{ijkl} \, g_{i\alpha} \, g_{k\beta} \, R^\alpha{}_{j\gamma\delta} \, R^\beta{}_{l\rho\sigma} \, \mathrm{d}x^\gamma \wedge \mathrm{d}x^\delta \wedge \mathrm{d}x^\rho \wedge \mathrm{d}x^\sigma \\
&= \frac{1}{4} \, \epsilon^{ijkl} \, R_{ij\gamma\delta} \, R_{kl\rho\sigma} \, \mathrm{d}x^\gamma \wedge \mathrm{d}x^\delta \wedge \mathrm{d}x^\rho \wedge \mathrm{d}x^\sigma
\end{aligned} \tag{3.23}$$

Using the following relation

$$\mathrm{d}x^i \wedge \mathrm{d}x^j \wedge \mathrm{d}x^k \wedge \mathrm{d}x^l = \epsilon^{ijkl}\sqrt{\det(g)}\mathrm{d}^4x \tag{3.24}$$

and identifying

$$\sqrt{\det(g)}\mathrm{d}^n x = \mathrm{d}\mu \tag{3.25}$$

as the n-dimensional covolume element, we conclude

$$\frac{1}{4}\epsilon^{ijkl}R_{ij\gamma\delta}R_{kl\rho\sigma}\epsilon^{\gamma\delta\rho\sigma}\mathrm{d}\mu = \frac{1}{4}\epsilon^{ijkl}\epsilon^{\gamma\delta\rho\sigma}R_{ij\gamma\delta}R_{kl\rho\sigma}\mathrm{d}\mu \tag{3.26}$$

In the last step we used the fact that in index notation, the elements are just real numbers and therefore commute. Rewriting, we get

$$\mathrm{Pf}(\Omega) = \frac{1}{32}\epsilon^{ijkl}\epsilon_{\gamma\delta\rho\sigma}R^{\gamma\delta}{}_{ij}R^{\rho\sigma}{}_{kl}\sqrt{det(g)}d^4x \tag{3.27}$$

where we used the symmetry under pairwise commutation of the Riemann tensor. Now, rewriting the two Levi-Civita symbols as determinant of Kronecker δ

$$\epsilon^{ijkl}\epsilon_{\gamma\delta\rho\sigma} = \mathrm{sgn}\left(\det(g)\right)\begin{vmatrix}\delta^i_\gamma & \delta^i_\delta & \delta^i_\rho & \delta^i_\sigma\\ \delta^j_\gamma & \delta^j_\delta & \delta^j_\rho & \delta^j_\sigma\\ \delta^k_\gamma & \delta^k_\delta & \delta^k_\rho & \delta^k_\sigma\\ \delta^l_\gamma & \delta^l_\delta & \delta^l_\rho & \delta^l_\sigma\end{vmatrix}. \tag{3.28}$$

As we want the metric to be Riemannian, its signature is Euclidean, meaning $\mathrm{sgn}(\det(g)) = +1$.

$$\epsilon^{ijkl}\epsilon_{\gamma\delta\rho\sigma} = \begin{vmatrix}\delta^i_\gamma & \delta^i_\delta & \delta^i_\rho & \delta^i_\sigma\\ \delta^j_\gamma & \delta^j_\delta & \delta^j_\rho & \delta^j_\sigma\\ \delta^k_\gamma & \delta^k_\delta & \delta^k_\rho & \delta^k_\sigma\\ \delta^l_\gamma & \delta^l_\delta & \delta^l_\rho & \delta^l_\sigma\end{vmatrix} \tag{3.29}$$

Writing this determinant out, we get

$$\begin{aligned}\epsilon^{ijkl}\epsilon_{\gamma\delta\rho\sigma} = {}&\delta^i_\gamma\Big[\delta^j_\delta(\delta^k_\rho\delta^l_\sigma - \delta^k_\sigma\delta^l_\rho) - \delta^j_\rho(\delta^k_\delta\delta^l_\sigma - \delta^k_\sigma\delta^l_\delta) + \delta^j_\sigma(\delta^k_\delta\delta^l_\rho - \delta^k_\rho\delta^l_\delta)\Big]\\ &-\delta^i_\delta\Big[\delta^j_\gamma(\delta^k_\rho\delta^l_\sigma - \delta^k_\sigma\delta^l_\rho) - \delta^j_\rho(\delta^k_\gamma\delta^l_\sigma - \delta^k_\sigma\delta^l_\gamma) + \delta^j_\sigma(\delta^k_\gamma\delta^l_\rho - \delta^k_\rho\delta^l_\gamma)\Big]\\ &+\delta^i_\rho\Big[\delta^j_\gamma(\delta^k_\delta\delta^l_\sigma - \delta^k_\sigma\delta^l_\delta) - \delta^j_\delta(\delta^k_\gamma\delta^l_\sigma - \delta^k_\sigma\delta^l_\gamma) + \delta^j_\sigma(\delta^k_\gamma\delta^l_\delta - \delta^k_\delta\delta^l_\gamma)\Big]\\ &-\delta^i_\sigma\Big[\delta^j_\gamma(\delta^k_\delta\delta^l_\rho - \delta^k_\rho\delta^l_\delta) - \delta^j_\delta(\delta^k_\gamma\delta^l_\rho - \delta^k_\rho\delta^l_\gamma) + \delta^j_\rho(\delta^k_\gamma\delta^l_\delta - \delta^k_\delta\delta^l_\gamma)\Big].\end{aligned} \tag{3.30}$$

As we will see, these products of Kronecker-δ symbols can be ordered into three different classes:

- $\delta^i_\gamma\delta^j_\delta\delta^k_\rho\delta^l_\sigma$, where the n-th upper index is contracted with the n-th lower indice
- $\delta^i_\delta\delta^j_\gamma\delta^k_\rho\delta^l_\sigma$, where only two index pairs match in position with respect to the indices of

the two Riemann tensors

- $\delta^i_\delta \delta^j_\gamma \delta^k_\sigma \delta^l_\rho$, where no index pairs are matching in position

Applying this to our remaining Riemann tensors, while taking into account the pairwise anti-symmetries of the Riemannian tensor, we get quadratic terms of

- full contractions resulting in the Ricci scalar: $\mathcal{R}^2 = R^{ij}{}_{ij} R^{\alpha\beta}{}_{\alpha\beta}$
- the Ricci tensor $R_{\gamma\delta} R^{\gamma\delta}$
- the Riemann tensor $R_{\gamma\delta\rho\sigma} R^{\gamma\delta\rho\sigma}$

We conclude that

$$\epsilon^{ijkl} \epsilon_{\gamma\delta\rho\sigma} R^{\gamma\delta}{}_{ij} R^{\rho\sigma}{}_{kl} = 4\mathcal{R}^2 - 16 R_{\gamma\delta} R^{\gamma\delta} + 4 R_{\gamma\delta\sigma\rho} R^{\gamma\delta\sigma\rho}. \tag{3.31}$$

The end result is

$$\mathrm{Pf}(\Omega) = \frac{1}{32} \epsilon^{ijkl} \epsilon_{\gamma\delta\rho\sigma} R^{\gamma\delta}_{ij} R^{\rho\sigma}_{kl} \sqrt{\det(g)} \mathrm{d}^4 x \tag{3.32}$$

$$= \frac{1}{8} (\mathcal{R}^2 - 4 R_{\gamma\delta} R^{\gamma\delta} + R_{\gamma\delta\sigma\rho} R^{\gamma\delta\sigma\rho}) \sqrt{\det(g)} \mathrm{d}^4 x. \tag{3.33}$$

And inserted into the theorem, we recover

$$\chi(\mathcal{M}) = \frac{1}{32\pi^2} \int_{\mathcal{M}} (\mathcal{R}^2 - 4 R_{\gamma\delta} R^{\gamma\delta} + R_{\gamma\delta\sigma\rho} R^{\gamma\delta\sigma\rho}) \mathrm{d}\mu. \tag{3.34}$$

3.2 Limits of the Euler Characteristic

Chern–Gauss–Bonnet is generally only defined for manifolds with even dimension (or in some forms for hypersurfaces in odd-dimensional Euclidean space). This is not because the different components fail to be well defined, but rather that the theorem turns vacuous in odd dimensions. Neither the Euler characteristic, nor the Euler class admit a non-vanishing value, when applied in odd rank. Heuristically, one can regard $\chi(\mathcal{M}) = 0$ for $\dim(\mathcal{M}) = 2n - 1$, where $n \in \mathbb{N}$, as an a priori definition and it can easily be seen that the Pfaffian in odd dimensions is not defined, as the determinant of a skew-symmetric matrix A in odd dimensions m vanishes

$$\det(A) = \det(-A^T) = (-1)^m \det(A) = -\det(A). \tag{3.35}$$

Integrating the vanishing Pfaffian therefore always equals a vanishing Euler characteristic, reducing the theorem to a tautological $0 = 0$. To prove this rigorously, one has to engage seriously with Chern's work on characteristic classes.

In addition to vanishing in odd dimensions, the Euler characteristic is not a bijection between topologies and integers. For closed orientable surfaces it happens to be complete, since $\chi = 2 - 2g$ determines the genus and therefore the surface up to homeomorphism, but

this is only true in two dimensions and for orientable surfaces. Dropping either assumption destroys injectiveness, as the torus and the Klein bottle both satisfy $\chi = 0$. In higher dimensions it is sufficient for some of the Betti numbers [30] to be equivalent, to have identical Euler characteristic. Finer invariants, as the fundamental group or the homology groups separate more cases, but none of them classify general manifolds completely and uniquely.

3.3 Building a Suitable Latent Space Formalism

Following Theorem 3.4, there are two main requirements for the Chern–Gauss–Bonnet theorem to be applicable to a manifold: compactness and even dimension. It is furthermore necessary to ensure a Riemannian metric, which is at least C^2, orientability and an empty or piecewise smooth boundary. As χ is additive, we are even able to cover cases where the domain is not connected, if the other requirements are met. The metric introduced below is at least C^2 and in our case non-degenerate.

In general, latent spaces are chosen to be $\mathbb{R}^n$, therefore being unbounded and non-compact, but there is reason enough to compactify them. As $\mathbb{R}^n$ itself is unbounded, countable encoder mappings of samples will never be able to fill the latent space. Although we will be using a statistical approach, which means that the decoder will always have support on the latent domain, there is a point where the density is so low that no sensible samples can be reconstructed. Introducing compactness by a boundary or by closing the latent space, would ensure that the decoder has been trained on the whole latent space.

Boundaries always lead to worse encoded and decoding stripes parallel to the boundary, as samples in there cannot have symmetric support. Therefore we will not compactify the conventional latent space by introducing borders to $\mathbb{R}^n$, but rather make the latent space compact after training, by choosing a suitable compact region of $\mathbb{R}^n$. Analogously to Eq. (3.3), Chern provided a generalization for the case where

$$\mathcal{M} \text{ compact}, \ \partial\mathcal{M} \neq \emptyset$$

$$\int_{\mathcal{M}} \frac{\mathrm{Pf}(\Omega)}{(2\pi)^n} \ + \ \int_{\partial\mathcal{M}} \Pi \ = \ \chi(\mathcal{M}), \tag{3.36}$$

where Π is the transgression or boundary form, a correction term built from the second fundamental form of $\partial\mathcal{M}$. This makes the theorem applicable to compact manifolds in general, enabling us to integrate over parts of conventional Euclidean latent spaces.

3.3.1 Enforcing Topology

All properties of the model that are not invariant under coordinate change are not to be interpreted as learned properties of the data. That is, distances and clustering in an arbitrary metric are not representative of the data. This is the heuristic motivation to investigate topology (alongside geometric invariants) as a characteristic latent diagnostic,

both by enforcing and measuring it. On a deeper level, as stated in the introduction, most data is either invariant or equivariant under some form of transformation. This could be through periodic variables, rotations, disconnected class clusters or the underlying symmetries and group structures of particle detector data for example. These observations of symmetries or periodicities in data lead to the conclusion that the underlying structure of data can admit non-trivial topology. As conventional latent spaces are isomorphic to $\mathbb{R}^n$, the encoder is forced to either tear the sub-data manifold, or cover it non-injectively, depending on available dimension. This creates artefacts or low-density regions in latent space, which lead to bad generalization and reconstruction. Furthermore, matching the latent dimension exactly to the intrinsic dimension of the represented data enables a global parametrization. Increasing the latent dimension means that the data becomes a learned submanifold of the latent space, which first has to be identified before an intrinsic interpretation is possible. This motivates matching topology of the latent space to the one underlying the data. In this case study, we will focus on two latent model-topologies, which we will compare to the flat Euclidean one. For reasons of visualization and computation, we will entirely work in two-dimensional spaces, although all discussed properties can be generalized.

Two-dimensional, closed orientable surfaces are completely classified by their genus. Here the 2-sphere $\mathbb{S}^2$ and the 2-torus $\mathbb{T}^2$ are the only surfaces[1] with non-negative Euler characteristic. All surfaces with higher genus are hyperbolic and admit negative Euler characteristic. Combining Gauss–Bonnet with the uniformization theorem, as formulated in [31], we can conclude that $\mathbb{S}^2$ and $\mathbb{T}^2$ are the only closed 2-surfaces which are not hyperbolic. The sphere is the unique elliptic surface and the torus the unique flat surface, admitting Euclidean geometry. This makes the torus the easiest compactification of $\mathbb{R}^n$ and the natural domain for describing periodic boundary conditions, as $\mathbb{T}^n = (\mathbb{S}^1)^n$, with every dimension representing a cyclic property. The general sphere $\mathbb{S}^n$ is the canonical way of representing directional data or orientations. Furthermore, the torus contributes nontrivial topology with zero intrinsic curvature, whereas the sphere contributes constant positive curvature with trivial topology. Applying Poincaré-Hopf, we can further follow, that the torus is the only closed orientable surface admitting a non-vanishing vector field. Additionally, $\mathbb{T}^2$ is the only surface that is isomorphic to a Lie group ($\mathbb{S}^1 \times \mathbb{S}^1$). The sphere on the other hand, is the unique simply connected closed surface with trivial fundamental group, therefore it is the "no holes" opposite of the "admitting one hole" topology of the torus. Or with mathematical rigor, they separate exactly the two cases non-trivial *curvature* (with trivial topology, i.e. simply connected) and non-trivial *topology* (with trivial curvature). Interesting for further studies, the sphere's Euler characteristic forbids a global chart, which would be interesting in network training, how this forced singularity of divergence and flow is dealt with. Furthermore, harmonic analysis is natural to both the torus and the sphere, as one can easily define eigenbases from Fourier series on $\mathbb{T}^2$ and from spherical harmonics on $\mathbb{S}^2$.

[1] surfaces in the sense that they are two-dimensional

Although this topological approach seems to be related to contrastive learning, there are conceptual differences, which are essential to the understanding of our application. Contrastive learning also normalizes n-dimensional data by projecting it onto $\mathbb{S}^{n-1}$, which resembles, what we will do. The difference is that contrastive learning operates in pairs. Representations are pulled together when the samples are similar and pushed apart, when they differ, thereby creating a contrast. This is a practice we will not be doing. We will simply enforce network architecture to encode into and decode from a latent space, which is spherical.

3.3.2 Dimensionality of Latent Spaces

Latent spaces of machine learning models usually range in dimensionality from around ten to a few hundred, depending on complexity or number of distinct characterization possibilities. Choosing a suitable dimension is often quite unique to the problem and both extremes have disadvantages. A too low latent dimension leads to bad reconstruction and blurring of the data. The mathematical reason is often simply that the data manifold cannot be embedded in the chosen dimension, which leads to systematic performance issues. Too high a latent dimension can lead to a model not learning enough structure, as complex structures might be represented as linear in higher dimensions (a concept which is used in reservoir computing [32]). Variational autoencoders, as the ones we will be using, tend to not use the redundant dimensions, so a slightly over-dimensionalized latent space costs little.

Generally, latent spaces can admit odd dimension, which would make them not suitable for the Chern–Gauss–Bonnet theorem. Therefore, reducing the latent space by one dimension might seem appropriate. There are a few algorithmic approaches to dimensionality reduction, as this has always been a problem in information processing, but in general, constructing a mapping into a manifold with lower dimensionality while keeping the topological structure is a non-trivial problem. Therefore, simply always choosing the next higher even dimension already in building the model's architecture is the easiest way to ensure applicability of Chern–Gauss–Bonnet.

3.4 Gauss–Bonnet and Gravity

As curvature of latent spaces is a gravity-type research question, there are conceptual parallels between trying to improve methods of geometric machine learning and improving the completeness of the theoretical framework and fundamental work of general relativity.

There are multiple approaches to improve Einstein's field equations, some motivated by observational reasoning, some by mathematical or philosophical considerations. Gauss–Bonnet gravity, as proposed by Lanczos [33, 34], proved and generalized by Lovelock [35, 36], is a modification of the field equations, done by adding a term which involves topology of the spacetime. This is done in such a way that no additional fields other than the ones of general relativity and no derivatives of the metric higher than second order are needed,

therefore creating a quite elegant extension, while remaining faithful to Einstein's original formulation. The result derived in Section 3.1.5 is exactly the geometric object, wich is known in the literature as the *Gauss–Bonnet gravity* term. The term derived in Eq. (3.32), which we will call $\mathcal{G}$, is simply added to the Lagrangian, motivated by involving a global curvature statement, which results in a topological invariant. Therefore, the Lagrangian can be formulated in the following way

$$\mathcal{L} = \sqrt{-g}[\alpha_0 + \alpha_1 \mathcal{R} + \alpha_2 \mathcal{G}], \tag{3.37}$$

where $\mathcal{L}$ denotes the Lagrangian and g the determinant of a Lorentzian[2] metric, $\mathcal{R}$ is the Ricci scalar, α_i constant prefactors and $\mathcal{G}$ part of the Pfaffian in $d = 4$. Integrated over a space time region with $d = 4$ has a variation with respect to the metric g^{ij} that vanishes identically, which is needed to fulfil sensible field equations.

$$\delta\mathcal{S} = \delta \int \mathcal{L} \mathrm{d}^4 x = 0 \tag{3.38}$$

The actual mathematical reason is as proven in Section 3.1.5. Therefore, this result seems trivial, as varying a constant always vanishes, but Lovelock showed that integrating this with respect to higher dimensions results in non-trivial dynamical field equations. Now, training a machine learning model is analogous to finding the global minimum of a potential, as

$$\mathcal{L}_{\text{loss}}(\theta) = \sum_i l(\theta, y_i), \tag{3.39}$$

with gradient flow solving

$$\dot{\theta} = -\nabla \mathcal{L}_{\text{loss}}. \tag{3.40}$$

The analogy to an action of a physical system becomes apparent if one looks at methods introducing momentum into loss optimization, recovering variational dynamics and Euler–Lagrange equations (we will not pursue a thermodynamic framing of machine learning, although it grants an even more natural description). Now, it might get obvious which analogue is built here, as one could think of adding the Chern–Gauss–Bonnet equation to the loss function of a machine learning model to try to include global curvature of the latent space into the learning process. Therefore, possibly regularizing the model's parameters towards a metric for which the integral agrees with the imposed topological invariant. This is in fact not the case, as neither the topological invariant nor the integral admits a non-vanishing gradient with respect to network parameters.

If we formulate the integral as a functional

$$\int_{\mathcal{M}} \mathrm{Pf}(\Omega[g_\phi]) \propto \int_{\mathcal{M}} \mathcal{G}(\Omega[g_\phi]) \sqrt{\det(g)} \mathrm{d}^4 x, \tag{3.41}$$

[2]The Lorentzian case only differs by the sign. Gauss–Bonnet is unaffected by this.

we can see that by using

$$\frac{\partial}{\partial \phi} = \sum_{ij} \frac{\partial g_{ij}}{\partial \phi} \frac{\partial}{\partial g_{ij}}, \tag{3.42}$$

the gradient

$$\nabla \mathcal{L}_{\text{loss, CGB}} = \frac{\partial g_\phi}{\partial \phi} \frac{\partial}{\partial g_\phi} \int_{\mathcal{M}} \mathcal{G}(\Omega[g_\phi]) \sqrt{\det(g)} \mathrm{d}^4 x = 0 \tag{3.43}$$

analogously to varying the action of Gauss–Bonnet gravity, vanishes identically.

This is due to the Chern–Gauss–Bonnet theorem embodying total independence of which metric is chosen for curvature and covolume, as long as it is Riemannian. Still, Chern–Gauss–Bonnet might be useful as a diagnostic tool, for example, when our representation is actually a double cover of our latent space instead of a single chart. This could happen if the neural network converges towards a local minimum, instead of the global one, where it might be reasonable to cover the latent space twice. Chern–Gauss–Bonnet would then exactly lead to double of what one would expect. Another case, which could occur in network training, is that the metric diverges, or generally becomes very large. This could justify a loss term that involves integrating over the absolute value of the curvature, to keep the model from inducing unnecessarily high curvature.

Chapter 4

Methodology of Statistical Latent Spaces

In the following, we will have to introduce or specify the geometric structure, onto which we will apply the theorem, by developing a statistical framework, which allows us to generate topological latent spaces and later sample from them. All of the above is still viable, as we will perform all relevant mathematical operations on the Riemannian parameter space using a Riemannian metric. As introduced, we will follow Amari's information geometric approach to continuously parametrized families of probabilities and likelihoods [37, 38].

4.1 Motivating Variational Autoencoders

A conventional autoencoder simply creates two maps E, D[1] (this would correspond to Dirac measures, staying in the framework developed in Section 2.3), one encoding data samples y to latent variables θ, the other reconstructing. This is done approximatively bijectively, meaning

$$E(y_1) = \theta_1 \tag{4.1}$$

$$D(\theta_1) \approx y_1 \tag{4.2}$$

$$D(E(y_1)) \approx \mathrm{id}(y_1) \tag{4.3}$$

This results in a cloud of points in latent space, whose shape is an artefact of the optimization. Between the points is empty space, from which the model cannot sample, as the points simply do not relate to each other, therefore decoding to nothing useful. Neither do we get a canonical metric, nor gauge-invariant observables, meaning latent coordinates do not have meaning and there is nothing coordinate independent, which could be interpreted either. Due to the lack of a metric, nothing fixes the resolution between two points and as the latent space is Hausdorff, there is no concept of identity of samples. Additionally, the model admits no density, meaning it cannot even be diagnosed to validity by solving an

[1] For both linear, this is simply the process of a principal component analysis, see [39].

integral regarding norm or marginal.

A *variational* autoencoder, as introduced by D. Kingma and M. Welling [40] in 2013, is a latent variable model with amortized Bayes, which shares the autoencoder's architecture in its basics. We introduce two conditional probability families, an encoder $\mathcal{E} = p(\cdot \mid y)$ and a decoder $\mathcal{D} = p(\cdot \mid \theta)$. The first learns a distribution of latent points θ, with given training data y and the second returns a distribution of recreated dataspace points y' with given latent parameters θ. Therefore, a variational autoencoder creates two distinct statistical manifolds, following the Bayes' theorem:

$$p(\theta|y)p(y) = p(y|\theta)\pi(\theta) \tag{4.4}$$

In theory, we could geometrically investigate both the likelihood and the posterior statistical manifold, but the decoder is the only one, for which we can guarantee parametrization and onto which it makes sense to introduce the Fisher metric. Marginalizing the latent variable, the generative model assigns to each observation the marginal likelihood

$$p_\phi(y) = \int p_\phi(y|\theta)\, \pi(\theta)\, \mathrm{d}\theta, \tag{4.5}$$

with a static prior $\pi(\theta)$. This integral is intractable for the nonlinear decoders considered here, which motivates the variational treatment below. Training maximizes a variational lower bound on the intractable marginal likelihood. Introducing an amortized inference model $q(\theta|y)$ and applying Jensen's inequality leads to the evidence lower bound (ELBO)

$$\log p(y) \;\geq\; \mathcal{L} \;=\; \mathbb{E}_{q(\theta|y)}[\log p(y|\theta)] \;-\; D_{\mathrm{KL}}(q(\theta|y) \,\|\, \pi(\theta))\,, \tag{4.6}$$

whose difference to the evidence is exactly $D_{\mathrm{KL}}(q(\theta|y)\|p(\theta|y))$. Maximizing $\mathcal{L}$ therefore fits the generative model and the posterior approximation at the same time. The first term is a reconstruction likelihood, thus favouring an injective encoder, the second regularizes the approximate posterior toward the prior and controls how the latent space is populated, therefore having a major influence on the represented topology.

The generative decoder is a map

$$\Theta \to \mathcal{P}(Y),\ \theta \mapsto p(\cdot|\theta) \tag{4.7}$$

and therefore introduces the canonical Fisher metric onto the latent space. Therefore, we have a gauge theoretic framework with diffeomorphism-invariant quantities. That is, the distributional encoder approach creates a continuous, statistical support, from which the decoder later can sample. Training data is not represented discretely, but rather through a mean and variance, therefore filling the latent space and creating a latent volume with density and topology, instead of a cloud of discrete datapoints with empty integral measure. The statistical decoder, then lets us introduce meaningful distances in latent space, by relating parameter coordinates according to the relative entropy of the

distributions, which they parametrize. As we can see, variational autoencoders admit the archetypical (meaning simplest possible) information geometric model architecture, while fulfilling all requirements this thesis needs. First of all, the decoder is an explicit, smooth map $\Theta \to \mathcal{P}(Y)$, so the Fisher information is well defined, without needing any further construction. Other generative models, for example deterministic decoders like GANs, do not induce a canonical, information metric, while models like normalizing flows just choose a different parametrization, which makes them geometrically invariant. Furthermore, variational autoencoders are explicitly parametrized and usually have a low-dimensional latent space, which makes them rather simple to mathematically describe and computationally feasible, while others like diffusion models do not have finite dimensional charts and need a metric functional. Last, the architecture allows us to directly enforce topology onto the latent space, by influencing the parametrization in an appropriate way.

It is worth making clear at this point, that variational autoencoders are obviously not the only statistical model, this applies to, they are just the perfect test object. Any method or conclusion which is formed in this thesis will be applicable to a generative model, which has a stochastic decoder and latent parametrization.

4.2 Topology-Inducing Architecture

To make a variational autoencoder differentiable, while creating an explicit latent space, we would usually sample from the encoder's distribution using the following reparametrization

$$\theta = \mu(x) + \sigma(x) \cdot \epsilon, \tag{4.8}$$

where ϵ is a Gaussian random variable. For implementing non-trivial topologies, we simply modify this parametrization. As a model's outputs only consist of tuples of real numbers, not points on manifolds, we do need to be careful to ensure differentiability. Furthermore, we also have to exchange the prior, which is used in general variational autoencoder training, as we explicitly want our latent representation to fill the entirety of our space, as both new topologies are closed and compact. Therefore, we will just use a uniform prior as basis for a network training which does not favor a distinct region of latent space ($\pi_{\mathbb{T}}(\theta) = 1/(2\pi)^2$ and $\pi_{\mathbb{S}}(\theta) = 1/4\pi$, as a uniform density needs to be normalized. In theory, this does not affect training, as it is constant, but for complete interpretability, we want to fix the lower bound of the KL-divergence). Calculating the explicit loss-ELBO function also needs modifying, as the posterior has to be approximated in a fitting way, which brings its own complications, as we cannot simply use a flat Gaussian. This won't be dealt with here, as we simply want to observe the static model's curvature and latent space, not the specific features and dynamics of training.

4.2.1 The Toroidal Latent Space

Building the $\mathbb{T}^2$ latent parametrization, naively one might suggest a map of the following form:

$$\mu \in \mathbb{R}^2 \to \theta \in \mathbb{S}^1 \times \mathbb{S}^1 \tag{4.9}$$

This is actually not realizable in a differentiable way, as $\mathbb{S}^1$ cannot be covered continuously by a single real input (precisely the reason, we introduce latent topology in the first place). Another approach would be to introduce a modulus function to identify equivalence classes of rectengular areas of $\mathbb{R}^n$ (therefore creating a quotient topology isomorphic to $\mathbb{T}^2$). This actually can be implemented in training, but can introduce difficulties with respect to guaranteeing differentiability up to the second degree of the metric, which is necessary for investigating curvature. Therefore, we have to alter the encoder in such a way that it outputs a four dimensional value for μ, which we will project onto $\mathbb{S}^1 \times \mathbb{S}^1$ (we basically introduce non-injectiveness of the encoder by an atlas of the now closed latent space). This is done by letting the encoder emit two planar vectors, both $\in \mathbb{R}^2$, each is radially projected onto its unit circle. Therefore, we get a map

$$\left(\mathbb{R}^2 \setminus \{0\}\right)^2 \longrightarrow \mathbb{T}^2 \cong \mathbb{S}^1 \times \mathbb{S}^1, \qquad (v_1, v_2) \longmapsto \left(\frac{v_1}{\|v_1\|}, \frac{v_2}{\|v_2\|}\right). \tag{4.10}$$

It might seem contrary to what we required from a latent embedding, as this is not fully continuous at $v = 0$, though we only require the *decoder* part of the variational autoencoder to be continuously differentiable. For the encoder it is sufficient, to ensure continuity almost everywhere. On the decoder's side, we need to embed the flat torus into $\mathbb{R}^4$, using the well known Clifford embedding.

$$\Phi: \ \mathbb{T}^2 = \mathbb{S}^1 \times \mathbb{S}^1 \longrightarrow \mathbb{S}^3 \subset \mathbb{R}^4, \qquad \Phi(\alpha, \beta) = \frac{1}{\sqrt{2}} \begin{pmatrix} \cos\alpha \\ \sin\alpha \\ \cos\beta \\ \sin\beta \end{pmatrix}, \qquad \alpha, \beta \in [0, 2\pi). \tag{4.11}$$

Therefore, we still recover a two dimensional latent space, the flat torus, to be precise, which parametrizes the decoder's distribution continuously and differentiably. One might wonder here, why we will not use the donut in $\mathbb{R}^3$, as this obviously has the same topological invariant $\chi = 0$. The reason for that is that we only want to investigate curvature introduced by the statistics of the decoder's distribution (the difference in two dimensions can be seen in the Appendix), as the torus' non-trivial topology is what we are interested in. The variance is simply mapped onto the flat torus, while ensuring periodic boundary conditions.

4.2.2 The Spherical Latent Space

The spherical latent space is built in the exact same manner, with the difference that $\mathbb{S}^2$ can be embedded into $\mathbb{R}^3$. Therefore, the outputs of the variational autoencoder have to

be modified to be elements of $\mathbb{R}^3$:

$$\mathbb{R}^3 \setminus \{0\} \longrightarrow \mathbb{S}^2, \qquad x \longmapsto \frac{x}{\|x\|}. \tag{4.12}$$

Furthermore, the variance is chosen to be a single scalar, as $\mathbb{S}^2$ is isotropic and admits no smooth global orthonormal frame[2]. The posterior is therefore an isotropic Gaussian with scalar variance σ around $\mu \in \mathbb{S}^2$. The decoder simply projects the unit vectors back into ambient $\mathbb{R}^3$, where the radial component identically vanishes in derivatives, as it is constant.

4.3 Preserving Topology

Now that we have fixed an encoder and introduced the latent space formalism, we can think about what happens to the data's topology when getting represented in the latent space. We start with the conventional case of a flat Euclidean latent space.

Topology is preserved exactly when the encoder is a homeomorphism onto its image, meaning for compact manifolds, when it is injective. By invariance of domain, no continuous, injective encoder between a compact d-dimensional manifold with empty boundary and $\mathbb{R}^d$ can exist. An encoder from a data manifold, e.g. $\mathbb{T}^2$ or $\mathbb{S}^2$ into a flat two-dimensional latent space therefore cannot preserve the topology. Since a network is guaranteed to be continuous and thus cannot tear, injectivity is lost. As non-injectiveness gets heavily penalized by the loss function, in practice, both mechanisms, stretching and folding, are presumably present. An arbitrarily large Lipschitz constant approximates a tear, neighbouring points on the data manifold are thereby separated by large distances in Θ, while encoding density drops to zero near these regions. Elsewhere, the map folds back onto itself, therefore violating injectivity, which is not observable, as the folded sheet carries near-zero statistical density. What we can observe instead, are gaps and abrupt transitions in the latent representation. Technically, these are not tears or discontinuities, as the encoder still remains continuous, but they are visually and practically indistinguishable from them. Therefore, we will refer to them as such from now on.

When introducing compact latent spaces, a faithful representation restricts both the representable data and the choice of latent space. A compact latent manifold can accommodate a data manifold only if the latter embeds into it and it can be fully covered only if the two are homeomorphic.

When applying this interpretation, we have to be aware that mathematical reasoning, like the one above, concerns the idealised continuous manifold. The structure we encode is based on a finite sample, resulting in a finite number of means and distributions over them. The topological obstruction applies only to the manifold the data is assumed to lie on, which we are approximating. Therefore, we are not able to verify preservation of the topology as a distinct statement, but rather by diagnosing the effective consequences.

[2]Also a consequence of global curvature being fixed by topology, or Poincaré-Hopf, to be precise.

4.4 Latent Metric

Before we start to compute the Fisher information metric for the variational autoencoder, we have to discuss the nature of the object we refer to as metric here. Intrinsically, a metric of distributions is a slowly varying, bilinear, symmetric tensor field on a topologically open set of distributions. Most of the time we use Eq. (2.11), where F_{ij} is defined as tensor field over the parameter (or latent) space Θ. As sketched in Eq. (2.9), this corresponds to the second order expansion of the Kullback-Leibler divergence. Actually, in the same step, we already choose to use densities and a precise basis, in order to operate on a differential manifold, which needs a coordinate basis. Equivalently, statistical manifolds are built on the premise that distributions admit the structure of a parametrized family. The KL-divergence [16] itself is defined on

$$\mathcal{P}(Y) = \Big\{ p : Y \to \mathbb{R}_{\geq 0} \;\Big|\; \int_Y p \,\mathrm{d}\mu = 1 \Big\}, \tag{4.13}$$

meaning over all probability measures, independent of parametrization. A second order expansion in δp, keeping the generality, would take the form of

$$D_{\mathrm{KL}}(p \,\|\, p + \delta p) = -\int_Y \delta p \,\mathrm{d}\mu + \frac{1}{2}\int_Y \frac{(\delta p)^2}{p}\,\mathrm{d}\mu + \mathcal{O}\Big(\|\delta p\|^3\Big) \tag{4.14}$$

$$= \frac{1}{2}\int_Y \frac{(\delta p)^2}{p}\,\mathrm{d}\mu + \mathcal{O}\Big(\|\delta p\|^3\Big) \;=\; \frac{1}{2}\, g_F(\delta p, \delta p) + \mathcal{O}\Big(\|\delta p\|^3\Big), \tag{4.15}$$

where δp is in $T_p\mathcal{P}(Y)$ and $\int_Y \delta p \mathrm{d}\mu = 0$ by definition[3]. g_F denotes the Fisher-Rao metric. As a rigorous proof includes properly defining signed, finite measures $\mathcal{P}$ over Y, μ as fixed, dominating measure (and much more) and calculations involve Radon-Nikodym, the reader is advised, to consult appropriate literature (e.g. [41, 42]). Now, choosing an explicit parametrization for a family of distributions

$$\theta \mapsto P(\cdot|\theta), \theta \in \Theta \tag{4.16}$$

we define a statistical manifold $\mathcal{S}(\Theta)$ as submanifold of $\mathcal{P}(Y)$. Pushing forward the coordinate frame of Θ along the decoder $\mathcal{D}$

$$\mathrm{d}\mathcal{D}(\partial_i) = \partial_i p(\cdot|\theta), \tag{4.17}$$

[3]This gets obvious, when defining the tangent space through velocities $\dot{p}$ of curves p_t through points $p \in \mathcal{P}(Y)$. $\int \dot{p}_t \mathrm{d}\mu = \frac{d}{dt}\int p_t \mathrm{d}\mu$, where p_t is normalized by definition.

we can write

$$(\mathcal{D}^* g_F)_{ij} = \frac{1}{2}\int \frac{\partial_i p \partial_j p}{p} \mathrm{d}y \tag{4.18}$$

$$= \frac{1}{2}\int p \partial_i \ln(p) \partial_j \ln(p) \mathrm{d}y \tag{4.19}$$

$$= \frac{1}{2}\mathbb{E}_{y\sim p(y|\theta)}[\partial_i \ln(p(y|\theta))\partial_j \ln(p(y|\theta))] \tag{4.20}$$

$$\propto F_{ij} \tag{4.21}$$

Therefore, pulling back the Fisher-Rao metric onto the parameter space using the distribution defined through our decoder $\mathcal{D}$ is the intrinsic[4] and unique (again, see [17]) way to represent the infinitesimal difference between distribution of the statistical manifold, based on their parametrization and introduces an intrinsic information geometric Riemannian metric onto the latent space Θ.

One caveat is that the Fisher information only endows Θ with a Riemannian metric, wherever $\mathrm{d}\mathcal{D}$ is injective. This is important, when wrongly enforcing a non-trivial topology, as introducing non-identical equivalence classes on latent space and statistical manifold would lead to a non-injective function. Non-injectiveness would create a degenerate metric, resulting in an ill defined connection and vanishing covolume. The Fisher information is for our purposes explicitly defined for a map from the latent space to a certain family of distributions through training, which means $P(\cdot|\theta)$ is always an injective parametrization of the statistical manifold and F_{ij} therefore well defined.

We can now precisely give reasons for the necessity of a non-arbitrary metric. The actual information the generative model delivers can be described through the marginal.

$$p(y) = \int_\Theta p(y|\theta)\pi(\theta)\mathrm{d}\theta \tag{4.22}$$

Now, taking an arbitrary diffeomorphism $\varphi : \Theta \to \Theta$, meaning a coordinate change in the latent space, such that

$$\mathcal{D}' = \mathcal{D} \circ \varphi^{-1} \quad \text{and} \quad \pi' = \varphi_* \pi. \tag{4.23}$$

This does not change the outcome of the model, as

$$p'(y) = \int_{\Theta'} p(y \,|\, \varphi^{-1}(\theta')) \, \pi(\varphi^{-1}(\theta')) \, |\det J_{\varphi^{-1}}(\theta')| \, \mathrm{d}\theta' \tag{4.24}$$

where we used, that $p'(y|\theta') = p(y|\varphi^{-1}(\theta'))$ and following $p'(y|\varphi(\theta)) = p(y|\theta)$. $p(\cdot|\theta)$ is only a function, not a density in θ and transforms as such. Here it gets useful, having introduced these properties as measures, as we are able to differentiate between the transformational

[4] One might also think of an extrinsically motivated case: pulling back the Euclidean metric, which lives on the generated data's space Y' through the decoder onto the latent space.

behaviour of functions and densities such as $\pi(\theta)$. Substituting $\theta' = \varphi(\theta)$

$$p'(y) = \int_\Theta p(y|\theta)\pi(\theta)|\mathrm{det}J_\varphi||\mathrm{det}J_{\varphi^{-1}}|\mathrm{d}\theta \tag{4.25}$$

$$= \int_\Theta p(y|\theta)\,\pi(\theta)\,\mathrm{d}\theta \;=\; p(y), \tag{4.26}$$

Where we used that $|\mathrm{det}J_{\varphi^{-1}}| = |\mathrm{det}J_\varphi|^{-1}$. Therefore, quantities computed in the latent space, such as Euclidean distance or straight line paths between samples, changing under the diffeomorphism φ are not actually meaningful properties of the model itself, just artefacts of a choice of coordinates. The geometry, introduced by the Fisher information metric is not a property of the latent space Θ, it lives on the decoder's image, which is a manifold of distributions. Therefore, computing geodesic distance or curvature in latent coordinates describes properties on the statistical manifold, which are independent of coordinate changes. If one argues using measures instead of densities in the latent space, this argument reduces to a fixed function and a measure on the statistical manifold, therefore being independent of coordinate choice on Θ. Following, the Fisher information metric transforms just as a $(0,2)$ tensor valued metric should.

$$\frac{\partial}{\partial\theta'^k}\mathrm{log}p'(y|\theta') = \frac{\partial\theta^i}{\partial\theta'^k}\frac{\partial}{\partial\theta^i}\mathrm{log}p(y|\theta) \tag{4.27}$$

And concluding

$$F'_{kl}(\theta') = \frac{\partial\theta^i}{\partial\theta'^k}\frac{\partial\theta^j}{\partial\theta'^l}F_{ij}(\theta) \tag{4.28}$$

Therefore, all elements, which include the fully contracted metric by construction are invariant under latent coordinate changes and are genuinally learned properties of the data. This includes the line element $\mathrm{d}s^2$, scalar curvature or gradient flow (which is precisely the reason for *natural* gradient descent, although this is a method used in *network parameter space*, as it optimizes the loss, not in the *latent parameter space*, see [7]).

It might not be relevant for an information geometric description, whether to parametrize measures, or to measure parametrizably, but it is necessary to understand why the statistically induced geometry is not arbitrary, but natural.

One might argue that choosing a prior $\pi(\theta)$ restricts diffeomeorphic gauge freedom, since in the standard variational autoencoder training, diffeomorphic changes must preserve $\mathcal{N}(0,\mathbb{1})$ and therefore restrict the gauge group to the stabilizer of π[5]. Although this is correct, the structure fixed by the prior is simply a modelling choice, not something the model has learned to represent. In addition, there are choices of prior, which allow more diffeomorphic gauge freedom, as for example a uniform prior, as imposed in Section 4.2 for the topological variational autoencoders.

The explicit use of terminology *pullback* instead of just parametrization or *chart*, was used deliberately to differentiate between statistical manifold and latent space, while

[5]For latent dimension of $d \geq 2$, $\mathrm{Diff}_\pi(\Theta)$ is infinite dimensional. Additionally, any prior π in $\mathbb{R}^n$ can be transformed to be a normalized Gaussian, implying generality.

especially keeping in mind that we might want to enforce topology onto the latter. A *chart* of a differentiable manifold $\mathcal{M}$ with dimension d is defined as a diffeomorphism

$$\psi \colon \mathcal{U} \subset \mathcal{M} \longrightarrow \mathcal{V} \subset \mathbb{R}^d \tag{4.29}$$

while for our purposes, a parametrization of a statistical manifold $\mathcal{S}$ could generally take the form

$$\Phi \coloneqq \mathcal{D} \circ \psi^{-1} \colon \ \mathcal{U} \subset \mathbb{R}^d \xrightarrow{\psi^{-1}} \mathcal{T} \subset \Theta \xrightarrow{\mathcal{D}} \mathcal{V} \subset \mathcal{S}, \tag{4.30}$$

where ψ is a chart of the parameter manifold Θ and $\mathcal{D}$ the decoder. Since Θ is locally Euclidean, a latent manifold with non-trivial topology is consistent with the definition of a statistical manifold.

4.5 Determinant of Latent Metric

In addition to the curvature, one can take information from the metric's determinant. Under reparametrization $\theta' = \varphi(\theta)$, the determinant transforms as follows

$$\det F'(\theta') = \left(\det J_{\varphi^{-1}}(\theta')\right)^2 \det F(\theta) = \frac{\det F(\theta)}{\left(\det J_{\varphi}(\theta)\right)^2}, \tag{4.31}$$

so $\det F$ is a scalar density rather than a scalar. The covolume element, on the other hand is invariant.

$$\sqrt{\det F'(\theta')}\,\mathrm{d}\theta' = \frac{\sqrt{\det F(\theta)}}{\left|\det J_{\varphi}(\theta)\right|}\left|\det J_{\varphi}(\theta)\right|\mathrm{d}\theta = \sqrt{\det F(\theta)}\,\mathrm{d}\theta, \tag{4.32}$$

Still, in comparison between datasets, this might appear to be useful. To maximize comparability, we choose to use unit radii for our latent parametrizations. As the metric determinant embodies the property of magnifying, to interpret this correctly, we need to determine the direction of implication. The Fisher information metric we are using here is defined by the decoder's distribution, which maps parameters onto data samples. Therefore a large metric determinant, implies that spacing between parameters corresponds to a strongly magnified spacing of distributions.

4.6 Latent Curvature

Choosing a suitable and meaningful Riemannian metric, as we just did, is often dependent on the context one wants to include into mathematical reasoning. In this case, our goal is to investigate the *learned curvature*, which is induced by the decoder's distribution. Furthermore, we want to deliberately use the Levi-Civita connection, as Chern-Gauss-Bonnet is a topologial theorem for Riemannian geometry. In the following we will use the standard procedure of calculating the Riemann curvature tensor using contractions and first derivatives of the Christoffel symbols calculated with contractions and first derivatives

of the metric. Using the Fisher information metric on the parameter space as derived in Eq. (4.18), we get the following

$$F_{ij} = -\int \frac{\partial^2 \log(p(y|\theta))}{\partial\theta^i \partial\theta^j} p(y|\theta) \mathrm{d}y \tag{4.33}$$

$$= \mathbb{E}_{y\sim p(\cdot|\theta)}[-\frac{\partial^2 \log(p(y|\theta))}{\partial\theta^i \partial\theta^j}] \tag{4.34}$$

or in a more useful form

$$F_{ij} = \mathbb{E}_{y\sim p(\cdot|\theta)}[\frac{\partial \log(p(y|\theta))}{\partial\theta^i} \frac{\partial \log(p(y|\theta))}{\partial\theta^j}] \tag{4.35}$$

(the equality of these equations can be seen when applying the chain rule in reverse.) In a machine learning context it is also reasonable to add indices of network parameters ϕ, to indicate which elements of the distribution evolve during a learning process, although the actual learning is not of interest in this thesis:

$$P(y|\theta) \longmapsto P_\phi(y|\theta) \tag{4.36}$$

Therefore, ${}^{\phi}F_{ij}$, ${}^{LC,\phi}\Gamma^k{}_{ij}$, ${}^{\phi}R_{ijkl}$ would be appropriate as well. Having mentioned this, for easier comprehension, we will dismiss the extra index wherever we find it appropriate, as we focus on a static model.

4.7 Gaussian Decoder

Assuming the decoder $\mathcal{D}$ to be Gaussian is the standard choice for continuous data, as it ensures an analytical Fisher metric. We do have to be aware, that this is not a general approximation, but rather a modelling choice, as a Gaussian is not universally the optimal fit. For a general Gaussian decoder, meaning

$$P_\phi(y, \theta) = \mathcal{N}(y; \mu_\phi(\theta), \sigma_\phi(\theta)), \tag{4.37}$$

so

$$p(y|\theta) = (2\pi)^{-n/2} \det(\Sigma(\theta))^{-1/2} \exp\left[-\tfrac{1}{2}(y-\mu(\theta))^\top \Sigma(\theta)^{-1}(y-\mu(\theta))\right], \tag{4.38}$$

we will start to calculate the latent curvature under the assumption of heteroscedasticity.

The log-likelihood follows as

$$\log p(y|\theta) = -\frac{1}{2}\sum_a \frac{(y_a - \mu_a(\theta))^2}{\sigma_a^2(\theta)} - \sum_a s_a(\theta) - \frac{n}{2}\log 2\pi. \tag{4.39}$$

with $s_a := \log \sigma_a$. Writing $r_a = y_a - \mu_a$, the score is

$$\partial_i \log p = \sum_a \left[\frac{r_a}{\sigma_a^2}\,\partial_i \mu_a + \left(\frac{r_a^2}{\sigma_a^2} - 1\right)\partial_i s_a\right]. \tag{4.40}$$

Taking the expectation and using $\mathbb{E}[r_a r_b] = \sigma_a^2 \delta_{ab}$, $\mathrm{Var}(r_a^2/\sigma_a^2) = 2$, and $\mathbb{E}[r_a(r_b^2 - \sigma_b^2)] = 0$, we conclude

$$F_{ij}(\theta) = \sum_a \frac{\partial_i \mu_a \, \partial_j \mu_a}{\sigma_a^2} + 2 \sum_a \partial_i s_a \, \partial_j s_a, \tag{4.41}$$

or in matrix form

$$F(\theta) = J_\mu^T \Sigma^{-1} J_\mu + 2 \, J_{\log \sigma}^T J_{\log \sigma}, \tag{4.42}$$

with J denoting the Jacobian. As we can see, areas with large uncertainty suppress the first part of the metric and distances become shorter. The second term is dependent on the derivative of the logarithm of the variance, therefore becoming large, when crossing between areas with large and small uncertianty.

4.7.1 Homoscedasticity as Numerical Necessity

To further simplify, we will assume the decoder to be homoscedastic

$$P_\phi(y, \theta) = \mathcal{N}(y; \mu_\phi(\theta), \sigma_\phi) \tag{4.43}$$

This means that the variance is constant in θ, but still learnable by the decoder. This results in

$$\partial_\theta \sigma = 0 \tag{4.44}$$

$$\partial_\phi \sigma \neq 0 \tag{4.45}$$

Therefore we can find simple analytical expressions for the metric and the Levi-Civita connection:

$$F_{ij}(\theta) = \frac{1}{\sigma_\phi^2} \partial_i \mu_\phi(\theta) \partial_j \mu_\phi(\theta) \tag{4.46}$$

$${}^{LC}\Gamma^k{}_{ij} = \frac{1}{\sigma_\phi^2} F^{kr} \partial_r \mu_\phi(\theta) \partial_i \partial_j \mu_\phi(\theta) \tag{4.47}$$

It follows from homoscedasticity that

$$C_{ijk} = 0, \tag{4.48}$$

as the variance is constant over the latent space, all directional skewness vanishes. This follows from the calculations above (Eq. (4.41)) and the definitions Eq. (2.13), Eq. (2.18), with $\partial_i = \frac{\partial}{\partial \theta^i}$.

Therefore, we notice that the Fisher information metric takes the form of the Euclidean pullback metric, with respect to $\mu(\theta)$

$$F \propto J_\mu^T J_\mu \tag{4.49}$$

This is interesting, as for this case (or for a Gaussian decoder in general, where we also need to involve $\log\sigma$) the Gaussianity of the distribution and the entropy way of measuring

distributional differences cancel and reduce to a simple change in variables.

From Eq. (4.46) and Eq. (4.47) we conclude

$$\partial_i \Gamma^l{}_{jk} = \frac{1}{\sigma^2}\Big[\partial_i F^{lm}\,\partial_m\mu(\theta)\,\partial_j\partial_k\mu(\theta) + F^{lm}\,\partial_i\partial_m\mu(\theta)\,\partial_j\partial_k\mu(\theta) + F^{lm}\,\partial_m\mu(\theta)\,\partial_i\partial_j\partial_k\mu(\theta)\Big] \tag{4.50}$$

$$\Gamma^l{}_{in}\,\Gamma^n{}_{jk} = \frac{1}{\sigma^4}\Big[F^{lm}\,\partial_m\mu\,\partial_i\partial_n\mu\cdot F^{np}\,\partial_p\mu\,\partial_j\partial_k\mu\Big]. \tag{4.51}$$

which results in

$$\begin{aligned} R^l{}_{kij} = {} & \frac{1}{\sigma^2}\Big[\partial_i F^{lm}\,\partial_m\mu(\theta)\,\partial_j\partial_k\mu(\theta) + F^{lm}\,\partial_i\partial_m\mu(\theta)\,\partial_j\partial_k\mu(\theta) \\ & + F^{lm}\,\partial_m\mu(\theta)\,\partial_i\partial_j\partial_k\mu(\theta)\Big] \\ & - \frac{1}{\sigma^2}\Big[\partial_j F^{lm}\,\partial_m\mu(\theta)\,\partial_i\partial_k\mu(\theta) + F^{lm}\,\partial_j\partial_m\mu(\theta)\,\partial_i\partial_k\mu(\theta) \\ & + F^{lm}\,\partial_m\mu(\theta)\,\partial_j\partial_i\partial_k\mu(\theta)\Big] \\ & + \frac{1}{\sigma^4}\Big[F^{lm}\,\partial_m\mu\,\partial_i\partial_n\mu\cdot F^{np}\,\partial_p\mu\,\partial_j\partial_k\mu\Big] \\ & - \frac{1}{\sigma^4}\Big[F^{lm}\,\partial_m\mu\,\partial_j\partial_n\mu\cdot F^{np}\,\partial_p\mu\,\partial_i\partial_k\mu\Big]. \end{aligned} \tag{4.52}$$

Applying Schwarz's rule for commutation of partial derivatives, we can see that

$$\frac{1}{\sigma^2}F^{lm}\,\partial_m\mu(\theta)\,\partial_i\partial_j\partial_k\mu(\theta) = \frac{1}{\sigma^2}F^{lm}\,\partial_m\mu(\theta)\,\partial_j\partial_i\partial_k\mu(\theta). \tag{4.53}$$

Therefore, the terms including third derivatives cancel identically, leaving

$$\begin{aligned} R^l{}_{kij} = {} & \frac{1}{\sigma^2}\Big[\partial_i F^{lm}\,\partial_m\mu(\theta)\,\partial_j\partial_k\mu(\theta) + F^{lm}\,\partial_i\partial_m\mu(\theta)\,\partial_j\partial_k\mu(\theta)\Big] \\ & - \frac{1}{\sigma^2}\Big[\partial_j F^{lm}\,\partial_m\mu(\theta)\,\partial_i\partial_k\mu(\theta) + F^{lm}\,\partial_j\partial_m\mu(\theta)\,\partial_i\partial_k\mu(\theta)\Big] \\ & + \frac{1}{\sigma^4}\Big[F^{lm}\,\partial_m\mu\,\partial_i\partial_n\mu\cdot F^{np}\,\partial_p\mu\,\partial_j\partial_k\mu\Big] \\ & - \frac{1}{\sigma^4}\Big[F^{lm}\,\partial_m\mu\,\partial_j\partial_n\mu\cdot F^{np}\,\partial_p\mu\,\partial_i\partial_k\mu\Big]. \end{aligned} \tag{4.54}$$

As we can see, this Riemannian curvature is generally non-vanishing and only identically 0, if $\partial_i\partial_j\mu(\theta) = 0$.

4.7.2 Curvature as Measure of Non-Linearity

We can hereby conclude that the curvature, or in general the absolute value or respectively the integrated absolute curvature

$$\int |K|\mathrm{d}V, \tag{4.55}$$

are a measure of how non-linearly the decoding function is learned to be. Referring again to principal component analysis Section 4.1 ([39]), which is a linear dimension reduction minimizing the variance loss. This would be an example, where the "latent"[6] space is Euclidean flat, as $\partial_i \partial_j f(\theta) = 0$, where f is the decoding function of the principal component analysis. This can also be deduced from

$$R^i{}_{jkl} = R^i{}_{jkl}(F,\, F^{-1},\, \partial F,\, \partial F^{-1},\, \partial^2 F), \tag{4.56}$$

while remembering that $F_{ij} \propto J_\mu^T J_\mu$ is the Fisher information metric of the decoder's distribution. Therefore, one would assume that trying to represent a non-trivial topology in a flat latent space of same dimension would necessitate areas with higher measure of non-linearity, which would result in non-vanishing curvature. Here, we have to differentiate between the two cases, we are investigating

- Non-trivial topology in a flat latent space: If embedding is not possible, as the data's topology is simply too rich to be represented in a flat latent space with this dimension, there exists no injective and continuous encoder, which necessarily results in latent tears or discontinuities, meaning areas with little statistical support for the decoder. This most likely results in large variance or high curvature.
- Mismatched topology in a latent space with non-trivial topology: As the architecture forces a certain symmetry onto the decoder's distribution, mismatched periodicity cannot be represented as such by the decoder. Therefore symmetry, or in the case of $\mathbb{T}^2$ periodicity in both dimensions, has to be ensured by non-linearly joining the discontinuities.

Of course, one can always construct cases in which the dimension of the latent space is high enough that, following the Whitney embedding theorem, an embedding is possible. This is probably the best solution when constructing a network that has to represent an unknown topology. Estimating the dimension of the underlying data structure and setting the latent dimension to twice that value always ensures that an embedding exists.

4.7.3 Metric Determinant

Assuming the two-dimensional homoscedastic case, we can see that

$$\sqrt{\det(F)} \propto \sigma^{-2} \tag{4.57}$$

Where σ is constant over Θ. Therefore, this can not be seen as a spatial measure for the uncertainty of the model. Writing the determinant out for a two dimensional, homoscedastic Gaussian decoder, we get

$$\det F \propto \|\partial_1 \mu\|^2 \, \|\partial_2 \mu\|^2 - \langle \partial_1 \mu, \partial_2 \mu \rangle^2. \tag{4.58}$$

[6] It is debatable, whether PCA actually creates a latent space, as it is generally not a feature space, but rather reduces dimensionality to those directions, which preserve the most variance.

We conclude from the Cauchy-Schwarz inequality that

$$\det F \geq 0. \tag{4.59}$$

4.8 Amari–Chentsov Duality and Non-Metricity

Until now, we have always talked about Riemannian curvature and Riemannian flatness, meaning curvature with respect to the unique, metric-compatible, torsionless Levi-Civita connection. To fully grasp the additional structure, which information geometry introduces, we have to further investigate metric compatibility under the α-connections.

$$\nabla_k^{(\alpha)} F_{ij} = \alpha C_{kij} \tag{4.60}$$

For the homoscedastic Gaussian decoder of Section 4.7.1, C_{ijk} vanishes identically, as the variance is constant, which results in vanishing skewness. Therefore information geometry contributes nothing but Riemannian geometry, as the dual connections collapse to Levi-Civita. Therefore, no duality is left and the non-metricity is identically 0.

4.9 Dealing with the Curse of Dimensionality

As there are applications, where very high dimensional latent spaces are necessary, one has to deal with the numerics of information geometry and needs to think of whether scaling of methods even makes it possible to apply them generally.

4.9.1 Generalization the Calculation of the Pfaffian

As we have deduced in Eq. (3.18), the Pfaffian form for a $2n$ dimensional manifold is proportional to

$$\begin{aligned}
\mathrm{Pf}(\Omega) &\propto \epsilon_{i_1 \cdots i_{2n}} \Omega^{i_1 i_2} \wedge \ldots \wedge \Omega^{i_{2n-1} i_{2n}} && (4.61)\\
&\propto \epsilon_{i_1 \cdots i_{2n}} R^{i_1 i_2}{}_{j_1 j_2} \, \mathrm{d}x^{j_1} \wedge \mathrm{d}x^{j_2} \wedge \ldots \wedge R^{i_{2n-1} i_{2n}}{}_{j_{2n-1} j_{2n}} \wedge \mathrm{d}x^{2n-1} \wedge \mathrm{d}x^{2n} && (4.62)\\
&\propto \epsilon_{i_1 \cdots i_{2n}} \epsilon^{j_1 \cdots j_{2n}} R^{i_1 i_2}{}_{j_1 j_2} \ldots R^{i_{2n-1} i_{2n}}{}_{j_{2n-1} j_{2n}} \sqrt{\det(F)} \mathrm{d}^{2n} x && (4.63)
\end{aligned}$$

Following the formalism developed in Section 3.1.5, we conclude that we have to calculate

- **The generalized Kronecker δ,**

  which results in $(2n)!$ terms which all consist of Kronecker δs with $2n$ upper and lower indices.

- **Contractions of the Riemannian tensor**

  We have to apply terms of the form

$$\delta^{i_1}{}_{\alpha} \delta^{i_2}{}_{\beta} \ldots \delta^{i_{2n}}{}_{\omega}, \tag{4.64}$$

where $\{\alpha, \beta, \ldots, \omega\}$ are permutations of $\{j_1, j_2, \ldots, j_{2n}\}$, onto products of

$$R^{i_1 i_2}{}_{j_1 j_2} \cdot \ldots \cdot R^{i_{2n-1} i_{2n}}{}_{j_{2n-1} j_{2n}}. \tag{4.65}$$

This implies, that we can calculate the

- **$2n$-dimensional Riemannian tensor**

  Which itself is no easy task, as it includes calculating the Christoffel symbols

  $$\Gamma^k{}_{ij} = \frac{1}{2} F^{kl} \left(\partial_i F_{jl} + \partial_j F_{il} - \partial_l F_{ij} \right), \tag{4.66}$$

  which is, dropping the Einstein sum convention

  $$\Gamma^k{}_{ij} = \frac{1}{2} \sum_{l=1}^{2n} F^{kl} \left(\frac{\partial F_{jl}}{\partial \theta^i} + \frac{\partial F_{il}}{\partial \theta^j} - \frac{\partial F_{ij}}{\partial \theta^l} \right). \tag{4.67}$$

  Although the Riemann tensor does not have $(2n)^4$ independent components, these still scale with dimension, to be $(2n)^2((2n)^2 - 1)/12$. One notices that Eq. (4.66) needs the inverse of the metric and the Riemann tensor also derivatives of the inverse.

- **Covolume of the decoder**

  In order to later integrate the Chern–Gauss–Bonnet theorem, we have to calculate the determinant of the $2n$ dimensional metric, which generally is a polynomial of degree $2n$.

- **Fisher information metric of the decoder**

  The decoder is a map

  $$\Theta \ni \theta \mapsto p(\cdot \mid \theta) = \mathcal{N}(\cdot\,; \mu_\phi(\theta) \in \mathbb{R}^m, \sigma_\phi \in \mathbb{R}) \tag{4.68}$$

  with m being the dimension of data space (obviously, we assume $m \gg 2n$). Where we used the convention with explicitly labeling network parameter dependence with ϕ. If we assume, that our decoder is homoscedastic and Gaussian, the metric reduces to

  $$F \propto J_\mu^T J_\mu \tag{4.69}$$

  Where we assume that the Chern–Gauss–Bonnet theorem is again independent of a constant prefactor of the metric (proof is analogous to Section 5.2.1).

High dimensionality also induces a lower bound for the resolvable curvature, which in reverse restricts how large σ is allowed to get, before we lose detail. As we assume the decoder to be homoscedastic, each element ϵ of a m dimensional tuple of a sample from

$p(\cdot \mid \theta)$ gets averaged over σ, which leads to

$$\mathbb{E}\|\epsilon\|^2 = m\sigma^2 \tag{4.70}$$

$$\|\epsilon\| \approx \sqrt{m}\sigma. \tag{4.71}$$

As only the translations perpendicular to our latent manifold smear the actual curvature (2n dimensions just result in translations parallel to the manifold), we modify the equation by subtracting the dimension of the latent space.

We conclude that if the curvature radius (inverse proportional to the scalar curvature) is smaller than $\sqrt{m-2n}\sigma \approx \sqrt{m}\sigma$ for large m, we cannot fully resolve it.

4.9.2 Numerical Advantages of Extrinsic Curvature

As all of this is computationally quite challenging, we propose a possible numerical improvement. This heavily relies on concepts, which were not properly introduced, therefore the reader is advised to consult literature (e.g. [21] or [43]). Instead of calculating the curvature of the latent space by using the intrinsic metric, which includes third derivatives of the decoder function, we exploit the fact that the latent manifold is already immersed in data space by the decoder itself. Crucially, this immersion is isometric, if we assume a *homoscedastic* Gaussian decoder. From this, we calculate the second fundamental form, which includes only derivatives up to second order and no derivatives of the inverse metric, therefore granting a computational advantage. From the second fundamental form, we are able to algebraically calculate the Riemann tensor of the latent space, as we assume ambient data space to be flat Euclidean.

Chapter 5

Experiment and Result

5.1 Setup

In the following we want to explore how the topology of latent space and data effect the curvature of variational autoencoders. Additionally, we want to rectify the Chern-Gauss-Bonnet theorem by numerically integrating over the curvature induced by the decoder mapping. To quantitatively estimate the differences between non-linearity and performance of the models, we will cross train and compare them. Herefore, we implement and train three different variational autoencoders, each with a different latent topology:

- a standard VAE, admitting a latent space of $\mathbb{R}^2$
- a spherical VAE, using a latent space $\Theta \cong \mathbb{S}^2$
- a toroidal VAE, using a latent space $\Theta \cong \mathbb{T}^2$

For each variational autoencoder, we create a "natural" training dataset, meaning a dataset admitting exactly the topology we will enforce onto the latent architecture. This means, we will construct data, where we expect a flat latent with trivial topology to work best, a dataset with a periodicity which is based on a topological $\mathbb{S}^2$ and a dataset which has an underlying $\mathbb{T}^2 = \mathbb{S}^1 \times \mathbb{S}^1$ symmetry.

5.1.1 Orbits of Group Action as Generator for Periodic Datasets

Our goal is to create data, which is high dimensional, but has an underlying structural manifold. The most obvious choice for such a task is to create an image of something simple, undergoing characteristic transformations of the group[1] one wants to represent. In mathematical terms, the underlying manifold, which we later want the model to recover in our latent space, is precisely the orbit of our identity, or first element. Speaking in terms of our chosen data type, this would be a (finite or infinite, depending on multiple aspects such as theoretical resolution and size of the picture and group action) set of images, which can all be transformed into each other by applying an appropriate transformation.

[1] or the topology, as not every transformation creating a topology is isomorphic to a Lie-group.

The model will learn to recover this structure and then sample from it to generate new data, which is theoretically also part of the complete set of images, positioned on this underlying submanifold. To guarantee as little noise as possible in our data, we choose to implement a deterministic approach. To make creation as simple as possible, we choose to plot a two dimensional Gaussian with maximal contrast over a dark background. We start by creating a set endowed with the topology of $\mathbb{R}^2$. This means the images have to be translationally symmetric, which leads us to restrict the Gaussian function to such a domain that its centre is always more than its variance away from the edges of the frame.

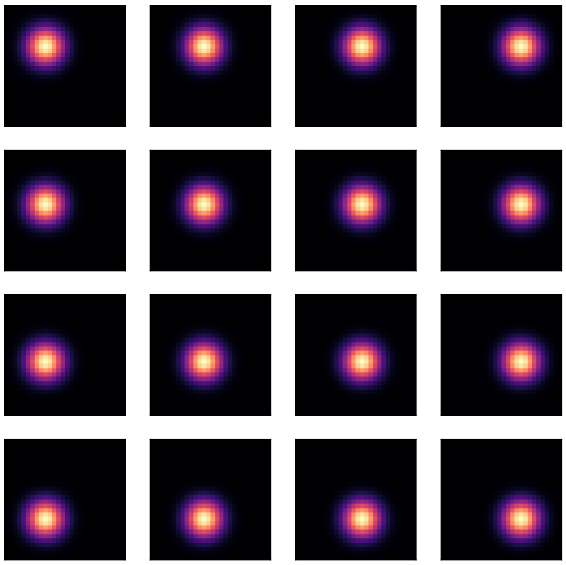


Figure 5.1: Visualization of samples from the flat dataset

Next, we want to endow the images of the Gaussian with a toroidal topology. Therefore, we simply identify the opposite sides of the image with each other, while ensuring, that the orientation stays the same. We basically create two wraparounds of the image, to make the Gaussian be able to exit one and enter on the opposite side. Visualizing some samples, we can observe this property. The translation is thereby chosen asymmetric, to display more variance.

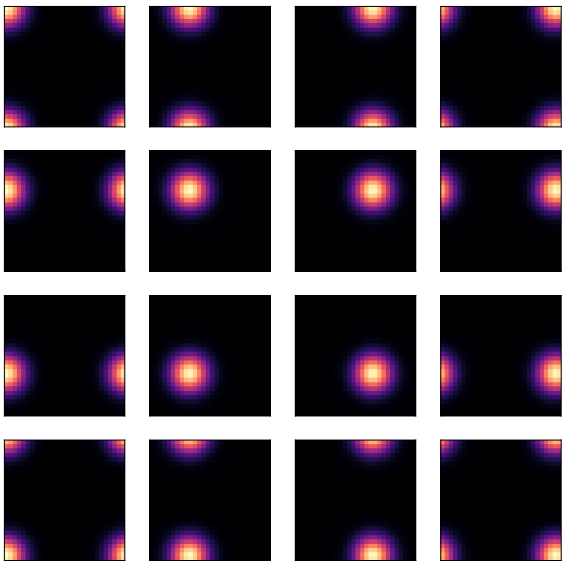


Figure 5.2: Visualization of samples from the toroidal dataset

Allowing the Gaussian function to analogously exit the sides of the frame for the flat data, with the difference of not-appearing on the opposite side, is a change which might be desirable, as to create perfect comparability between datasets, but it breaks translation symmetry, which one would require from a truly *flat* dataset. Due to computational restrictions, we cannot present a flat and continuous dataset without any borders, which is why we have to extrapolate meaning and result of the used, restricted flat dataset. In addition, one can argue that by moving the Gaussian out of frame, information is lost, which means the model has less data to interpret and recreate. As expected after these explanations, all models have a harder time in correctly generating new samples of these border cases. Still, analytics and plots of the different models running on this dataset can be found in the Appendix, as it was quite interesting to add this dataset into the comparison.

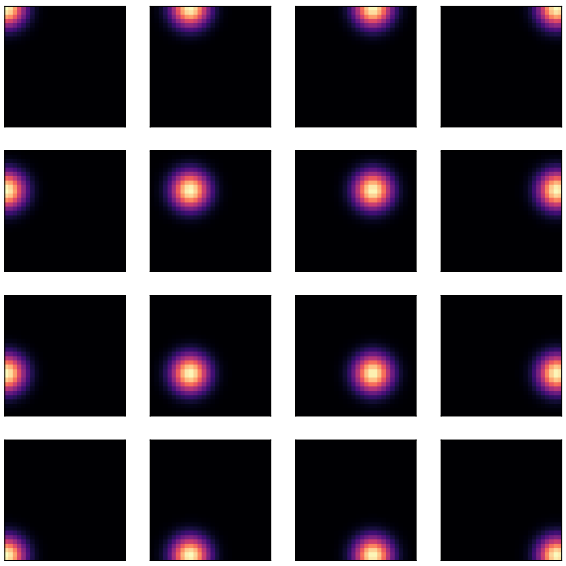


Figure 5.3: Visualization of samples from the flat dataset with edge crossing enabled

As there is no such thing as a *flat* sphere, we have to take another route to create an image set, which has $\mathbb{S}^2$ as underlying structure. Also, there is no 2 dimensional group whose orbit is the $\mathbb{S}^2$. We therefore create randomly generated centres on the surface of the sphere, render the Gaussian on it and compress them into two dimensional images. At first, performing a stereographic projection seems appropriate, but since we want the dataset to resemble the sphere's symmetry as perfect as possible, taking out a point of the domain is not optimal, which was also noticeable in training (one could obviously join datasets of stereographic projection from opposite poles). Therefore, instead of projecting or rendering a picture of the sphere with the Gaussian on it, we basically describe the image as a scalar field over $\mathbb{S}^2$, lay the trivial polar, azimuthal angle grid onto its surface and reshape the scalar field's values at the grid points into an image. Therefore, we keep rotational symmetry of the sphere and can display the full surface, without losing any detail, as the grid only has to be as fine as the chosen resolution of the images. We obviously cannot represent the Gaussian without distorting it, but we don't have to, as we simply want to create any dataset, which has $\mathbb{S}^2$ as underlying structure.

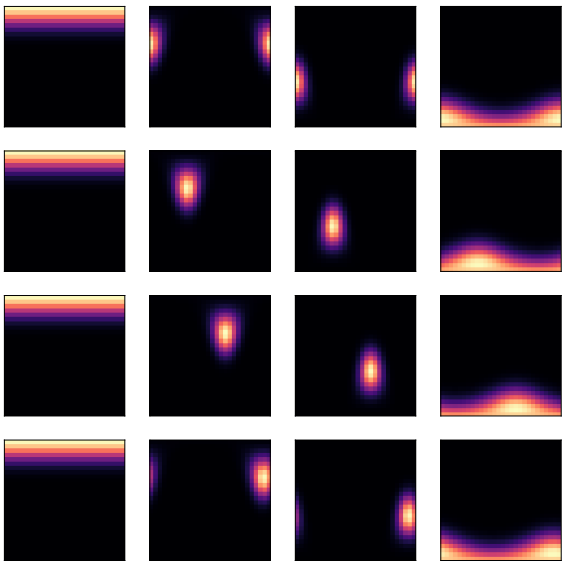


Figure 5.4: Visualization of samples from the spherical dataset

To create a training dataset, we will randomly choose a translation on the appropriate topological manifold and save the resulting image together with the two dimensional translation values $\{\gamma_1, \gamma_2\}$, which act as ground truth in training.

5.2 Technical Implementation

As we computationally benefit from simple analytical expressions, we will assume our decoders to be homoscedastic Gaussian.

For the actual computation we will use an autograd feature, which is provided with PyTorch.

As already mentioned in Section 4.2, we will not only have to ensure differentiability for network training, but also to calculate the curvature, as it includes differentiating the metric twice. Therefore, we are forced to use smooth activation functions.

For diagnosing and interpreting latent structure and curvature, we will use two different ways of displaying the data. First, we investigate, how the training data is represented in the latent space. For this, we will plot the mean of each training sample in the latent space. To be able to diagnose, how neighboring data samples (neighboring in the sense that they lie next to each other on the structural manifold underlying data space. For our purposes, neighboring can be assumed to be pictures with small changes in position of the Gaussian.) are being represented, we will continuously colour the means regarding their translation γ_i. As the translation is two dimensional, we will need each diagram twice. As one might suggest, a decoder, which is able to represent a continuous colour spectrum in both dimensions over all plotted samples, is able to correctly represent the data. As we will be able to see, this is exactly where topology of the latent space will get necessary, as representing two-manifolds with curvature in $\mathbb{R}^2$ is generally not possible.

The colouration of angles is chosen as such that 0 and 2π admit the same colour, to

be able to locate a possible seam. It is important to note, that this colouration does not *suggest* periodicity, as it is not a learned property. We expect the flat data to therefore have opposite edges in the same colour. If a structure is truly periodic, the colouration will seem continuous throughout.

From experience, we learn that to ensure encoder stability, we additionally have to restrict σ from getting to small or to large in training.

5.2.1 Integration

To integrate over the latent space, we simply estimate the Riemann sum by evaluating both the volume element and the curvature on a discrete, regular grid with fixed size.

$$\int_{\Theta} K(\theta)dV = \int\int K(\theta)V(\theta)\mathrm{d}\theta_1\mathrm{d}\theta_2 \approx V_{total} \cdot \frac{1}{N}\sum_{i=1}^{N} K(\vec{\theta_i}) \tag{5.1}$$

The assumption that this converges to the true value for a vanishing grid spacing, holds, if we can assure smoothness of the curvature as well as the volume element. For an appropriate fit and to ensure validity of our result, the functions are assumed only to vary sharply in scales larger than the grid spacing, to only restrict in resolution, not miss structure.

In future projects, it would be reasonable to include a Monte Carlo, or quasi-Monte Carlo, approach, especially while investigating high dimensional latent spaces, as the computational advantages scale with integration domain dimension. For this application, which can be regarded as proof of concept, a simple grid approach suffices.

Another approach includes triangulating the single grid spaces, by slicing them by the diagonal and calculating geodesic length of the sides to get the angles, then using Theorem 2.2. As this allows a direct calculation of discretely parametrized geometry without integrating, no error arises from approximating a grid.

In integrating, we obviously have to differentiate between the topologically flat and the toroidal or spherical cases. Latter are closed and compact, but the first has to be closed manually. For the flat torus, we use a rectengular, regular grid, for the sphere we have to apply one which matches the geometry, therefore, we integrate using a latitude-longitude grid.

Regarding the unbounded case, using Eq. (3.3), we conclude that the naive chose of just drawing a rectangle around the space, which the mapped mean sample points inhabit, is not computationally reasonable, as we would also have to calculate the outer angles, in addition to the geodesic curvature of the border. A more reasonable approach is to compactify $\mathbb{R}^2$, by integrating over a disc D with centre in the middle of the latent data points and radius R large enough to include all. Therefore, the border is smooth and the outer angles drop out from the equation. If we look back at Theorem 3.3 in combination

with Eq. (3.3), we can see that

$$\int K \mathrm{d}A + \kappa_\partial(R) = 2\pi\chi(D). \tag{5.2}$$

We then calculate the total geodesic curvature κ_∂ of the border

$$\kappa_\partial(R) = \oint_{\partial D} k_g \mathrm{d}s, \tag{5.3}$$

where k_g is the geodesic curvature of the curve ∂D. We choose

$$\gamma(t) = c + R(\cos(t), \sin(t)) \tag{5.4}$$

with $t \in [0, 2\pi)$. c and R are constants, which we have to determine from the spread of the encoded samples. From [21] we conclude that

$$k_g = \frac{F_{ij} a^i N^j}{v^2}, \tag{5.5}$$

with the covariant acceleration a

$$a^k = \ddot{\gamma}^k + \Gamma^k{}_{ij} \dot{\gamma}^i \dot{\gamma}^j, \tag{5.6}$$

the covariant velocity v

$$v = \sqrt{F_{ij} \dot{\gamma}^i \dot{\gamma}^j} \tag{5.7}$$

and the unit normal N (obviously $F_{ij} N^i N^j = 1$ holds)

$$N^i = -\sqrt{\det(F)} F^{ij} \epsilon_{jk} T^k, \tag{5.8}$$

where we used the unit tangent

$$T^i = \frac{\dot{\gamma}^i}{v}. \tag{5.9}$$

As we know that the Euler characteristic of a two dimensional disk is $\chi(D) = 1$, we have to verify that

$$\int_0^{2\pi} \int_0^R K(r, \phi) \sqrt{\det(F)} r \mathrm{d}r \mathrm{d}\phi + \int_0^{2\pi} k_g(t) v(t) \mathrm{d}t = 2\pi. \tag{5.10}$$

From this simple approach, we can also generalize to the case where $r \longrightarrow \infty$, therefore approximating the real valued two dimensional plane.

As we argued in Section 4.7.1, we are able to dismiss a constant in the metric, while

still integrating the curvature. We now want to explicitly prove that this holds.

$$\int_\Theta K dA = \int_\Theta \frac{R_{1212}}{\det(F)} \sqrt{\det(F)} \mathrm{d}^2\theta \tag{5.11}$$

$$= \int_\Theta \frac{R_{1212}}{\sqrt{\det(F)}} d^2\theta \tag{5.12}$$

$$= \int_\Theta \frac{F_{1i} R^i{}_{212}}{\sqrt{\det(F)}} d^2\theta \tag{5.13}$$

Now, we transform $F_{ij} \to \tilde{F}_{ij} = \sigma^2 \cdot F_{ij}$. $R^i{}_{212}$ does not transform, as it is fully defined by Γ and

$$\Gamma^k_{ij} = \tfrac{1}{2} F^{kl} \left(\partial_i F_{lj} + \partial_j F_{li} - \partial_l F_{ij}\right) \tag{5.14}$$

$$= \tfrac{1}{2} \sigma^2 \tilde{F}^{kl} \sigma^{-2} \left(\partial_i \tilde{F}_{lj} + \partial_j \tilde{F}_{li} - \partial_l \tilde{F}_{ij}\right) \tag{5.15}$$

$$= \tilde{\Gamma}^k_{ij}, \tag{5.16}$$

the integrand transforms as invariant tensor density

$$\frac{F_{1i} R^i{}_{212}}{\sqrt{\det(F)}} = \frac{\sigma^{-2} \tilde{F}_{1i} R^i{}_{212}}{\sqrt{\det(F)}}. \tag{5.17}$$

The determinant of a matrix A with dimension n transforms under a constant prefactor as such

$$\det(\tilde{A}) = \lambda^n \det(A). \tag{5.18}$$

Therefore the integrand admits

$$= \frac{\sigma^{-2} \tilde{F}_{1i} R^i{}_{212}}{\sqrt{\sigma^{-4} \det(\tilde{F})}} \tag{5.19}$$

$$= \frac{\sigma^{-2} \tilde{F}_{1i} R^i{}_{212}}{\sigma^{-2} \sqrt{\det(\tilde{F})}} \tag{5.20}$$

and we conclude

$$\int_\Theta K dA = \int_\Theta \tilde{K} d\tilde{A}. \tag{5.21}$$

The same applies for the integral over the geodesic curvature of the border, as Christoffel symbols do not transform and

$$v \to \tilde{v} = \sigma v, \qquad a^k \to \tilde{a}^k = a^k, \tag{5.22}$$

$$T^k \to \tilde{T}^k = \sigma^{-1} T^k, \qquad N^i \to \tilde{N}^i = \sigma^{-1} N^i. \tag{5.23}$$

The geodesic curvature therefore picks up a prefactor

$$\tilde{k}_g = \frac{\tilde{F}_{ij} \tilde{a}^i \tilde{N}^j}{\tilde{v}^2} = \frac{\sigma F_{ij} a^i N^j}{\sigma^2 v^2} = \sigma^{-1} k_g, \tag{5.24}$$

which is exactly compensated by the rescaling of the line element, $\mathrm{d}\tilde{s} = \tilde{v}\,\mathrm{d}t = \sigma \mathrm{d}s$. Therefore

$$\int \tilde{k}_g(t)\,\tilde{v}(t)\,\mathrm{d}t = \int k_g(t)\,v(t)\,\mathrm{d}t. \tag{5.25}$$

which rectifies us, simplifying calculations by using

$$F = J_\mu^T J_\mu. \tag{5.26}$$

5.3 Results

As the encoder basically tries to optimize a basis representation for the data in the imposed dimensionality, we notice that linear dependence can distort the visual representation. Take the example of a two dimensional, flat dataset, the obvious expectation would be, for the encoder to identify the ground truth basis γ_i with its two dimensions θ_j, meaning

$$\{\gamma_1, \gamma_2\} \cong \{\theta_1, \theta_2\}. \tag{5.27}$$

However,

$$\gamma_1 \cong \theta_1, \tag{5.28}$$

$$\gamma_2 \cong \theta_2 + \frac{1}{2}\theta_1 \tag{5.29}$$

would also be a linearly independent basis representation. Therefore, the latent basis must be identical to the ground truth basis, up to an affine transformation.

5.3.1 Canonical Latent Geometry and Topology Mismatch

We will now explicitly look at the different variational autoencoders cross trained on the various sets of data. We will both look at the *encoder's* placement of the training samples, as well as the curvature and metric determinant with respect to the *decoder*. We start with the conventional, or flat, variational auto encoder with a two dimensional latent space. First, we train the encoder with the flat dataset.

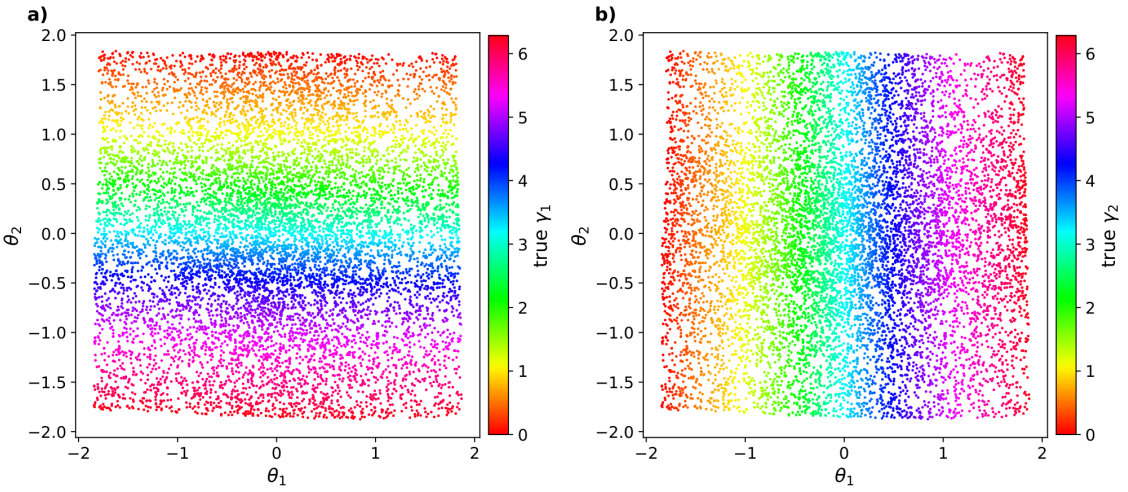


Figure 5.5: Flat latent representation of a flat dataset. a) and b) represent the two ground truth angles of variation.

As we can see, the dataset gets represented as a square, with both ground truths showing a continuous colour flow. This means that images with small translations of the Gaussian do in fact lie next to each other in the latent space. Contrary to what we stated above, it seems as though the encoder identified the ground truth's dimensions exactly with the directions of the latent space's axes. From other runs, we can verify that depending on random seed and training epochs, the coloured square sometimes is depicted rotated with respect to the latent coordinate axes. To truly represent a dataset with underlying topology homeomorphic to $\mathbb{R}^2$, we would need infinite translations in infinitely large images. Although we can not demonstrate that, we can extrapolate this result to look like a continuously coloured infinite plane in latent $\mathbb{R}^2$. Next, we want to investigate the geometry, which the decoder introduces.

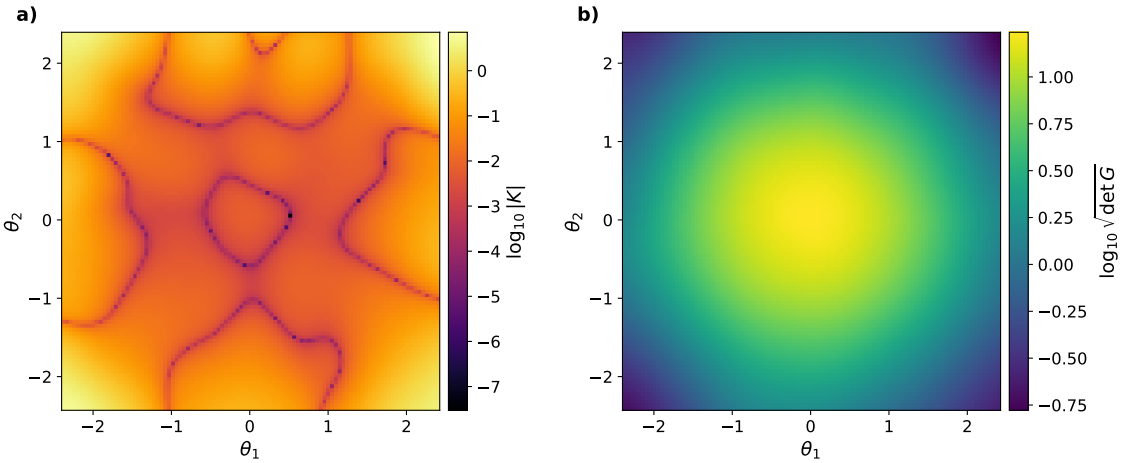


Figure 5.6: a) absolute curvature and b) magnification (both logarithmically scaled) of the Fisher metric of the decoder of a flat dataset in a flat latent space.

As expected, there is not much variation in the curvature, other than a symmetric

structure, which probably is due to the borders of the dataset, where the decoder looses a properly defined environment. The magnification appears to be perfectly homogeneous, when imagining the dataset to be infinite. Next, we will look at a more intriguing case, by representing the toroidal dataset in the two dimensional latent space. As $\mathbb{T}^2$ cannot be embedded into $\mathbb{R}^2$, we expect discontinuities[2] in the encoder.

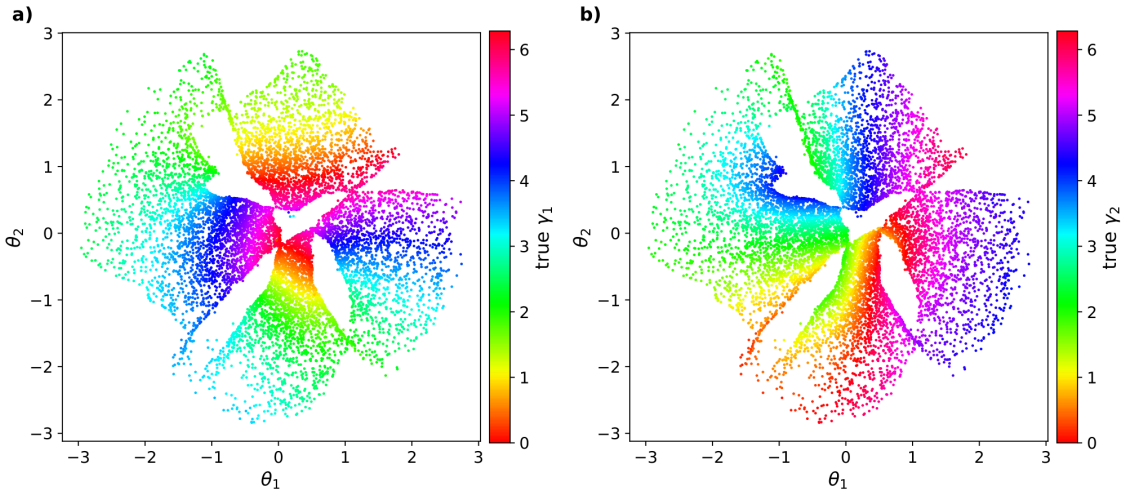


Figure 5.7: Flat latent representation of a toroidal dataset. a) and b) represent the two ground truth angles of variation.

We can identify multiple tears of the toroidal structure, as the encoder seems to converge to a form of cutting the torus into at least two parts and folding all of them open. Discontinutities are also quite noteable, when looking at the differences in color opposite of an area, where no samples are placed. This means that samples in close surrounging of each other do not necessarily correspond to datasamples with similar translations. This will also mean that sampling from a point next to a tear, will probably result in bad performance, as newly generated samples are not guaranteed to be similar to other samples in this area.

[2]See Section 4.3 for a comment on use of “discontinuity”.

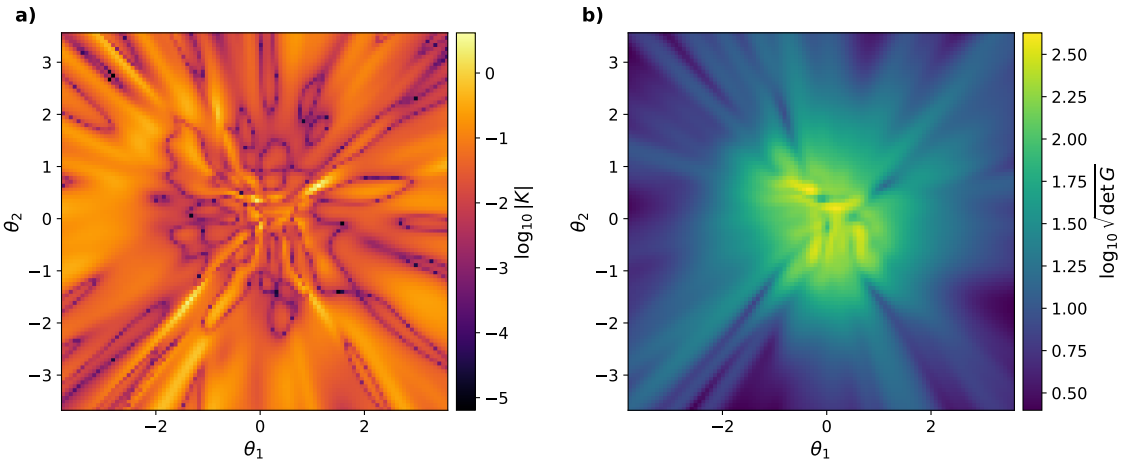


Figure 5.8: a) absolute curvature and b) magnification (both logarithmically scaled) of the Fisher metric of the decoder of a toroidal dataset in a flat latent space.

We can now see directly that neither the curvature nor the magnification are homogeneous. Both seem to vary quite drastically for the toroidal dataset. To make precise statements about how curvature and magnification act around topological tears of the encoder, one would have to invest time and computational cost into plotting a finer grid. As this is not the goal of this survey, we will stay with qualitative statements. It seems as though the curvature does stay stable in continuous areas, similar to what we saw with the flat autoencoder in Fig. 5.6, while both the curvature and the magnification in and around areas with little support and discontinuities, vary and increase strongly.

Next, we train the flat latent variational autoencoder on the spherical dataset. As the sphere is parametrized by $\psi \in [0, \pi]$, $\varphi \in [0, 2\pi)$, the colour of one of the ground truth angles of translation obviously does not cover the same range as the other.

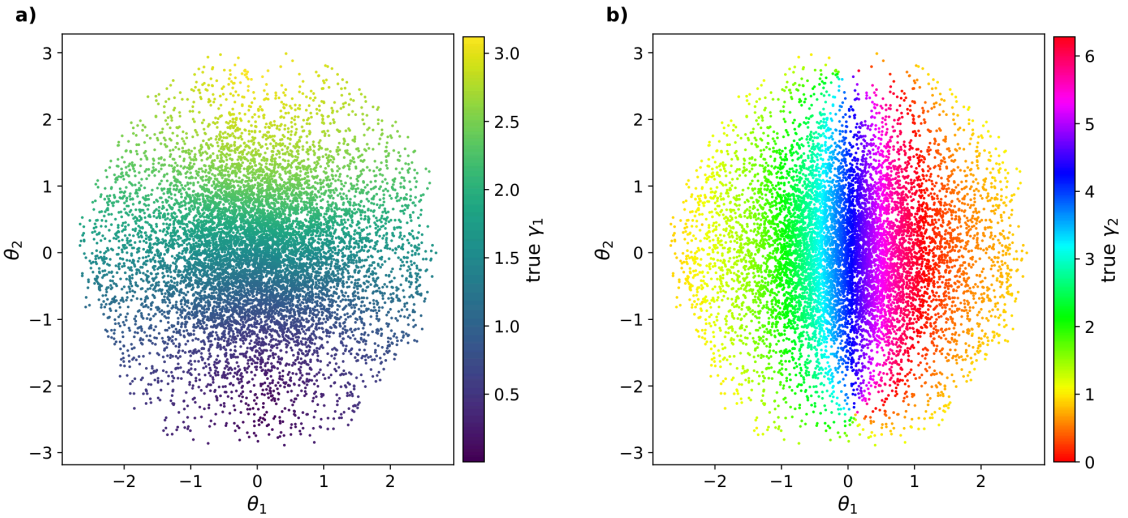


Figure 5.9: Flat latent representation of a spherical dataset. a) and b) represent the two ground truth angles of variation.

As we can see, the encoder manages to continuously map ψ, but fails with φ, as the sphere has to be torn open somehwere to be able to be represented in two dimensions.

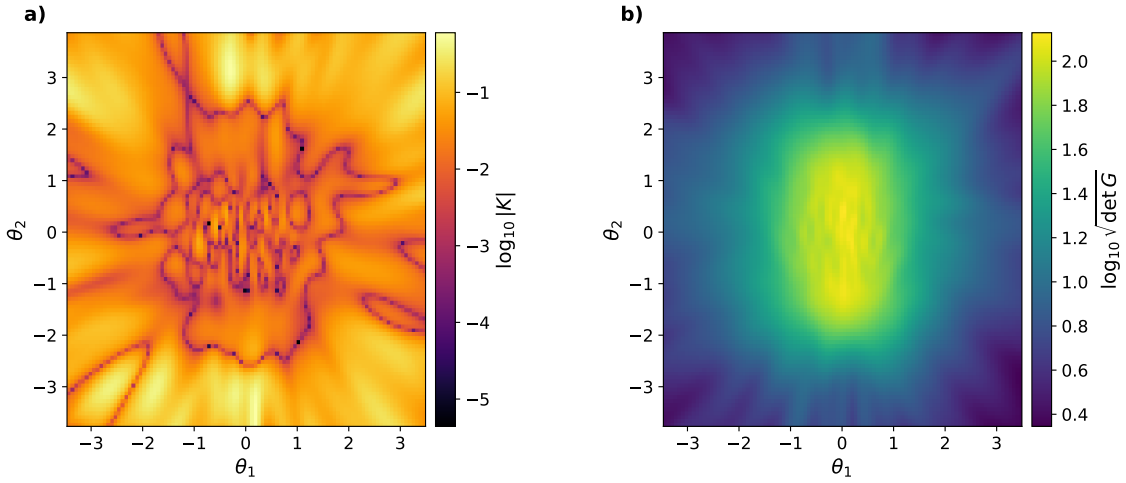


Figure 5.10: a) absolute curvature and b) magnification (both logarithmically scaled) of the Fisher metric of the decoder of a spherical dataset in a flat latent space.

As we identified the topological tear of the spherical dataset to reach around the whole spherical point cloud, we interpret it analogously to the toroidal set. Along the tear, we see a drop in absolute curvature to nearly zero, while in the area of little support, here the outside of the round cloud, the absolute curvature rises. The continuous interior, again shows random variations with little magnitude. The magnification seems to vary slightly inside the cloud and drops immediately outside.

Now, we will introduce the first variational autoencoder with altered latent space topology. We will start with the toroidal latent model and compare how the encoder and decoder handle the flat and the toroidal data. Although we can visualize the flat torus in 2d, it is also quite helpful to visualize the distribution of the encoder mappings of the data on a 3 dimensional model of the donut. This makes it easier to understand the topology, although this way of representing a torus introduces curvature. Therefore, we will only use this 3 dimensional model to display the placement of the data, curvature is displayed on the flat domain.

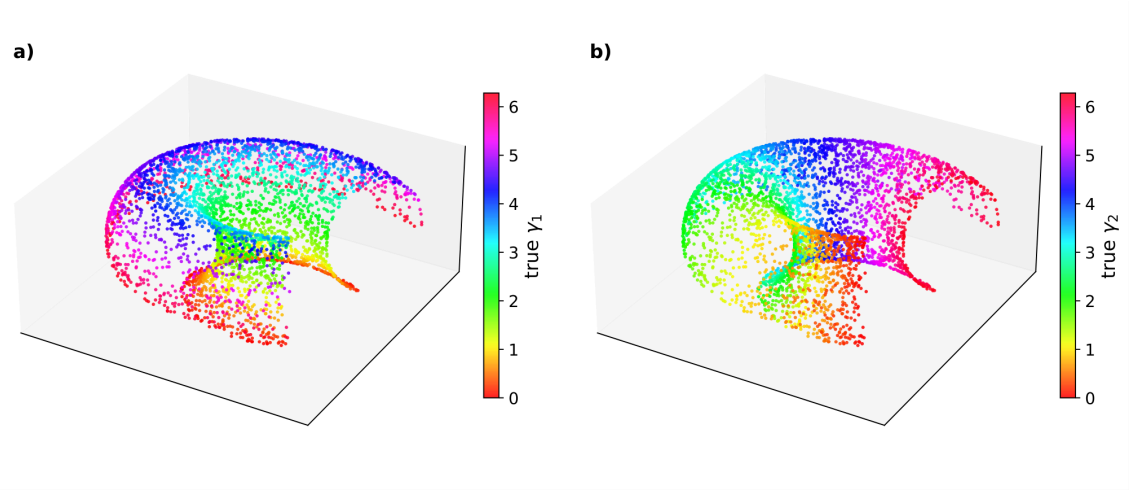


Figure 5.11: Toroidal latent representation of a flat dataset. a) and b) represent the two ground truth angles of variation, 3 dimensional plot.

As we can see, the encoder simply tries to position the plane onto the torus, which leads to a large empty area of the latent space, where absolutely no data samples are placed. This is only true for the means, statistically, the latent space is filled by the encoder.

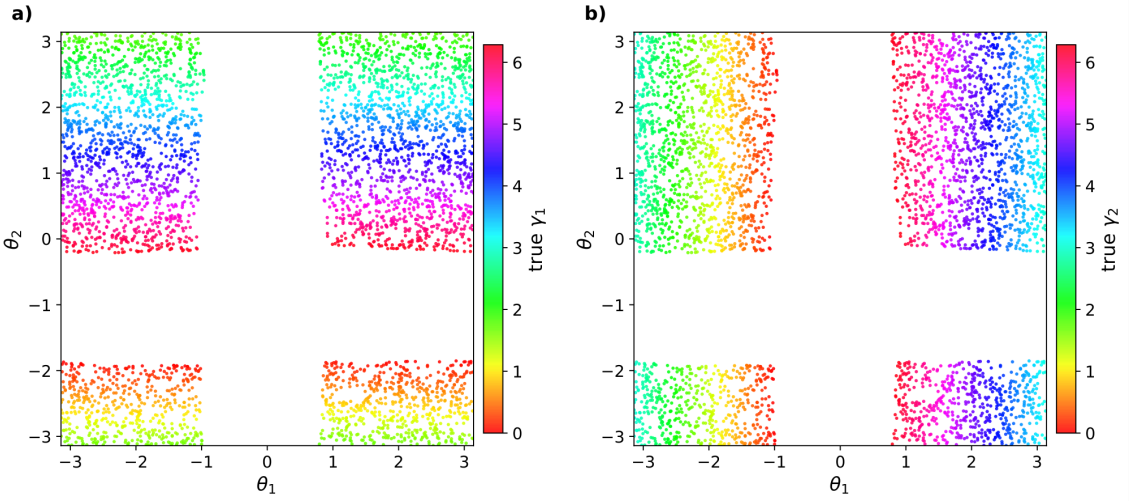


Figure 5.12: Toroidal latent representation of a flat dataset. a) and b) represent the two ground truth angles of variation, 2 dimensional plot.

Here we can see the two objectives of the training working against each other, as the prior suggests filling the whole torus, while the encoder simultaneously benefits from being as injective as possible. We therefore identify the empty strips between borders of the dataset not as tears, but as spacing, created by the encoder, because the borders of the flat set obviously can't be identified with each other, unlike the toroidal set.

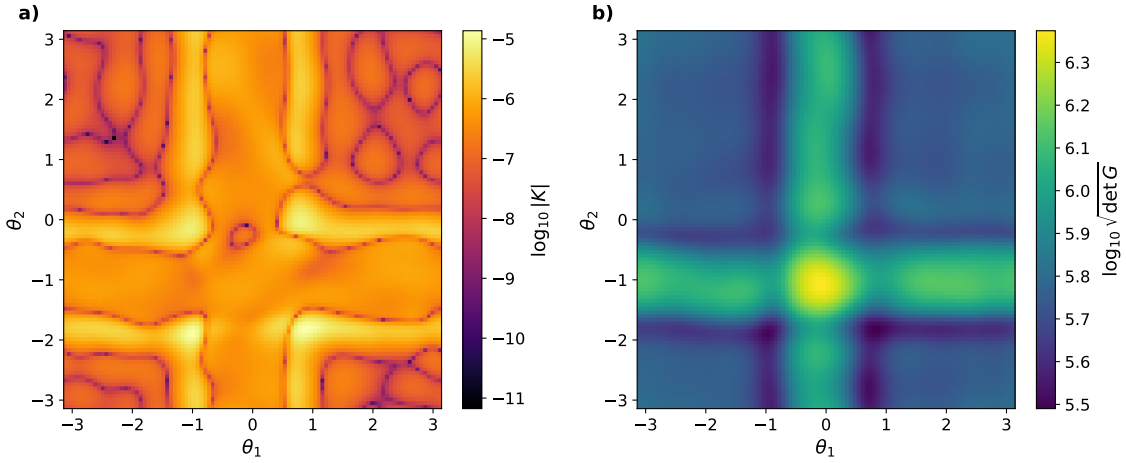


Figure 5.13: a) absolute curvature and b) magnification (both logarithmically scaled) of the Fisher metric of the decoder of a flat dataset in a toroidal latent space.

At closer investigation, it is apparent, that curvature approaching an area of low statistical support first rises, then falls to a minimum at the border, then rises inmidst the tear. The magnification just shrinks approaching the tear, then rises to a global maximum in the middle of the tear. The latter further justifies our interpretation, that the decoder increases the space between the borders of the flat dataset, as they are not connected.

Next, we will train the toroidal latent variational autoencoder on the toroidal dataset. As the topology and dimension of dataset and latent space match, we assume that there will be no discontinuities and the latent space is filled homogeneously, as the data's periodicities benefit from the toroidal latent space.

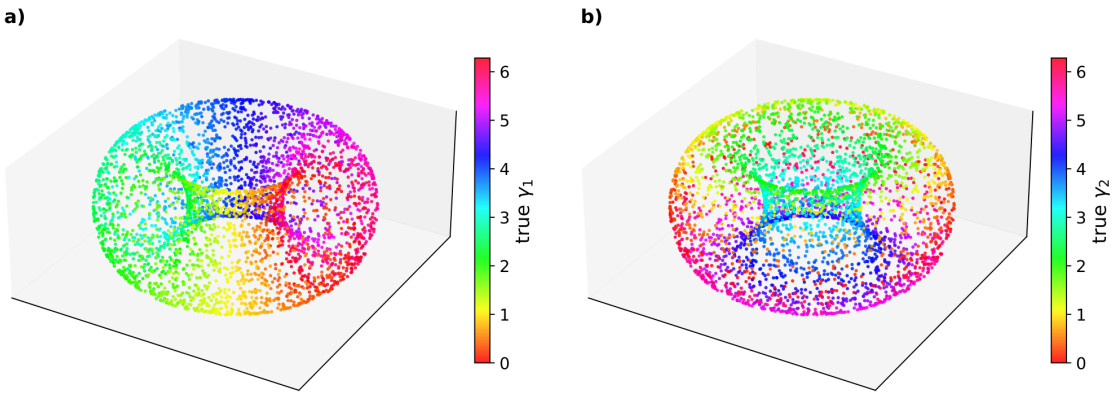


Figure 5.14: Toroidal latent representation of a toroidal dataset. a) and b) represent the two ground truth angles of variation, 3 dimensional plot.

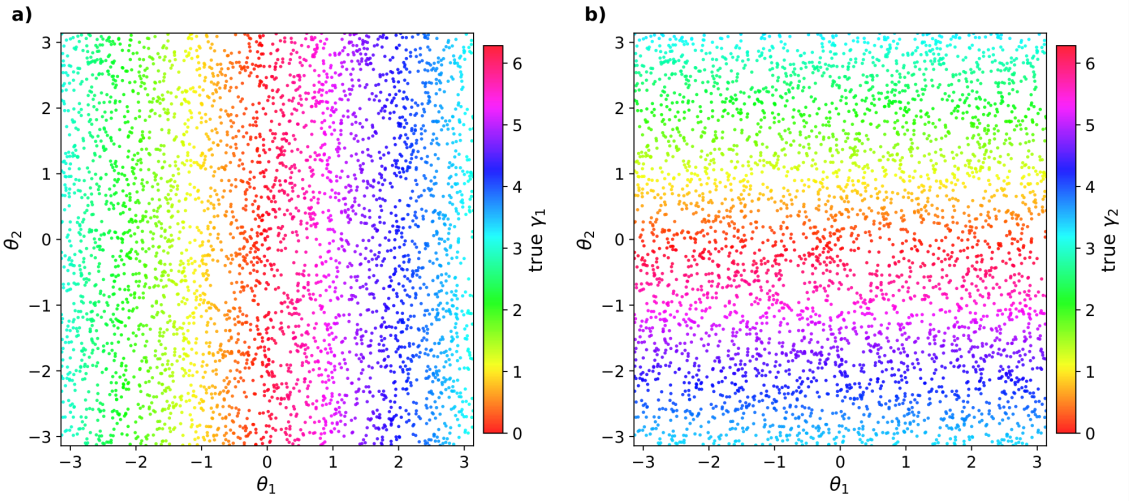


Figure 5.15: Toroidal latent representation of a toroidal dataset. a) and b) represent the two ground truth angles of variation, 2 dimensional plot.

As expected, the latent space is filled continuously and homogeneously by the encoder, without introducing nonlinearities in identifying the dataset with the latent space, as there is simply no necessity to do so. Training can simultaneously fill the latent space according to the uniform prior, while still ensuring injectivity of the encoder (This dataset and latent space introduce non-injectivity through periodic boundary conditions. The encoder is injective with respect to equivalence class partitioning.).

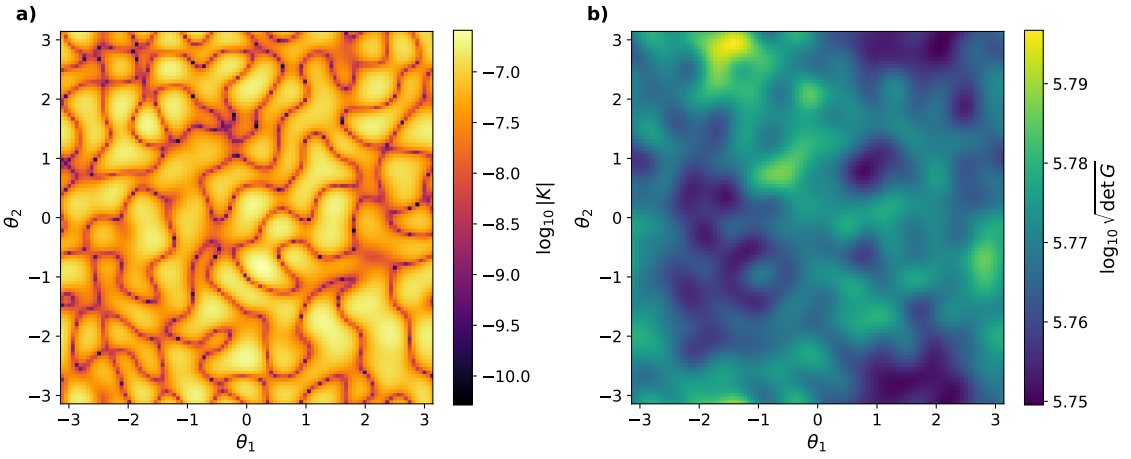


Figure 5.16: a) absolute curvature and b) magnification (both logarithmically scaled) of the Fisher metric of the decoder of a toroidal dataset in a toroidal latent space.

The curvature and magnification validate this conclusion, as they appear to be homogeneous as well, while varying only slightly, compared to the cases, where the encoder introduces nonlinearities.

Next, we will do the same procedure for the variational autoencoder with the topologically spherical latent space, starting with the flat dataset. Similar to the toroidal latent model, we are able to visualize the latent space in two different ways, to improve

understandability. First, we visualize the points on the surface of the three dimensional sphere, then we use the Mollweide projection, to display the encoder mapping flatly.

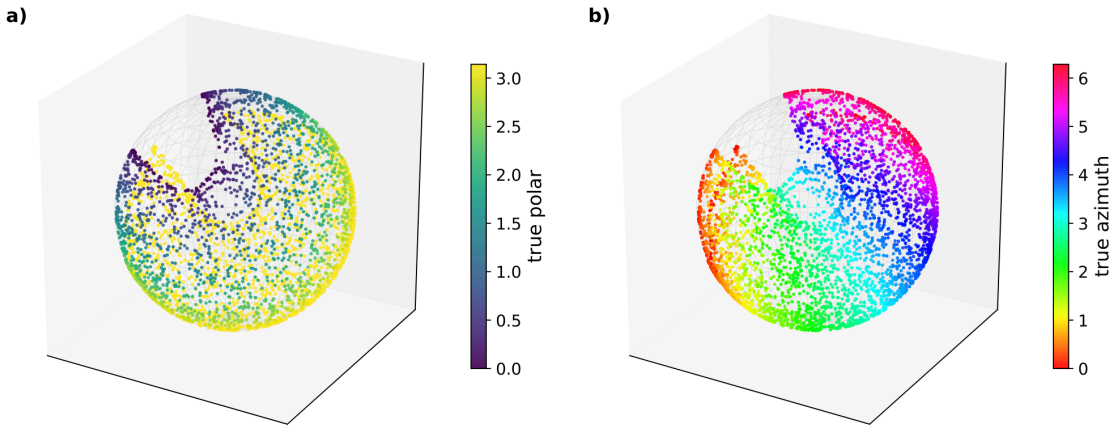


Figure 5.17: Spherical latent representation of a flat dataset. a) and b) represent the two ground truth angles of variation, 3 dimensional plot.

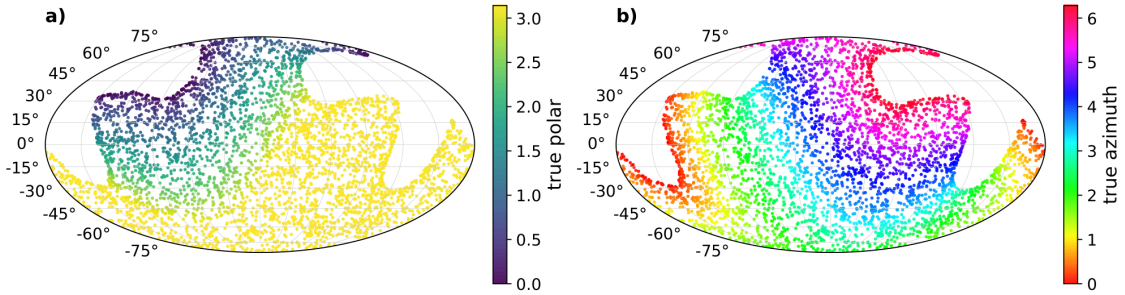


Figure 5.18: Spherical latent representation of a flat dataset. a) and b) represent the two ground truth angles of variation, 2 dimensional plot.

As we can see, training the spherical latent model with the flat dataset basically reproduces the result of the toroidal latent model. As the flat dataset does not have any periodic properties, the encoder simply tries to spread out the flat plane onto the surface of the sphere, while keeping space between the edges.

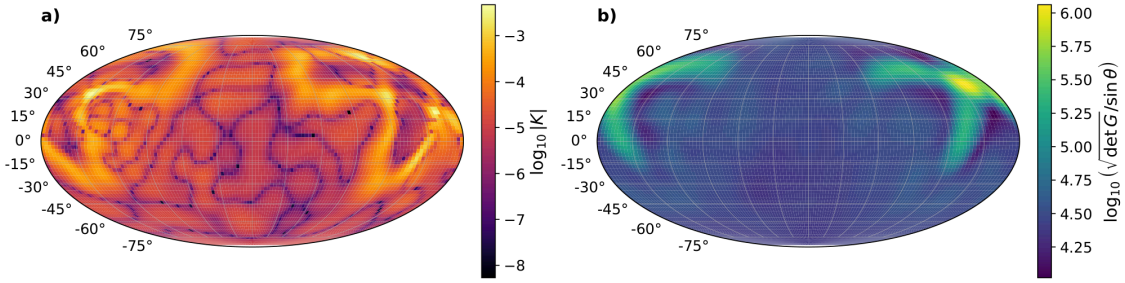


Figure 5.19: a) absolute curvature and b) magnification (both logarithmically scaled) of the Fisher metric of the decoder of a flat dataset in a spherical latent space.

The curvature and magnification also follow the observations we made when analysing the toroidal model on the flat dataset. Absolute curvature rises approaching the empty space created by the encoder, while admitting a minimum, which is a continuous curve, surrounding the area with little support. The area, which can be identified with the interior of the flat dataset, appears to be homogeneous both in curvature and in magnification and admits the same random dark laces as in Fig. 5.16. Also, similar to the flat dataset in the toroidal latent model, the magnification rises between borders of the dataset (see Fig. 5.13).

Next, we will train the spherical latent model on the spherical dataset.

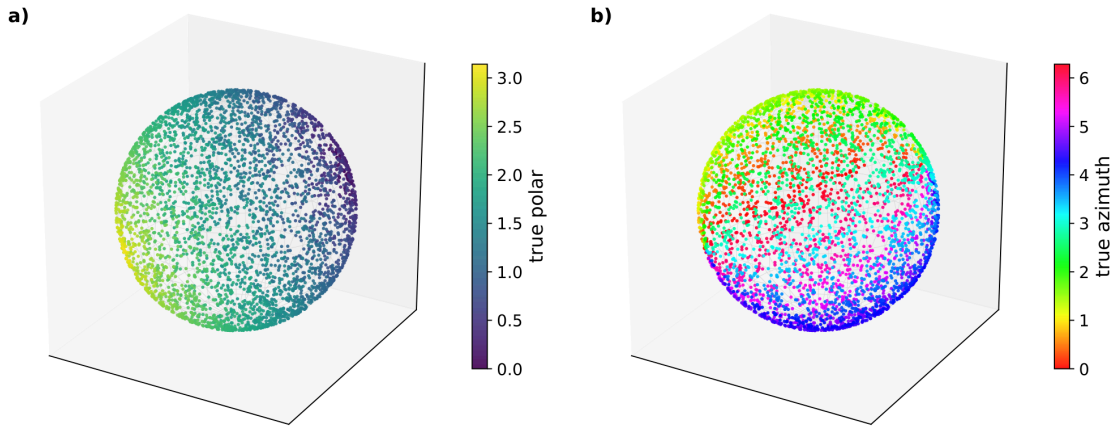


Figure 5.20: Spherical latent representation of a spherical dataset. a) and b) represent the two ground truth angles of variation, 3 dimensional plot.

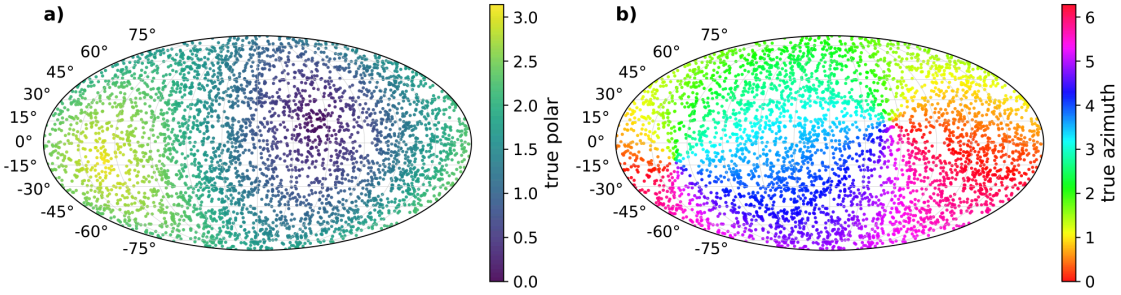


Figure 5.21: Spherical latent representation of a spherical dataset. a) and b) represent the two ground truth angles of variation, 2 dimensional plot.

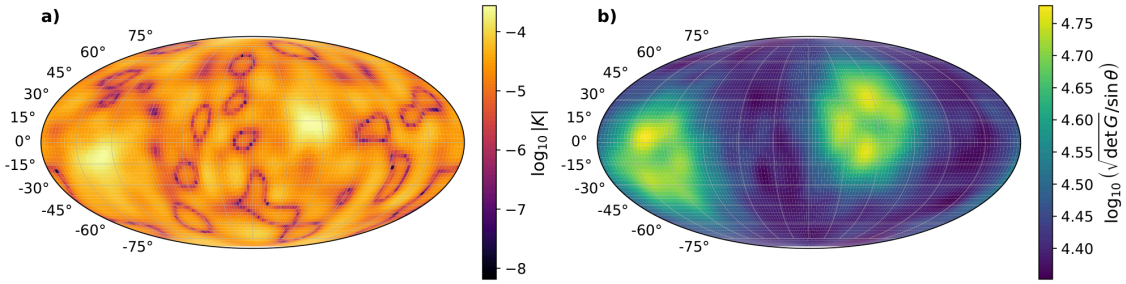


Figure 5.22: a) absolute curvature and b) magnification (both logarithmically scaled) of the Fisher metric of the decoder of a spherical dataset in a spherical latent space.

As we are able to see, similar to the toroidal latent model with the toroidal dataset, the latent space gets filled homogeneously, while representing the data linearly. Both the absolute curvature and the magnification appear nearly homogeneous and do not vary strongly. To definitively verify a homogeneous distribution of samples, we would need to look at a spatial histogram.

Next, we will state some general observations regarding the curvature in combination with the different topologies and try to interpret these results. Especially important seem to be the curves, on which the absolute curvature vanishes. We will refer to them as laces or strings, due to their appearance.

- Generally, we can differentiate between two kinds of laces: The ones surrounding discontinuities and the ones filling areas which are continuous.
- If areas are flat with respect to the latent space (meaning both the flat dataset on the flat latent and the toroidal dataset on the torus latent space), the laces or curves are not closed, but exit the frame of the plot.
- Areas bounded by tears are enclosed by laces (or pairs of laces) of vanishing absolute curvature.
- The spherical dataset induces randomly distributed, circular in the sense of $\mathbb{S}^1$, laces of vanishing absolute curvature (maybe this is due to the constant curvature of the sphere, where random variation of the curvature favour closed loops, instead of unclosed wiggling strings.).

We can probably dismiss the possibility that a diverging metric determinant[3], leads to curves of near zero absolute curvature, as we don't see corresponding structures in the magnification plot. It would actually make sense, to plot the curvature, instead of its absolute, as it is almost certain that the absolute curvature vanishes on curves due to a sign flip. From the Chern–Gauss–Bonnet theorem we can clearly deduce that these changes between positive and negative curvature have to appear somewhere in the latent space (at least for the torus, as the integral has to vanish. We can state something similar for

[3]As it happens, when observing the statistical geometry of a latent space model performing a categorizing task, as can be seen in [9]

the sphere and the disk in the flat latent, when comparing the scales of curvature, volume element and Euler characteristic.). This also explains, why the magnification plots do not admit such curves of vanishing value, as we deduced Section 4.7.3, the determinant of the homoscedastic Gaussian metric is always positive. We interpret the absolute curvature plots as following:

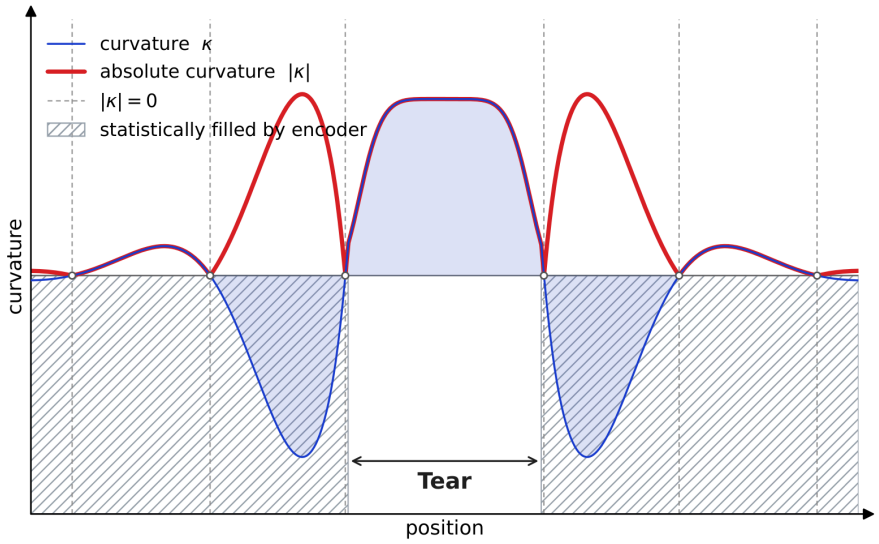


Figure 5.23: Simple visualization of the curvature surrounding a tear of the encoder. Sign is chosen arbitrarily. Area under the middle three peaks is supposed to cancel identically. (Claude.ai was used to implement this plot based on a sketch done by the author.)

In areas with no topological defects, the curvature slowly varies around zero. This is homogeneous for the flat datasets and ordered in circles for the spherical, which is probably a consequence of the sphere's constant curvature. Approaching a topological tear of the encoder, or an area with low statistical support for the decoder, the curvature flips its sign twice while crossing into the tear (these analyses were done on Fig. 5.13, as it is the simplest to interpret. They can be generalized for the other curvature plots, although the precise curvature profiles might be more complex, while still following the proposed framework.). On the example of the torus we can try to build a hypothesis, on why we observe exactly this. As the decoder has to get non-linear, to generate a continuous function, which might also need to be periodic, curvature increases in a tear, as this is an area of low support and discontinuous in colour (colour standing for correct placement of samples with respect to their neighbouring samples). To compensate the increase in curvature, while fulfilling the differential geometric identity of Chern–Gauss–Bonnet, every tear also introduces an area of negative curvature around its border. Choosing width and value of both curvature areas is an optimization problem, which gets solved by training the model.

After accidentally training the flat variational autoencoder on the toroidal dataset for only a tenth of the duration of recent runs, the encoder recovered a structure resembling the Moebius strip. As this is topologically the same simplex as the torus with only two

sides identified under different orientation, this could be the model recognizing only one periodic feature of the data.

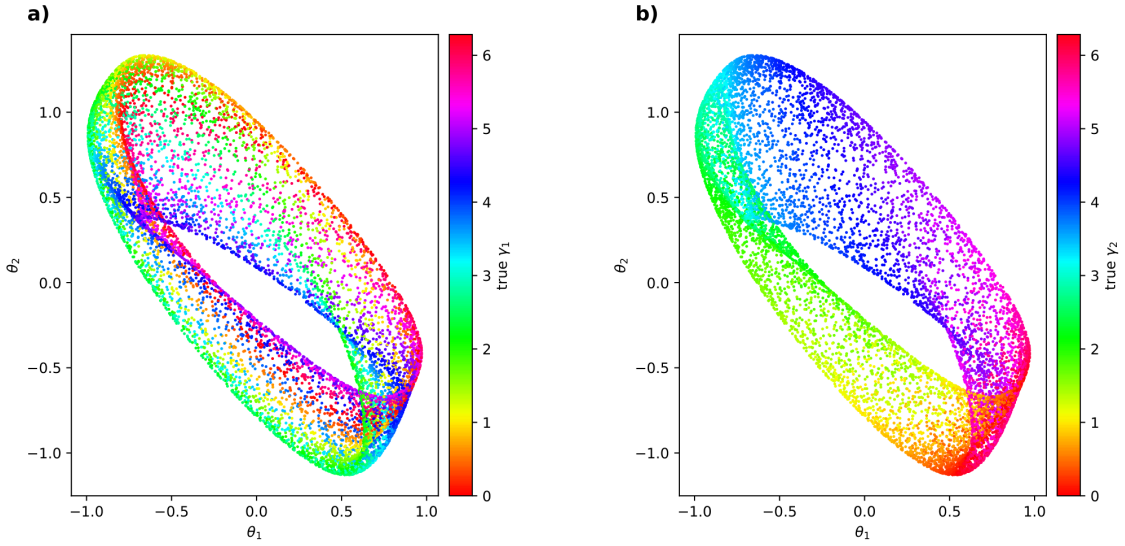


Figure 5.24: Latent space heavily undertrained flat variational autoencoder on the toroidal latent space

This raised the question, whether the models were overfitting, an aspect of network training we have not touched upon yet, as we wanted to focus on the latent space and not on the performance of models. For a proper interpretation of the plotted curvature, we have to guarantee that the model is in fact not overtrained. Although this is not as present in variational autoencoders and especially unlikely for our datasets, as there is practically no noise in the data by construction, an analysis of this problem is still beneficial for all of the above. Therefore, we added a newly generated validation set and compared its metrics to the model performing on the training dataset, to analyse, how well the trained models can generalize. Luckily, we did not find typical signs of overfitting, but realized something different: From the performance of the various models on unseen data, we can directly compare, how *well* a variational autoencoder with a specific topology performs on data, which admits some other topology.

5.3.2 Validating Topology

To estimate how well a model performs in validation in comparison to training, we simply calculate the mean squared error between the input and the reconstructed sample, average over all and plot it against the epochs. Overfitting would show itself in a steadily falling training error and a validation error which falls to a minimum and starts rising again. To guarantee complete independence of the two datasets, we will use a different random seed. As we specifically take interest in the topology and want to look for structural benefits or disadvantages of a certain topology of the latent space, we also plot the samples that are performing the worst. Here, they have two different coordinate systems in which they

can be located, both informative. On the one hand, we can investigate where they are placed in the latent space, therefore being able to make a connection between curvature, magnification and topology, and on the other hand, we can observe in which areas of the ground truth (translation of the Gaussian) they are situated.

As we suspect a tear in the data manifold leading to weak performance at said topological defect, both visualizations are necessary. We start with the flat autoencoder and the flat dataset.

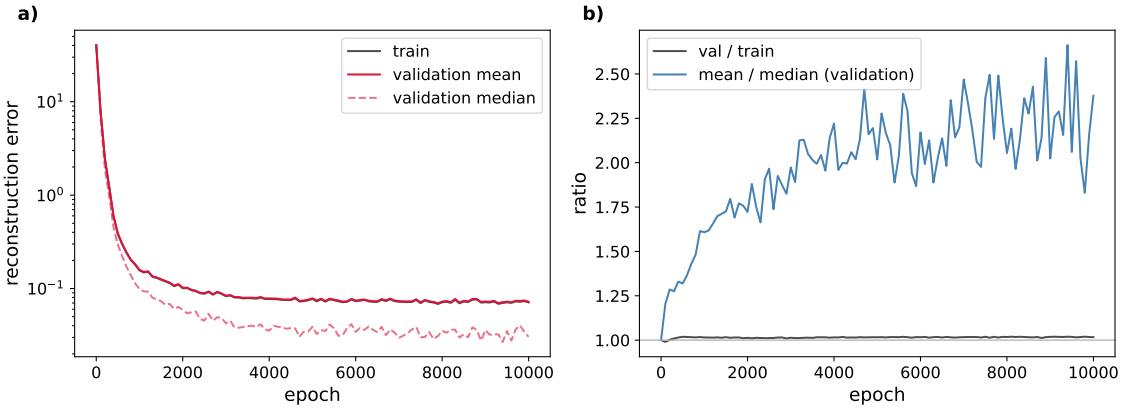


Figure 5.25: Training of the flat autoencoder on the flat dataset. a) Reconstruction error against epoch (logarithmic ordinate): training error (grey), validation mean (solid red), validation median (dashed red). b) Ratios against epoch: validation to training (grey) and mean to median of the validation error (blue).

As we can see, the model does not overtrain, as both the median and mean of the validation error drop quickly and converge. Therefore we can follow that the model itself has converged. We can see that the validation error in relation to the train error is at exactly 1.00. Although the mean over median rises and varies strongly, from interpreting the left plot, we can see that this is a result of the denominator shrinking, not of a bad fit.

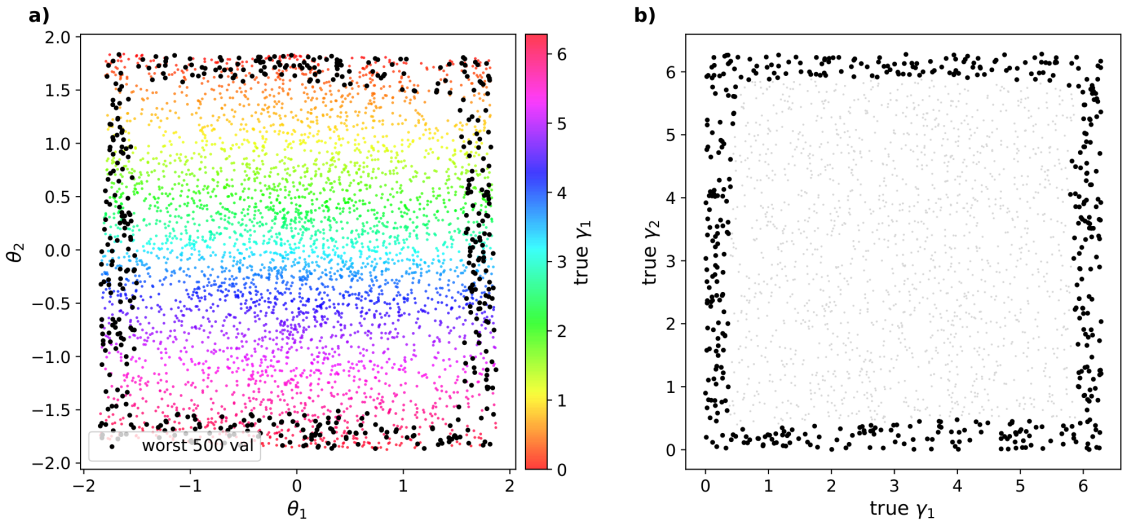


Figure 5.26: Localization of the reconstruction error of the flat variational autoencoder with the flat dataset. a) Latent coordinates (θ_1, θ_2) of the validation set, coloured by the ground truth γ_1, the 500 points with the largest reconstruction error are marked in black. b) The same points in ground-truth coordinates (γ_1, γ_2), with the worst 500 in black and the remainder in grey.

As expected, the worst samples of the flat dataset are positioned on the borders, as these are the only topological "defects", as we cannot create a borderless, flat dataset. They are distributed evenly and in the ground truth coordinates we can see the same distinct borders, where the validation performed worst. This is also expected, as there is a one-to-one map between the sample coordinates and the latent space.

Next, we will analyse the performance of the toroidal dataset in the flat latent space.

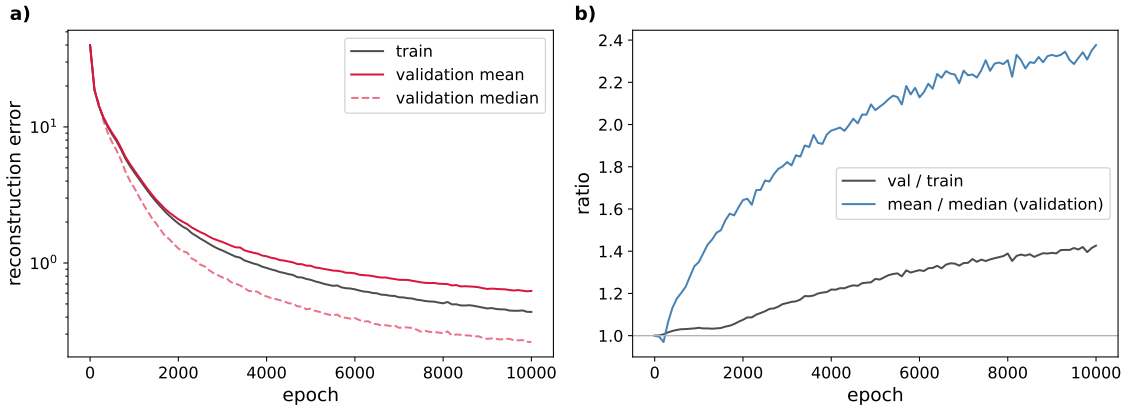


Figure 5.27: Training of the flat autoencoder on the toroidal dataset. a) Reconstruction error against epoch (logarithmic ordinate): training error (grey), validation mean (solid red), validation median (dashed red). b) Ratios against epoch: validation to training (grey) and mean to median of the validation error (blue).

We can see that the error of validation samples falls way slower than in Fig. 5.25 (a). Although they do not converge, telling us that further training would improve performance,

it is clear that the model does not overfit. After the training, the mean squared error reaches approximately 10^0, which means that on average the error is ten times as high as for the flat dataset. Furthermore, we can see that in image Fig. 5.27 (b), both the ratio of validation and training and the mean over median are still rising, telling us that the model is rather undertrained on this dataset, or is not able to perform better.

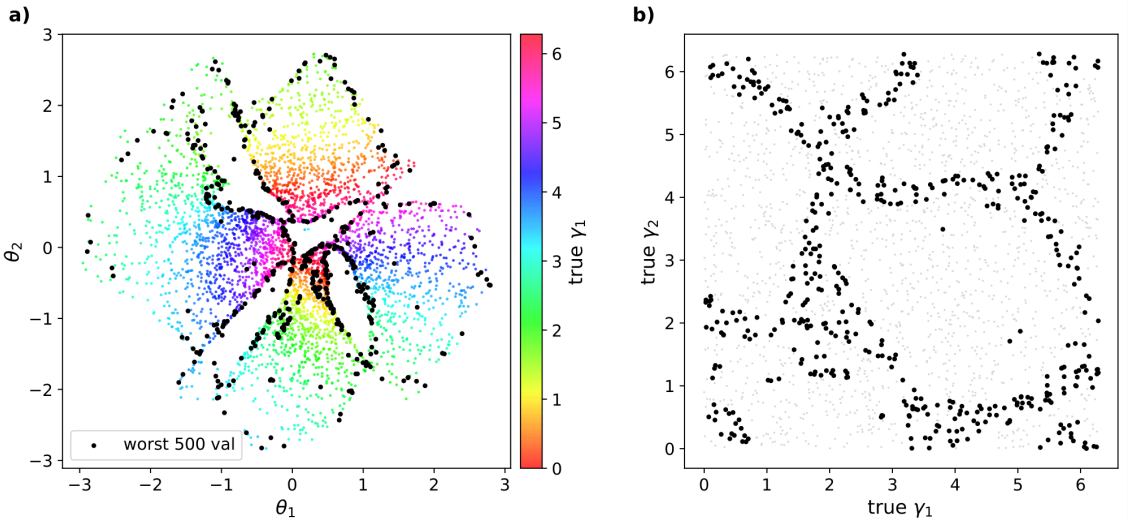


Figure 5.28: Localization of the reconstruction error of the flat variational autoencoder with the toroidal dataset. a) Latent coordinates (θ_1, θ_2) of the validation set, coloured by the ground truth γ_1, the 500 points with the largest reconstruction error are marked in black. b) The same points in ground-truth coordinates (γ_1, γ_2), with the worst 500 in black and the remainder in grey.

Here, we can perfectly validate our prediction, that the model will perform worst at topological tears of the encoder, which leads to continuous strips with respect to the ground truth coordinates. This means that the flat variational autoencoder has distinct and connected sets of data on which it can't perform very well, as the encoder has to separate the data at these lines.

Now, training and validating the flat variational autoencoder on the spherical dataset.

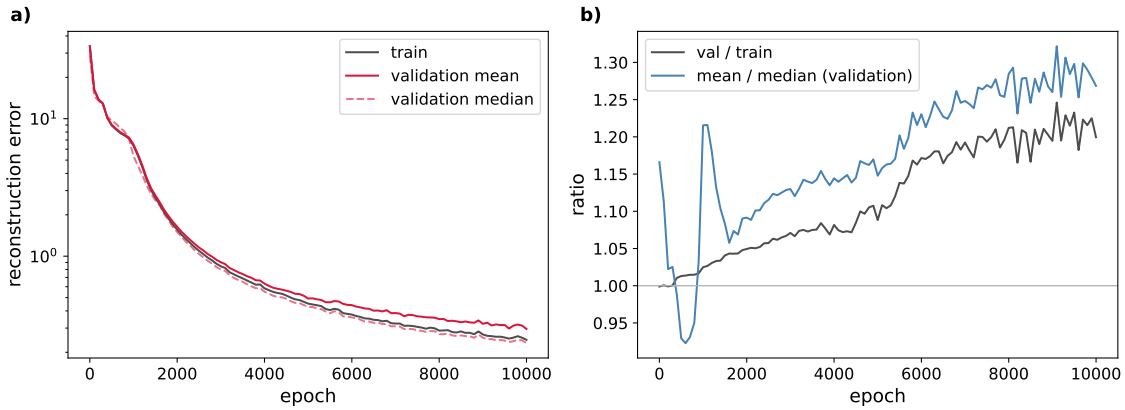


Figure 5.29: Training of the flat autoencoder on the spherical dataset. a) Reconstruction error against epoch (logarithmic ordinate): training error (grey), validation mean (solid red), validation median (dashed red). b) Ratios against epoch: validation to training (grey) and mean to median of the validation error (blue).

The error of training and validation drops faster than for the toroidal dataset and seems to plateau at around 10^{-1}. Therefore, the performance is better than that of the toroidal dataset but worse than the flat dataset. This is exactly what we expected, as the spherical dataset can be embedded without creating multiple tears and discontinuities inside the dataset. As we can see, the curve representing the ratio of validation and training error does not behave as the others, as the mean over median has two distinct local extrema. Even after multiple runs and a change in seed, this does not change. Therefore, we will shortly look into what changes around these two sharp kinks.

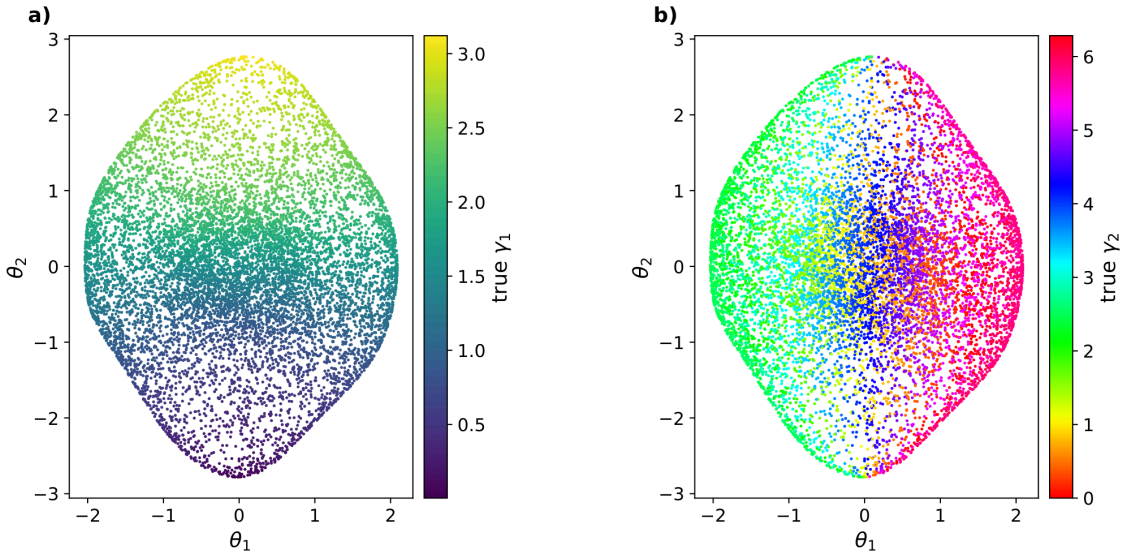


Figure 5.30: Latent space representation of the spherical dataset in a flat latent space, investigated at the minimum of mean/median in Fig. 5.29 (b)

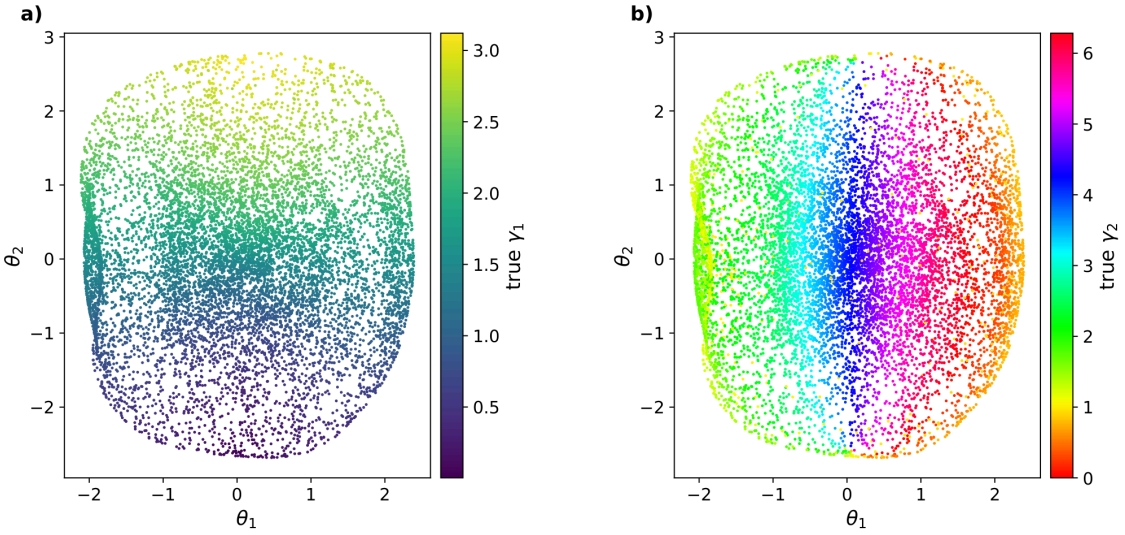


Figure 5.31: Latent space representation of the spherical dataset in a flat latent space, investigated at the local maximum of mean/median in Fig. 5.29 (b)

It seems as though the model switches between representing the spherical dataset as a squished sphere, which gets penalized, as there are multiple colors mixed into each other, which contradicts injectivity. The model reacts by overcorrecting into a square, which gets really noticable, when comparing to the fully trained latent space in Fig. 5.9. The two spikes in the diagram can be interpreted as the process of unfolding the data sphere into a two dimensional latent representation.

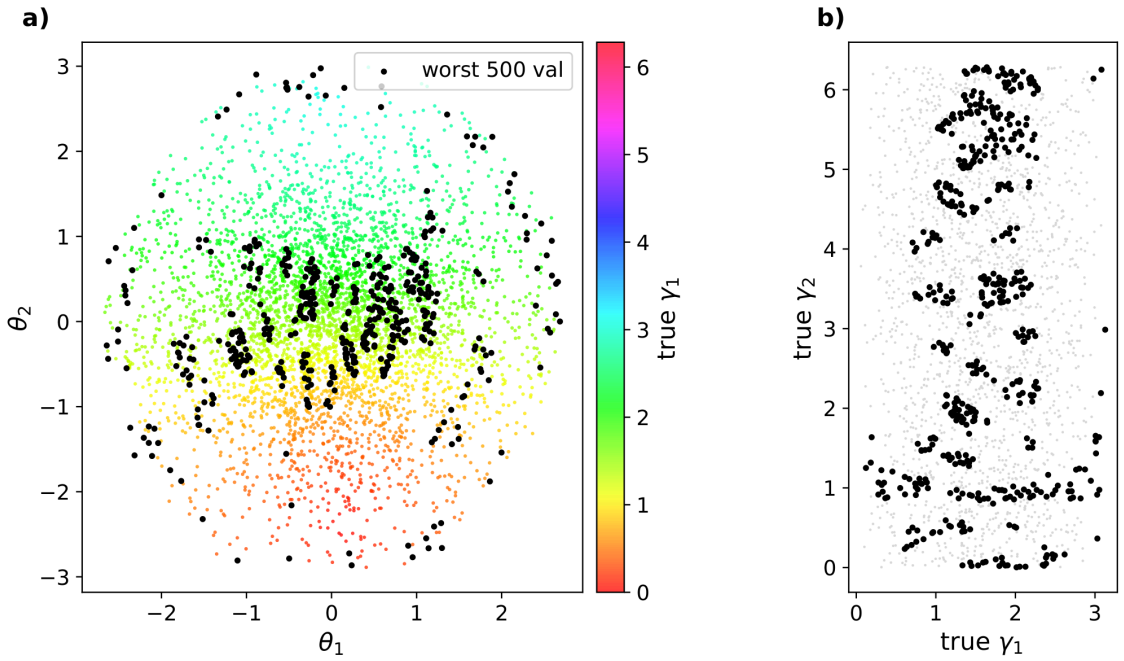


Figure 5.32: Localization of the reconstruction error, of the flat variational autoencoder with the spherical dataset, variant 1. a) Latent coordinates (θ_1, θ_2) of the validation set, coloured by the ground truth γ_1, the 500 points with the largest reconstruction error are marked in black. b) The same points in ground-truth coordinates (γ_1, γ_2), with the worst 500 in black and the remainder in grey.

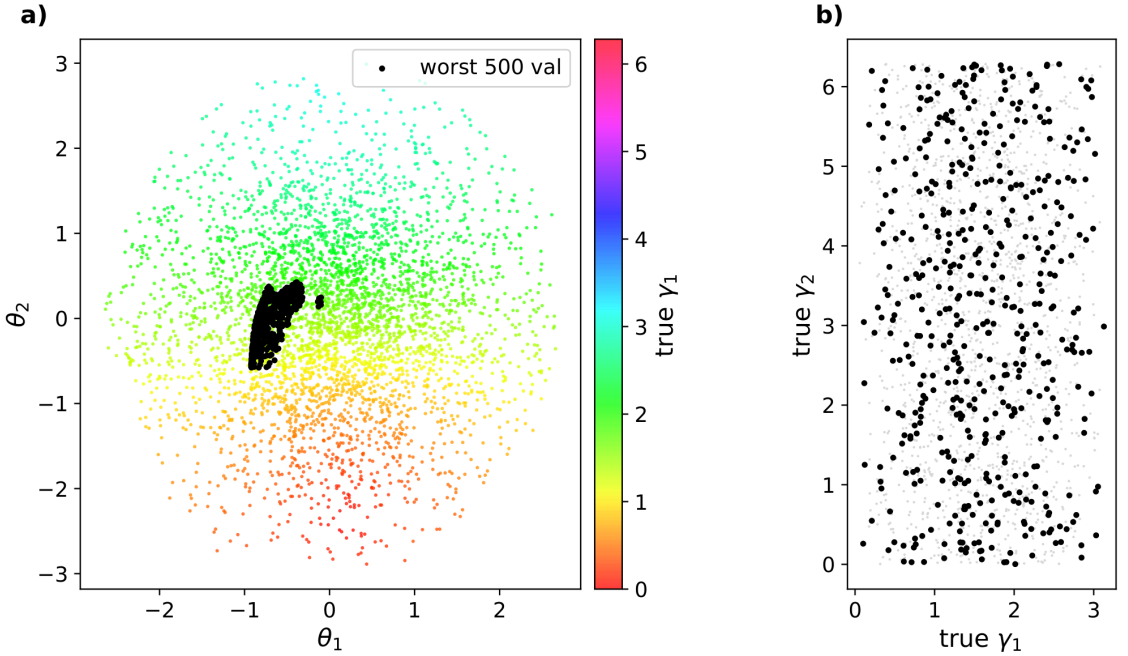


Figure 5.33: Localization of the reconstruction error, of the flat variational autoencoder with the spherical dataset, variant 2. a) Latent coordinates (θ_1, θ_2) of the validation set, coloured by the ground truth γ_1, the 500 points with the largest reconstruction error are marked in black. b) The same points in ground-truth coordinates (γ_1, γ_2), with the worst 500 in black and the remainder in grey.

As we can see, different runs and random seeds produce different positions of the worst replicated validation samples. We will interpret this as two different versions of mapping the $\mathbb{S}^2$ into $\mathbb{R}^2$. The first uses multiple cuts along the longitudes, the second version takes out a single point (as for example the stereographic projection does).

To now motivate non-trivial latent spaces, we will do the same for the toroidal and spherical latent models. We will focus on the performance of each of the datasets with their according canonical variational autoencoder, as it is obvious, that the autoencoders will perform worse with the other datasets. It is necessary, to note that the amount of training periods of the flat encoder and the topologically non-trivial encoders cannot be compared directly, as they are batched differently. The chosen amounts should correspond to a similar order of magnitude and therefore be comparable qualitatively. We start by validating the toroidal variational autoencoder on the toroidal dataset.

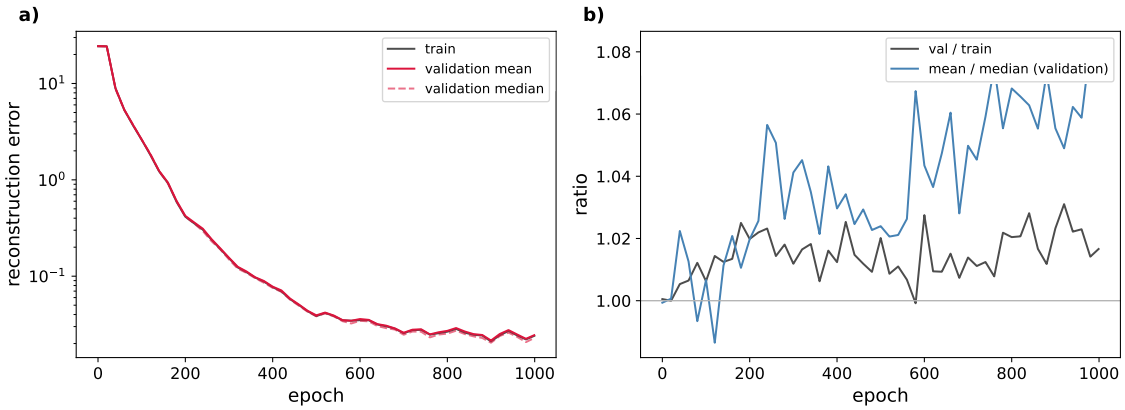


Figure 5.34: Training of the toroidal autoencoder on the toroidal dataset. a) Reconstruction error against epoch (logarithmic ordinate): training error (grey), validation mean (solid red), validation median (dashed red). b) Ratios against epoch: validation to training (grey) and mean to median of the validation error (blue).

As we can see, the error of both training and validation drop fast and level off, meaning the model converges. Also the ratio and the mean over median stay below 1.10. The flat autoencoder did not achieve either on the toroidal dataset.

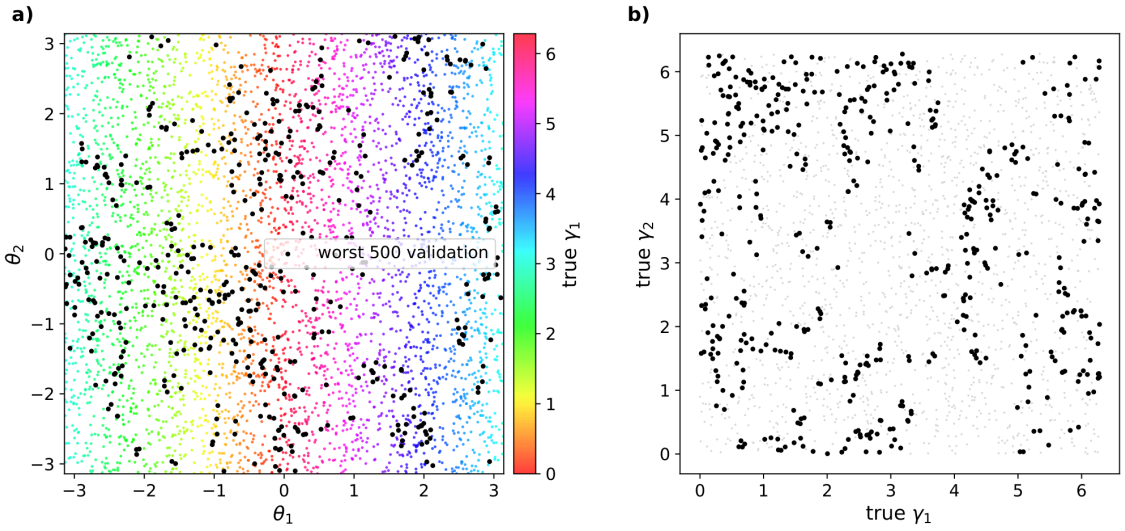


Figure 5.35: Localization of the reconstruction error, of the toroidal variational autoencoder with the toroidal dataset. a) Latent coordinates (θ_1, θ_2) of the validation set, coloured by the ground truth γ_1, the 500 points with the largest reconstruction error are marked in black. b) The same points in ground-truth coordinates (γ_1, γ_2), with the worst 500 in black and the remainder in grey.

The worst performing points seem to be distributed randomly on both the latent as well as the ground truth domain. This tells us that both the encoder and the decoder do not have problems with the topology of the dataset with respect to the latent space. No tears have to be cut and no discontinuities lead to bad performance. Again, to actually *prove* homogenity, we would have to look at histograms of the worst performing samples.

Next, we will train and validate the spherical autoencoder on the spherical dataset.

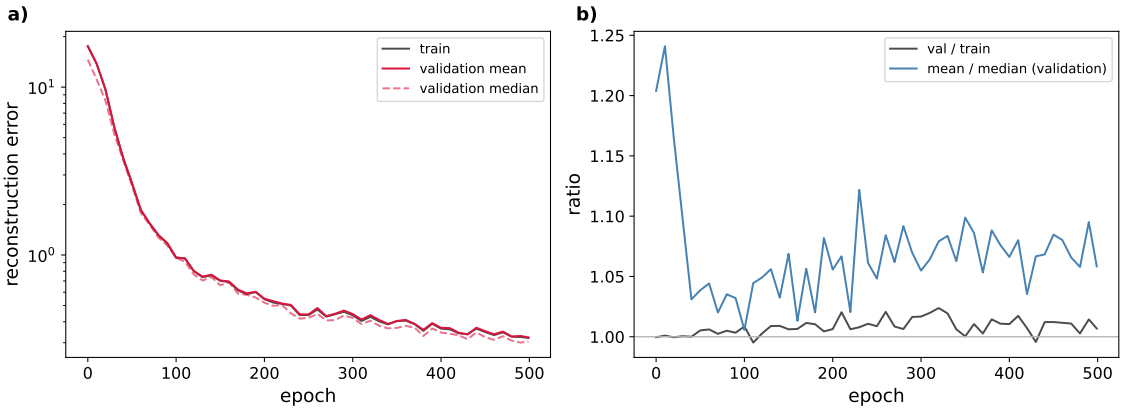


Figure 5.36: Training of the spherical autoencoder on the spherical dataset. a) Reconstruction error against epoch (logarithmic ordinate): training error (grey), validation mean (solid red), validation median (dashed red). b) Ratios against epoch: validation to training (grey) and mean to median of the validation error (blue).

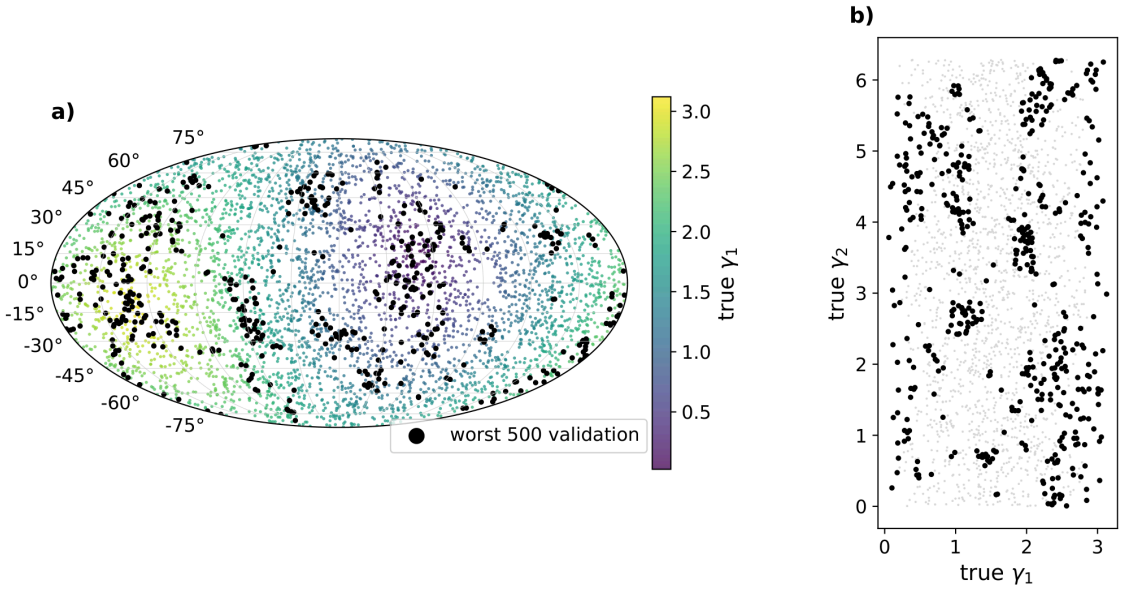


Figure 5.37: Localization of the reconstruction error, of the spherical variational autoencoder with the spherical dataset. a) Latent coordinates (θ_1, θ_2) of the validation set, coloured by the ground truth γ_1; the 500 points with the largest reconstruction error are marked in black. b) The same points in ground-truth coordinates (γ_1, γ_2), with the worst 500 in black and the remainder in grey.

As we can see, the worst points are not centred around any structure, they seem randomly distributed in their domain (actually, not quite randomly, but this clustering seems to be correlated with the learned metric, as, looking back at Fig. 5.22, the samples concentrate in areas of high curvature and magnification). This is exactly what we expected and clearly mirrors what one would expect for an infinitely large, flat dataset in a flat

latent: an approximately homogeneous distribution of the worst reconstructed samples of a validation set.

As of this point the advantage of variational autoencoders should get clear, as the decoder's domain is by construction statistically restricted to the learned representation and its topology. This can be seen, as all of the worst performing samples are placed at borders of tears. None of them are positioned in empty areas and samples inmidst the filled areas generally do not perform as badly.

Technically, as both the encoder and the decoder are continuous functions, they cannot alter topology through tears or discontinuities. As we can see from the results, phenomenons like these seem apparent in our latent space plots. Therefore, it might be appropriate to introduce a concept of *effective* topology. By identifying low statistical support, meaning areas left empty by the encoder, and the resulting high curvature and magnification of the decoder, with *effectively* separated or disjoint subsets of the latent space. In this effective framework it is justified, to speak of necessarily encoded tears to be able to represent a dataset with non-trivial topology.

5.3.3 Verifying Chern–Gauss–Bonnet

We now want to numerically verify the Gauss-Bonnet theorem by integrating over the different latent spaces, as described in Section 5.2.1. The actual statement we are trying to verify here is the one proposed in Section 3.4, namely the independence of the theorem from the decoder. To perfectly hold, meaning for the integral to always converge, we would have to sustain a very large training set and orders of magnitude more computational power than the one this survey is based on. Additionally, the integration grid would necessarily have to converge to the continuum limit. Therefore, we will quantitatively compare the different datasets and latent topologies by how fast the integral converges towards the by Gauss–Bonnet predicted value.

Table 5.1: Comparison of calculated Gauss–Bonnet values on different latent spaces with expected values, dependent on the dataset

Dataset and latent topology	expected value	calculated value	deviation
Flat (e.c.) data in $\mathbb{R}^2$	$2\pi \approx 6.283185$	6.283781	0.010%
Flat data in $\mathbb{R}^2$	2π	6.284022	0.013%
Torus data in $\mathbb{R}^2$	2π	6.709860	6.79%
Spherical data in $\mathbb{R}^2$	2π	6.285565	0.04%
Flat (e.c.) data in $\mathbb{T}^2$	0	1.821424×10^{-2}	-
Flat data in $\mathbb{T}^2$	0	2.498182×10^{-9}	-
Torus data in $\mathbb{T}^2$	0	$-7.451074 \times 10^{-16}$	-
Flat (e.c.) data in $\mathbb{S}^2$	$4\pi \approx 12.566371$	12.426868	1.11%
Flat data in $\mathbb{S}^2$	4π	12.519879	0.37%
Spherical data in $\mathbb{S}^2$	4π	12.525181	0.33%

Although a proper comparison through standard variance of the values is not possible, as we do not have an error for our integral, it can be noticed that each topological dataset performs best on its latent space counter part, therefore confirming the result we reported from the validation evaluation. Furthermore, we can see that the two flat datasets perform equally well on the flat latent space, while the one where the Gaussian function is allowed to cross edges[4] of the image always performs worse than the truly flat dataset on the toroidal variational autoencoders. Additionally, the integral of the toroidal dataset converges the worst on the flat latent, which is what we would expect, as we have multiple spots where the curvature admits nearly diverging behaviour (see Fig. 5.8). The spherical dataset performs actually quite well on the flat latent space, which was also foreseeable, as the latent represenation in Fig. 5.9 nearly resembles the one of the flat dataset in Fig. 5.5. The toroidal latent space probably has the most significant result, as the integral values vary with a factor of 10^7 between each dataset. The sign of the integral over the toroidal latent space with the toroidal dataset can be easily explained, as by Gauss-Bonnet the curvature has to be partly negative, if not identically vanishing.

Therefore, we have to modify the statement from Section 3.4. From this results we follow that in a grid coarse enough (as there is no real comparison between convergence of the integral dependent on grid spacing, *coarse enough* cannot be specified further), a manifold not embeddable into the latent space will converge slower than a manifold which is (for a proper formulation of this statement, we would also need to quantify how *well* something can be represented, if not embeddable. From this limited pool of datasets and latent spaces it seems as though the difference in Euler characteristics could create such a metric, but we do not want to generalize.). This follows directly from our investigations regarding the network parameter gradient of this integral, but due to numerical reasons, having more absolute curvature does result in slower convergence.

Table 5.2: Comparison of median and maximum of absolute curvature and magnification, all [a.u.]

Dataset and latent topology	median $\lvert K\rvert$	max $\lvert K\rvert$	median $\sqrt{\det(F)}$	max $\sqrt{\det(F)}$
Flat (e.c.) data in $\mathbb{R}^2$	5.440×10^{-2}	3.253×10^{0}	1.007×10^{1}	2.039×10^{2}
Flat data in $\mathbb{R}^2$	1.863×10^{-2}	7.106×10^{0}	2.753×10^{0}	1.730×10^{1}
Torus data in $\mathbb{R}^2$	3.717×10^{-2}	4.149×10^{0}	1.262×10^{1}	4.233×10^{2}
Spherical data in $\mathbb{R}^2$	2.994×10^{-2}	6.003×10^{-1}	9.242×10^{0}	1.345×10^{2}
Flat (e.c.) data in $\mathbb{T}^2$	3.951×10^{-6}	7.385×10^{-3}	2.385×10^{5}	5.514×10^{5}
Flat data in $\mathbb{T}^2$	2.923×10^{-7}	1.365×10^{-5}	5.808×10^{5}	2.367×10^{6}
Torus data in $\mathbb{T}^2$	5.079×10^{-8}	2.568×10^{-7}	5.867×10^{5}	6.262×10^{5}
Flat (e.c.) data in $\mathbb{S}^2$	3.424×10^{-5}	1.992×10^{-3}	2.032×10^{4}	7.131×10^{4}
Flat data in $\mathbb{S}^2$	9.406×10^{-6}	4.715×10^{-3}	2.277×10^{4}	8.688×10^{5}
Spherical data in $\mathbb{S}^2$	2.206×10^{-5}	2.745×10^{-4}	2.043×10^{4}	5.978×10^{4}

[4] abbreviation e.c. in table stands for e*dge c*rossing

If we now compare absolute curvature and magnification, we can see clear trends between datasets and latent topology. First, it seems as though the curvature of the flat latent space is the largest and not really dependent on the dataset. The magnification of the flat latent heavily benefits from the flat set, as it is ten times lower. Second, on the variational autoencoders with $\mathbb{S}^2$ and $\mathbb{T}^2$ topology, the flat set, where the Gaussian can touch the borders, has always the highest curvature and lowest magnification. Furthermore, the matching topology cases admit the lowest median curvature and median magnification. Furthermore, there seems to be a antiproportional trend between curvature and magnification, which could result from the definition of K through the determinant.

5.3.4 Allowing Higher Dimensionality

Instead of enforcing a topology onto the latent space, one would suspect that an ideal flat variational autoencoder with enough training data, epochs and a high enough dimension of the latent space, would be able to fully reconstruct the underlying data structure. It would not only represent the data manifold, but find an actual embedding function, therefore preserving the topology. As discussed multiple times throughout this thesis, this is obviously not possible for non-trivial topologies, with identical latent space and data dimension, which is suggestively why the filled latent space of e.g. Fig. 5.8 looks as it does. So now one[5] might suggest to increase the dimension of the flat latent space to $d = 3$ and see, where the encoder places the means of the training set. Or, in more precise words, which topology the encoder learns to represent from the data. As a sanity check, we will start by looking at the flat dataset. Again, we have to distinguish between the two flat datasets. We first plot how the truly flat set is represented.

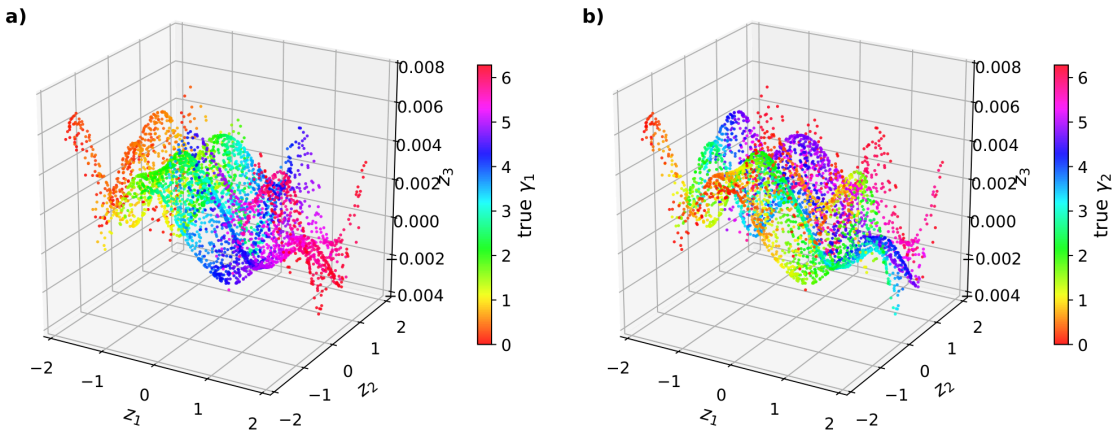


Figure 5.38: The flat dataset in a flat, 3 dimensional latent space. a) and b) again denote the two dimensions of ground truth translation.

[5]Matti Weber, to be precise

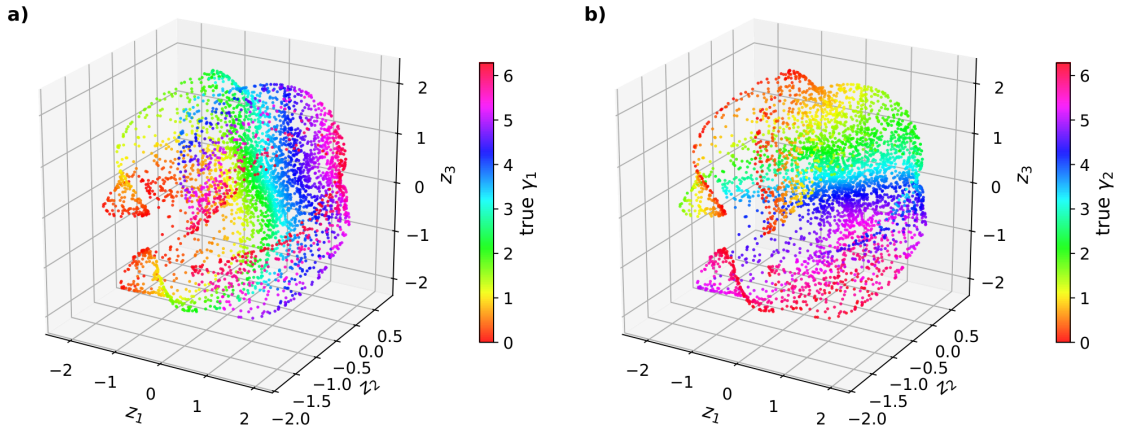


Figure 5.39: The flat dataset with edge crossing in a flat, 3 dimensional latent space. a) and b) again denote the two dimensions of ground truth translation.

As $\mathbb{R}^2$ is a hypersurface of $\mathbb{R}^3$, there is nothing forcing a constraint onto the third coordinate of the representation. We can observe that both representations are topologically equivalent, as they should be. In the second diagram, where the edge-case dataset is used, we can see that the encoder assumes a closed topology, by trying to connect the edges. In regard of the underlying dataset this makes sense, as nothing points at translational symmetry farther than the borders. In the first diagram we can see that the sheet is basically embedded flat into $\mathbb{R}^3$. Enlarging the dataset's images, while keeping resolution constant, would lead to a larger sheet in $\mathbb{R}^3$, as one would expect from a dataset, with a flat[6] underlying topology. Looking closer at the axes, we can see that the Euclidean z-axes is magnitudes smaller than the others. We can therefore assume that all other seemingly smooth surfaces of the latent plots admit some form of oscillation. In this particular case this is due to the third degree of freedom not being needed to describe the data, therefore this coordinate periodically moves around a mean. Which part of the network architecture is responsible for the amplitude of this oscillation is not known to the author. Next, we train on the toroidal dataset.

[6]flat obviously meaning the topology of $\mathbb{R}^2$, as flat is a geometric property.

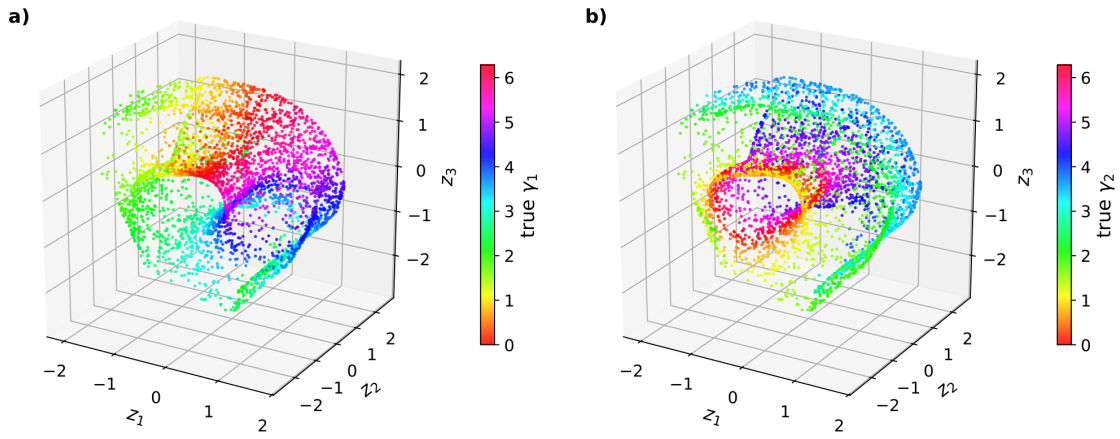


Figure 5.40: The toroidal dataset in a flat, 3 dimensional latent space. a) and b) again denote the two dimensions of ground truth translation.

As we can see, the representation definitively admits a non-trivial topology, although it looks more like a mixture of Klein's bottle and the typical three dimensional depiction of a torus, the donut. This is not even that far off, because the two topological objects just differ in orientability. So, translation instances where the Gaussian is in the middle of an edge or exactly on a corner would look identical for a torus' and a Klein's bottle's topology. Furthermore, from looking at reconstructed samples it got clear that diffusion due to reconstruction error occured, which would make confusing a toroidal with a Klein's bottle topology even more reasonable, as a Gaussian touching the according border would look like a smeared bright spot, nearly independent from the translation parallel to the edge. One might wonder whether, with more training data and more training time, the flat variational autoencoder might actually depict a torus. Additionally, forcing the latent space to dimensionality $d = 3$ would introduce more curvature, as the two dimensional torus can only be embedded flatly into $\mathbb{R}^4$, which could hinder the decoder from correctly representing the underlying symmetry, as the training data consists of a *flat* torus, not a *donut*. It would be interesting, to set the latent dimension of the flat variational autoencoder to $d' = 4$, train with the two dimensional toroidal dataset and *then* project the $4d$ latent points back into $3d$, to enable the variational autoencoder to actually learn a flat torus.

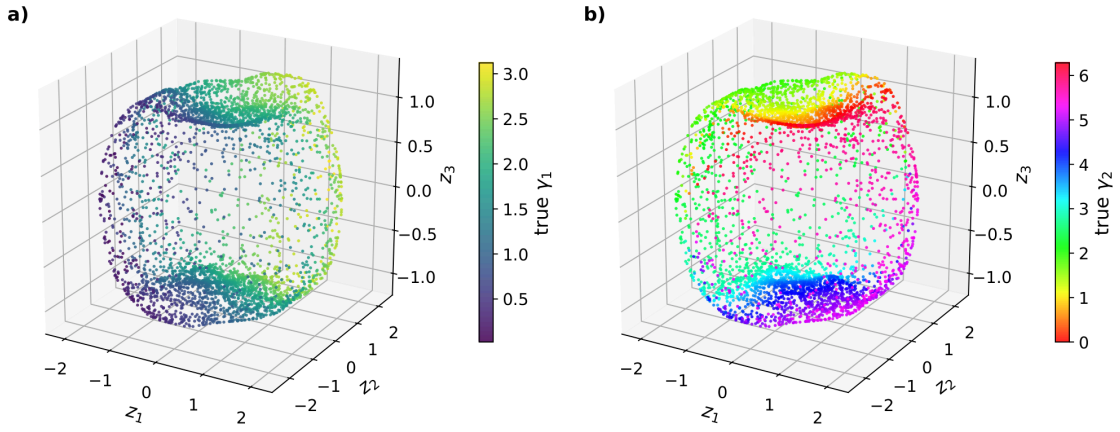


Figure 5.41: The spherical dataset in a flat, 3 dimensional latent space. a) and b) again denote the two dimensions of ground truth translation.

Contrary to the toroidal dataset, we are definitively able to identify the topological sphere. This is not surprising, as $\mathbb{S}^2$ can be embedded into $\mathbb{R}^3$. The clustering of points at the poles could be explained by the way, we project the $\mathbb{S}^2$ surface. The farther away a point is from the poles, the more distinguishable it gets. While converging towards the poles, the differences between images get progessively smaller (It is again necessary to remember that this depiction only involves the encoder's mapping, and the plot is Euclidean, so distances are not representative.), leading to clustering. The next logical step, in a more general application, with possibly unknown data space topology, would be to integrate over the developing representation's curvature in latent space (although it has to be clear that even dimensionality is necessary for that), thereby possibly identifying the data's underlying structure with the Euler characteristic (see Section 3.2 for the problems of this topological invariant).

Chapter 6

Discussion and Conclusion

The main objective of this thesis was to explore the possibility of using the Chern–Gauss–Bonnet theorem in the context of latent spaces of generative machine learning models. We furthermore asked the question whether it was possible to identify the topology of the underlying manifold of a dataset by numerically integrating over its curvature. Additionally, we investigated topological latent spaces and their advantages over Euclidean ones, where we used methods of differential geometry and statistical physics to verify the assumptions we had regarding continuity and curvature of a representation whose topology is too complex to be represented in a Euclidean latent space of the same dimension.

To reach that point, we investigated the foundational motivation and meaning of the theorem, while verifying that it was suitable for the notion of latent spaces. We furthermore derived the theorem in dimensions 2 and 4 and discussed its generalization for higher dimensional latent applications. We then noted the limitations of the Euler characteristic and concluded that we had to build our latent spaces around these limitations, also that it was a differential geometrically and numerically accessible topological invariant. After defining the necessary prerequisites, we created a latent space formalism, onto which we could apply the theorem, thereby motivating compactness and accepting the even dimensionality as an architectural premise. We further discussed and motivated the 2-dimensional sphere and torus as our toy-model latent spaces. As standard in machine learning, we then focused on using the Chern–Gauss–Bonnet theorem as a possible regularization in a loss function. By drawing the analogy to Gauss–Bonnet gravity, we falsified this idea, as the theorem is only dependent on topology, not on a learned metric and can therefore contribute nothing to a parameter gradient.

We then built an intrinsic information geometric description of latent spaces by introducing the Fisher information metric as the canonical choice of geometry, therefore making curvature and magnification quantities of the learned representation itself instead of describing latent spaces with arbitrarily chosen and therefore meaningless coordinates. Since encoder and decoder are continuous by construction of the network architecture, neither of them can create actual tears or discontinuities. Additionally, any regular differentiable decoder induces a Riemannian metric over the whole latent space, therefore

always fulfilling the requirements for Chern–Gauss–Bonnet. On this basis, we constructed variational autoencoders whose latent spaces carry the topology we motivated above through modifying sampling of a conventional variational autoencoder to parametrize the two-dimensional torus and sphere. These two, alongside the conventional Euclidean latent space baseline were then trained on datasets we created to precisely mirror these topologies, meaning datasets which each had a different underlying topology. We thereby had created a possibility to not only visualize unfaithful Euclidean representations, but also to numerically verify the Gauss–Bonnet theorem on imposed topological latent spaces and compare the performance of the models with respect to matching or mismatching topology of dataset to latent space.

Training revealed that the Euclidean latent space fails to produce faithful continuous representations of the toroidal and spherical datasets. This does not contradict the continuity of the encoder and the decoder. The maps themselves remain continuous, although the encoder leaves regions of the latent space with low statistical support and the decoder develops high curvature and magnification precisely there. These regions act as effective separations, necessary to encode 2-manifolds with curvature in a two-dimensional latent space. We further observed that the topological latent spaces correctly represent the topological datasets. Qualitatively, this establishes the necessity of non-trivial topological latent spaces when representing data of non-trivial topology.

We then validated the various variational autoencoders quantitatively by comparing their performance and observing where the worst performing samples were situated. This confirmed our assumptions, as the conventional flat latent variational autoencoder was not able to represent the spherical and toroidal datasets faithfully. We could observe that the tears and discontinuities in the representation of non-trivial topology in a flat latent space led to distinct areas in the latent space and subsets of the data, on which the model performed worst. Conversely, we could show that the toroidal latent model could represent the toroidal dataset not only continuously, but with worst-performing samples that showed no discernible structure in either the latent or the ground-truth domain. The same held for the spherical variational autoencoder on the spherical dataset.

Integrating the calculated curvature showed that the Chern–Gauss–Bonnet theorem holds numerically on the constructed latent spaces and reproduces the correct values with deviations below one percent. Within each latent space the dataset whose topology matches that of the latent space converged closest to the predicted value, which led us to identify the deviation of the Euler characteristic as confirmation of the validation performance.

Finally, allowing a third latent dimension showed what the encoder did when the topology was not imposed. The flat dataset remained a flat sheet and the edge-crossing flat dataset closed its topology by connecting the edges, which is what its translational structure suggests. The representation of the spherical dataset can be clearly identified as a sphere, while the toroidal dataset did not produce a toroidal topology, plausibly because the flat torus admits an isometric embedding only into four dimensions. Concluding, the encoder recovered the topology of the data where the ambient dimension allowed it and

approximated it otherwise.

These conclusions are obviously bounded by several limitations. First, we do not have an error estimate for the integrals, so the quantitative results can only be compared qualitatively, not by statistical significance. Furthermore, the number of training epochs between the flat and the topological latent spaces differs substantially, while the validity of evaluation rests on the assumption that batching restores comparability, which only holds qualitatively. Most of the analysis is based on visual assessments. A stronger claim would require analysis through histograms. We examined only a small number of datasets and latent topologies, while the Euler characteristic itself is a weak invariant.

Last, concerning the question of whether data topology can be verified, we only verified known topologies of our own datasets by comparing how well they can be represented. We did not use the Euler characteristic to identify the topology of a representation, as for our setup the theorem always recovers the Euler characteristic of the latent space by construction.

Concluding from this, we propose two methods for finding the topology of a dataset. Either construct latent spaces which match different hypotheses of the data's manifold and compare performance and Euler characteristic deviation, or increase the dimension to be able to actually represent the topology by embedding it, then integrate over the submanifold to recover the topological invariant. Whether this is numerically feasible, or whether a formalism exists that would allow this to be implemented easily, remains for future work.

We can therefore clearly state that through introducing the Fisher information metric, the global Chern–Gauss–Bonnet theorem can be applied to compact latent spaces. While the value of the integral is fixed by the latent topology by construction, the closeness and rate of its convergence indicate whether the topology of the data matches that of the latent space.

Chapter 7

Outlook and Generalization

7.1 On Identifying Symmetries

With the goal in mind, to uniquely identify underlying symmetry of data, we would formalize the problem as such[1]:
Fundamental physical symmetry shall be a Lie group G, acting smoothly and properly on data space Y as such

$$\Phi : G \times Y \to Y, \qquad \Phi(e, y) = y, \quad \Phi(g, \Phi(h, y)) = \Phi(gh, y). \tag{7.1}$$

A dataset is in general a union of orbits, so that Y fibres over Y/G with fibres G/H.

$$\mathcal{O} = G \cdot y_0 \cong G/H_{y_0}, \tag{7.2}$$

where $H_{y_0} = \{g \in G \mid g \cdot y_0 = y_0\}$ is the stabiliser of y_0. In the ideal case, the model would be able to create a G-equivariant map $Y \to \Theta$ such that for a single orbit

$$\Theta \cong G/H. \tag{7.3}$$

The problem now arising, is that even if we could uniquely identify Θ topologically, we still could not generally uniquely identify G, only a class of pairs (G, H)[2]. If we instead impose a symmetry through explicit equivariant properties in the architecture of our encoder, we would be able to test different hypotheses by goodness of fit, therefore justifying the whole act of enforcing topology onto a latent space.

Throughout, "representation" was used in a machine learning context, not in the sense of representation theory, as the latent space is no linear representation of a possible group creating the dataset. The decoder though, should admit a genuine decomposition into irreducible representations of G.

From this point onward, everything should be taken as highly speculative, as only

[1] Thereby completely ignoring both the stochasticity of the data and the model.

[2] A simple example would be $\mathbb{S}^2$, which can be represented as quotient space of $SO(3)/SO(2)$ as well as $SU(2)/U(1)$.

heuristically motivated, derived with minimal prerequisites and theoretically situated largely outside the author's area of expertise. Future projects regarding these topics therefore heavily rely on comments or corrections of colleagues, reviewing this section.

7.2 Generalization of Chern–Gauss–Bonnet

There are actually two ways of generalizing the Chern–Gauss–Bonnet theorem. We will shortly define and discuss both. First, as defined in Eq. (2.13), Eq. (2.18) one could think of generalizing a topological integral theorem for general connections, meaning connections which are not metric. Through this, we would be able to, instead of using the Levi-Civita connection and the standard Riemannian curvature tensor to calculate the Pfaffian form, use a more general Riemann curvature tensor. Second, coming from an operator theory standpoint, Chern–Gauss–Bonnet is a theorem relating the index the exterior derivative and coderivative on a manifold fulfilling the requirements, with the topologically invariant Euler characteristic. This generalization is stated for elliptic operators on manifolds by the Attiyah-Singer theorem

$$\ker(D) - \ker(D^*) = \chi(\mathcal{M}), \tag{7.4}$$

where $D = \mathrm{d} + \mathrm{d}^*$ [44].

7.2.1 Amari Curvature

For an information geometric α-connection, the corresponding generalized Riemannian tensor would take the following form

$$R^{(\alpha)\,l}{}_{kij} = \partial_i \Gamma^{(\alpha)\,l}{}_{jk} - \partial_j \Gamma^{(\alpha)\,l}{}_{ik} + \Gamma^{(\alpha)\,l}{}_{im}\Gamma^{(\alpha)\,m}{}_{jk} - \Gamma^{(\alpha)\,l}{}_{jm}\Gamma^{(\alpha)\,m}{}_{ik}, \tag{7.5}$$

which we can write it in dependence of the Levi-Civita derived Riemannian and the Amari-Chentsov tensor

$$\begin{aligned} R^{(\alpha)\,l}{}_{kij} = R^{(0)\,l}{}_{kij} &- \frac{\alpha}{2}\left(\nabla_i^{(0)} C^l{}_{jk} - \nabla_j^{(0)} C^l{}_{ik}\right) \\ &+ \frac{\alpha^2}{4}\left(C^l{}_{im} C^m{}_{jk} - C^l{}_{jm} C^m{}_{ik}\right). \end{aligned} \tag{7.6}$$

7.2.2 Operators on Statistical Manifolds

As one can think of non-Riemannian curvature using an information geometry specific connection, we could also think of information geometry specific operators, as the ones defined below, using the most general definition of each operator in combination with dual connections and Fisher information metric:

- **Exterior differential d**
  We define ([45])

$$\mathrm{d} = \sum_i \epsilon^i \wedge \nabla_{e_i}, \tag{7.7}$$

where $\{\epsilon\}$ is a coframe. When we now assume torsion freeness and look at the expression in indices, of d applied on a k-form, we get

$$(\mathrm{d}\omega)_{i_0\ldots i_k} = (k+1)\nabla_{[i_0}\omega_{i\ldots i_k]}. \tag{7.8}$$

Now, swapping in ∇^α, as defined in Section 2.2, we can clearly see that antisymmetrizing the correction, which is precisely the Amari-Chentsov contraction, vanishes identically, as C is totally symmetric.

$$\nabla^{(\alpha)}_{i_0}\omega_{i_1\ldots i_k} = \nabla^{LC}_{i_0}\omega_{i_1\ldots i_k} + \frac{\alpha}{2}\sum_{m=1}^{k} C^{l}{}_{i_0\ldots i_m=l\ldots i_k} \tag{7.9}$$

Therefore, we conclude

$$\mathrm{d}^{(\alpha)} = \mathrm{d}^{(0)}. \tag{7.10}$$

- **Codifferential d***

  Analogously ([46])

  $$(\mathrm{d}^*\omega)_{i_2\ldots i_k} = -g^{ji_1}\nabla_j\omega_{i_1 i_2\ldots i_k}, \tag{7.11}$$

  replacing the Levi-Civita covariant differential with the information geometric pendant and the metric by the Fisher information metric, we recover

  $$\mathrm{d}^{*(\alpha)} = d^*\omega_{i_2\ldots i_k} - \frac{\alpha}{2}F^{ji_1}[C^{l}{}_{ji_1}\omega_{li_2\ldots i_k} + \sum_{m\geq 2} C^{l}{}_{ji_m}\omega_{i_1\ldots i_m=l\ldots i_k}]. \tag{7.12}$$

  We follow

  $$F^{ji_1}C^{l}{}_{ji_1} = F^{ji_1}F^{lm}C_{ji_1 m} = F^{lm}\tau_m = \tau_l \tag{7.13}$$

  $$\Sigma_{m\geq 2}C^{l}{}_{ji_m}\omega_{i_1\ldots i_m=l\ldots i_k} = 0 \tag{7.14}$$

  where τ is the trace of the Amari-Chentsov tensor, also known as the Chebyshev field (see [47, 48]). In the second argument, we used that

  $$F^{ji_1}F^{la}C_{ji_m a} = F^{jl}F^{i_1 a}C_{ji_m a}, \tag{7.15}$$

  as C is absolute symmetric. Contracting against a k-form ω, which is totally antisymmetric in its indices, leads the terms to cancel pairwise. We conclude

  $$\mathrm{d}^{*(\alpha)} = \mathrm{d}^{*(0)} - \frac{\alpha}{2}\iota_{\tau^\sharp} \tag{7.16}$$

  where ι denotes the interior product.

Whether a theorem in the form of Chern–Gauss–Bonnet for α-curvature is actually equivalent to the Atiyah-Singer theorem applied to an information geometric Chern–Gauss–Bonnet operator is to be investigated.

Other interesting operators in an information geometric framework:

- **Dirac-type[3] operators**

  If we choose to use the general definition of a Dirac operator (following Atiya-Singer's convention and notation), we get

  $$D = \sum_i e_i \cdot \nabla_{e_i}, \tag{7.17}$$

  where $\{e_i\}$ is a local frame of a general basis. We now replace the partial derivative by the information geometrically covariant derivative

  $$\begin{aligned} D^{(\alpha)} &= g^{jn} e_n \cdot \nabla_j^{(\alpha)} && (7.18)\\ &= D^{(0)} + \frac{\alpha}{2} F^{jn} e_n \cdot \left(\textstyle\sum_m C^l{}_{j\,i_m} \right) && (7.19)\\ &= D^{(0)} - \frac{\alpha}{2} \iota_{\tau^\sharp} && (7.20)\\ &= \mathrm{d}^{(\alpha)} + \mathrm{d}^{*(\alpha)}, && (7.21) \end{aligned}$$

  where we used the definition of the dual covariant derivative using the Amari-Chentsov tensor C and the Levi-Civita covariant derivative $\nabla^{(0)}$ (a proper analysis of this operator would require defining $\cdot$ as multiplication in a Clifford algebraic sense and many things more, see [49].)

- **Laplace-Beltrami operators**

  Again, defining a Laplace-Beltrami operator in its most general form, we get

  $$\triangle = \mathrm{d}\mathrm{d}^* + \mathrm{d}^*\mathrm{d}. \tag{7.22}$$

  Plugging in our recently derived information geometric definitions,

  $$\begin{aligned} \triangle^{(\alpha)} &= \mathrm{d}^{(\alpha)}\mathrm{d}^{*(\alpha)} + \mathrm{d}^{*(\alpha)}\mathrm{d}^{(\alpha)} && (7.23)\\ &= \mathrm{d}\Big(\mathrm{d}^{*(0)} - \tfrac{\alpha}{2}\iota_{\tau^\sharp}\Big) + \Big(\mathrm{d}^{*(0)} - \tfrac{\alpha}{2}\iota_{\tau^\sharp}\Big)\mathrm{d} && (7.24)\\ &= \triangle^{(0)} - \frac{\alpha}{2}\big(\mathrm{d}\,\iota_{\tau^\sharp} + \iota_{\tau^\sharp}\,\mathrm{d}\big), && (7.25) \end{aligned}$$

  where the exterior differential could be left unchanged, as shown above. Identifying the Lie derivative of the Chebyshev field, we conclude that

  $$\triangle^{(\alpha)} = \triangle^{(0)} - \frac{\alpha}{2}\mathcal{L}_{\tau^\sharp}. \tag{7.26}$$

  Applying this operator onto a function, we get

  $$\triangle^{(\alpha)} f = -F^{ij}\nabla_i^{(0)}\nabla_j^{(0)} f - \frac{\alpha}{2} F^{ij}\tau_i \partial_j f, \tag{7.27}$$

[3]The information geometric connection is not Clifford compatible, as not metric compatible, therefore this is not a true Dirac operator.

which basically recovers a Laplacian modified by a drift term along the Chebyshev field.

7.3 Diffusion Models

An application for elliptic operators in a machine learning and information theory field is the diffusion model. A diffusion model destroys data by a forward stochastic process and learns to reverse it. Diffusion of the data is done by a stochastic differential equation whose marginals obey the Fokker–Planck equation. Replacing the Euclidean Laplacian with the above derived information geometric operator would make the generator of this process intrinsic to the learned Fisher geometry.

Chapter 8

Appendix

8.1 Explicit Proof of Chern–Gauss–Bonnet for Low Dimensionality

8.1.1 $\mathbb{S}^2$

For $d = 2$ we can either choose to calculate the Pfaffian form and use the general theorem as proved by Chern, or we can calculate the Gaussian curvature

$$K = \frac{R^1_{212}}{\det(g)}, \tag{8.1}$$

as suggested by the original Gauss-Bonnet theorem. In the following, we will do both. For $\mathcal{M} = \mathbb{S}^2$ with $r = 1$, we get

$$\mathrm{d}s^2 = \mathrm{d}\theta^2 + \sin^2(\theta)\mathrm{d}\phi^2. \tag{8.2}$$

We choose the coframe

$$\epsilon^1 = \mathrm{d}\theta, \tag{8.3}$$

$$\epsilon^2 = \sin(\theta)\mathrm{d}\phi. \tag{8.4}$$

Following from this, we get

$$\mathrm{d}\epsilon^1 = 0, \tag{8.5}$$

$$\mathrm{d}\epsilon^2 = \cos(\theta)\mathrm{d}\theta \wedge \mathrm{d}\phi. \tag{8.6}$$

Under the assumption that ∇ is torsion free, it holds that

$$\mathrm{d}\epsilon^a + \omega^a{}_b \wedge \epsilon^b = 0. \tag{8.7}$$

For $a = 1$, using $\omega^1{}_1 = 0$ by antisymmetry and $\mathrm{d}\epsilon^1 = 0$,

$$\omega^1{}_2 \wedge \epsilon^2 = \omega^1{}_2 \wedge \sin(\theta)\mathrm{d}\phi = 0, \tag{8.8}$$

therefore

$$\omega^1{}_2 = f(\theta,\phi)\mathrm{d}\phi. \tag{8.9}$$

For $a = 2$, using $\omega^2{}_2 = 0$,

$$\cos(\theta)\mathrm{d}\theta \wedge \mathrm{d}\phi + \omega^2{}_1 \wedge \mathrm{d}\theta = 0, \tag{8.10}$$

which, using $\mathrm{d}\phi \wedge \mathrm{d}\theta = -\mathrm{d}\theta \wedge \mathrm{d}\phi$, is solved by

$$\omega^2{}_1 = \cos(\theta)\mathrm{d}\phi \tag{8.11}$$

and therefore

$$\omega^{21} = -\omega^{12} = \cos(\theta)\mathrm{d}\phi. \tag{8.12}$$

The curvature two-form follows as

$$\begin{aligned}\Omega^{12} &= \mathrm{d}\omega^{12} + \omega^1{}_i \wedge \omega^i{}_2 \\ &= \mathrm{d}\omega^{12} + \omega^1{}_1 \wedge \omega^1{}_2 + \omega^1{}_2 \wedge \omega^2{}_2 \\ &= \mathrm{d}(-\cos(\theta)\mathrm{d}\phi) \\ &= \sin(\theta)\mathrm{d}\theta \wedge \mathrm{d}\phi = \mathrm{Pf}(\Omega).\end{aligned} \tag{8.13}$$

Comparing with

$$\Omega^{12} = K\epsilon^1 \wedge \epsilon^2 = K\sin(\theta)\mathrm{d}\theta \wedge \mathrm{d}\phi, \tag{8.14}$$

we obtain $K = 1$.

Variant 1: Gauss–Bonnet. With $\mathrm{d}A_{\mathbb{S}^2} = \sin(\theta)\mathrm{d}\theta\mathrm{d}\phi$

$$\int_{\mathbb{S}^2} K\mathrm{d}A = \int_0^\pi \int_0^{2\pi} \sin(\theta)\mathrm{d}\theta\mathrm{d}\phi = 4\pi = 2\pi\chi(\mathbb{S}^2), \tag{8.15}$$

therefore $\chi(\mathbb{S}^2) = 2$.

Variant 2: Chern's generalization. With $n = \frac{1}{2}d = 1$ and $\mathrm{d}\theta \wedge \mathrm{d}\phi$ the chosen orientation of $\mathbb{S}^2$,

$$\begin{aligned}\chi(\mathcal{M}) &= \int_{\mathcal{M}} \mathrm{eu}(\Omega) = \frac{1}{(2\pi)^n}\int_{\mathcal{M}} \mathrm{Pf}(\Omega) \\ &= \frac{1}{2\pi}\int_{\mathcal{M}} \Omega^{12} \\ &= \frac{1}{2\pi}\int_0^{2\pi}\int_0^{\pi} \sin(\theta)\mathrm{d}\theta\mathrm{d}\phi \\ &= \frac{4\pi}{2\pi} = 2 = \chi(\mathbb{S}^2)\end{aligned} \tag{8.16}$$

8.1.2 $\mathbb{T}^2$

As apparent, we can define the abstract torus in $d = 2$ either as intrinsically flat

$$\mathbb{T}^2 = S^1 \times S^1 = [0, 2\pi) \times [0, 2\pi), \tag{8.17}$$

where we canonically use the flat Euclidean metric

$$\mathrm{d}s^2 = \mathrm{d}\theta^2 + \mathrm{d}\varphi^2 \tag{8.18}$$

choosing the basis $\epsilon^1 = \mathrm{d}\theta$ and $\epsilon^2 = \mathrm{d}\varphi$. Now calculating the curvature form, it is obvious, that the necessary curvature tensors are identically zero. Therefore

$$R^1{}_{212} = 0, \tag{8.19}$$

meaning the Gaussian curvature vanishes. Therefore, we have proven that

$$\frac{1}{2\pi}\iint \mathrm{Pf}(\Omega) = 0 = \chi(\mathbb{T}^2), \tag{8.20}$$

as we defined the Euler characteristic such that the torus has $\chi(\mathbb{T}^2) = 0$.

Now, the more complex and calculation intense case, $\mathbb{T}^2$ as hyper surface of $\mathbb{R}^3$. We state

$$\mathrm{d}s^2 = (R + r\cos(\phi))^2\mathrm{d}\theta^2 + r^2\mathrm{d}\phi^2. \tag{8.21}$$

We choose

$$\epsilon^1 \equiv (R + r\cos(\phi))\mathrm{d}\theta, \tag{8.22}$$

$$\epsilon^2 \equiv r\mathrm{d}\phi. \tag{8.23}$$

Following, we get

$$\mathrm{d}\epsilon^1 = -r\sin(\phi)\mathrm{d}\phi \wedge \mathrm{d}\theta, \tag{8.24}$$

$$\mathrm{d}\epsilon^2 = 0. \tag{8.25}$$

Further assuming torsion freeness

$$\mathrm{d}\epsilon^a + \omega^a{}_b \wedge \epsilon^b = 0, \tag{8.26}$$

holds again. For $a = 2$, using $\mathrm{d}\epsilon^2 = 0$ and $\omega^2{}_2 = 0$,

$$\omega^2{}_1 \wedge \epsilon^1 = 0, \tag{8.27}$$

therefore

$$\omega^2{}_1 = f(\theta, \phi)\mathrm{d}\theta. \tag{8.28}$$

For $a = 1$, using $\omega^1{}_1 = 0$ and $\omega^1{}_2 = -\omega^2{}_1$,

$$-r\sin(\phi)\mathrm{d}\phi \wedge \mathrm{d}\theta - r f(\theta, \phi)\mathrm{d}\theta \wedge \mathrm{d}\phi = 0 \tag{8.29}$$

$$r\sin(\phi)\mathrm{d}\theta \wedge \mathrm{d}\phi - r f(\theta, \phi)\mathrm{d}\theta \wedge \mathrm{d}\phi = 0 \tag{8.30}$$

therefore $f(\theta, \phi) = \sin(\phi)$ and

$$\omega^1{}_2 = -\sin(\phi)\mathrm{d}\theta. \tag{8.31}$$

The curvature two-form follows as

$$\Omega^1{}_2 = \mathrm{d}\omega^1{}_2 + \omega^1{}_a \wedge \omega^a{}_2 \tag{8.32}$$

$$= \mathrm{d}(-\sin(\phi)\mathrm{d}\theta) = -\cos(\phi)\mathrm{d}\phi \wedge \mathrm{d}\theta. \tag{8.33}$$

Using $\epsilon^1 \wedge \epsilon^2 = r(R + r\cos(\phi))\mathrm{d}\theta \wedge \mathrm{d}\phi$, we get

$$\Omega^{12} = \frac{\cos(\phi)}{r(R + r\cos(\phi))}\epsilon^1 \wedge \epsilon^2 = K\epsilon^1 \wedge \epsilon^2 = \mathrm{Pf}_{2d}(\Omega). \tag{8.34}$$

Applying the theorem of Chern–Gauss–Bonnet

$$\int_{\mathcal{M}} \mathrm{Pf}(\Omega) = 2\pi\chi(\mathcal{M}), \tag{8.35}$$

with $\mathrm{Pf}(\Omega) = K\mathrm{d}A$ and $\mathrm{d}A = r(R + r\cos(\phi))\mathrm{d}\theta\mathrm{d}\phi$,

$$\int_0^{2\pi}\int_0^{2\pi} \cos(\phi)\mathrm{d}\theta\mathrm{d}\phi = 0, \tag{8.36}$$

therefore $\chi(\mathcal{M}) = 0$.

Further consideration of the Gaussian curvature reveal

$$K = \frac{\cos(\phi)}{r(R + r\cos(\phi))}. \tag{8.37}$$

As $\mathbb{T}^2$ is symmetric under θ, K is independent of θ, and we conclude that $K \propto \frac{1}{r^2}$, $K \propto \frac{1}{R}$. For

$$\phi = 0: \quad K = \frac{1}{rR + r^2} > 0, \tag{8.38}$$

$$\phi = \frac{\pi}{2}: \quad K = 0, \tag{8.39}$$

$$\phi = \pi: \quad K = \frac{-1}{rR - r^2}. \tag{8.40}$$

Therefore $\mathbb{T}^2$ is positively curved on the outer side like $\mathbb{S}^2$ and as $R > r$ we have $K < 0$ at $\phi = \pi$, giving a saddle point of curvature on the inner side.

8.2 Edge Crossing Flat Data Modelling

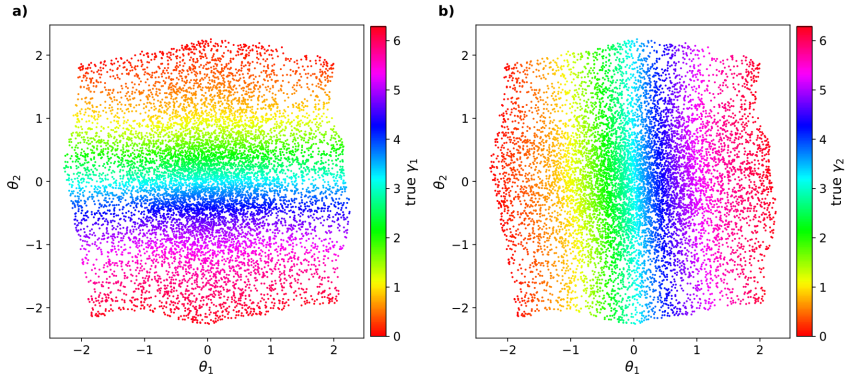


Figure 8.1: Flat latent representation of a flat dataset with edge crossing. a) and b) represent the two ground truth angles of variation.

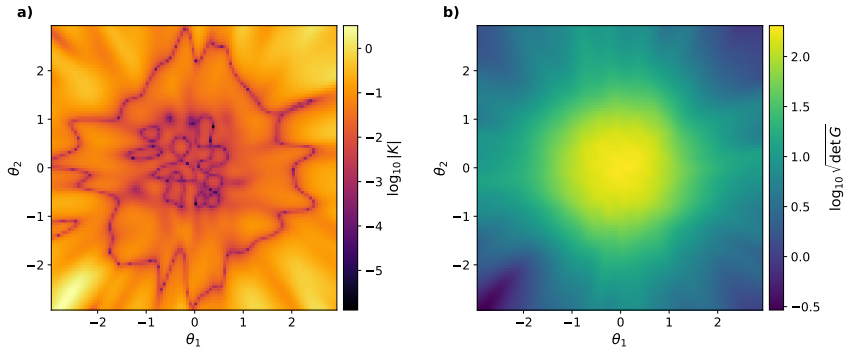


Figure 8.2: a) absolute curvature and b) magnification (both logarithmically scaled) of the Fisher metric of the decoder of a flat dataset with edge crossing in a flat latent space.

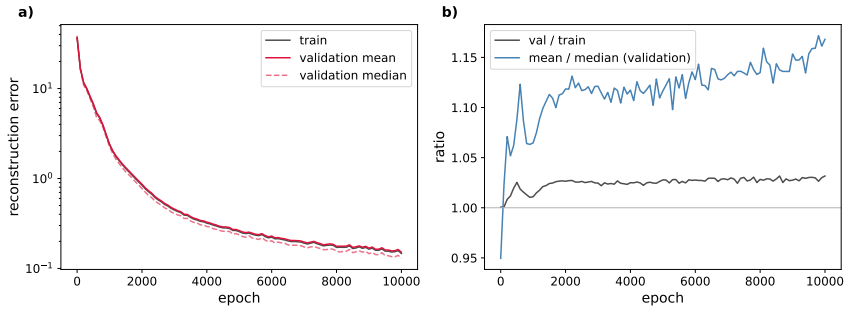


Figure 8.3: Training of the flat autoencoder on the flat dataset with edge crossing. a) Reconstruction error against epoch (logarithmic ordinate): training error (grey), validation mean (solid red), validation median (dashed red). b) Ratios against epoch: validation to training (grey) and mean to median of the validation error (blue).

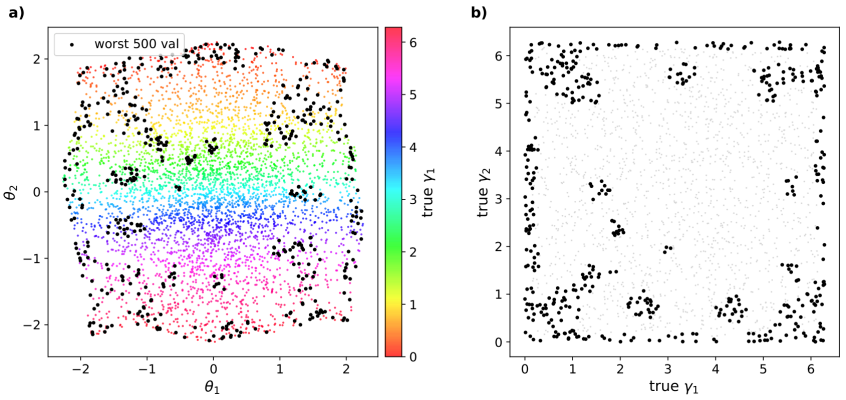


Figure 8.4: Localization of the reconstruction error, of the flat variational autoencoder with the flat dataset with edge crossing. a) Latent coordinates (θ_1, θ_2) of the validation set, coloured by the ground truth γ_1; the 500 points with the largest reconstruction error are marked in black. b) The same points in ground-truth coordinates (γ_1, γ_2), with the worst 500 in black and the remainder in grey.

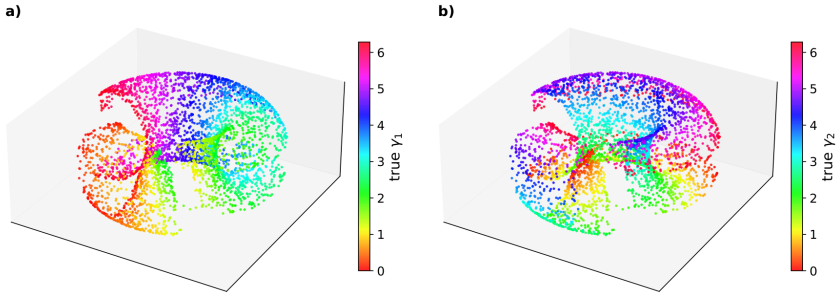


Figure 8.5: Toroidal latent representation of a flat dataset with edge crossing. a) and b) represent the two ground truth angles of variation, 3 dimensional plot.

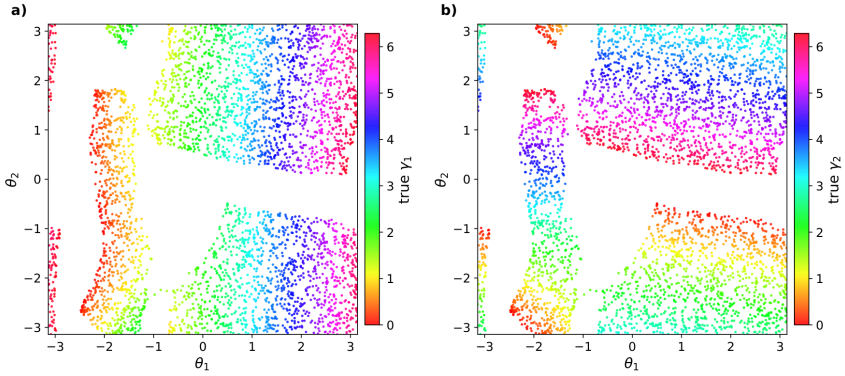


Figure 8.6: Toroidal latent representation of a flat dataset with edge crossing. a) and b) represent the two ground truth angles of variation, 2 dimensional plot.

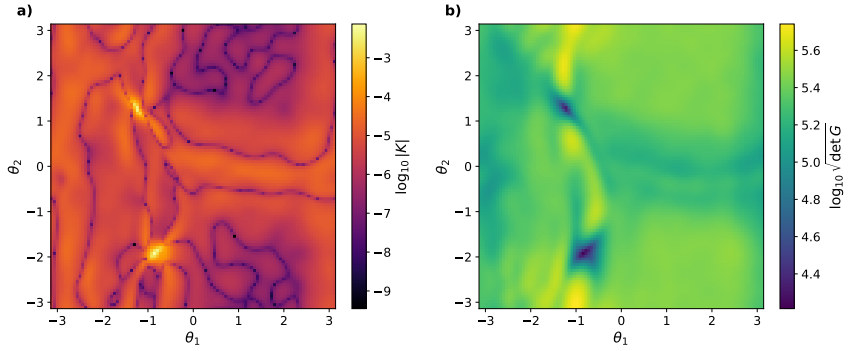


Figure 8.7: a) absolute curvature and b) magnification (both logarithmically scaled) of the Fisher metric of the decoder of a flat dataset with edge crossing in a toroidal latent space.

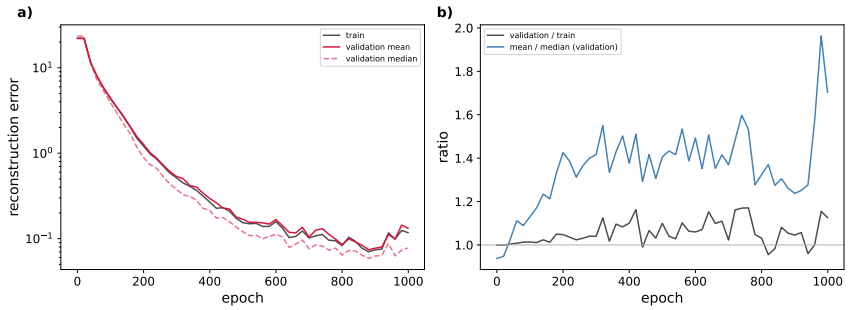


Figure 8.8: Training of the toroidal autoencoder on the flat dataset with edge crossing. a) Reconstruction error against epoch (logarithmic ordinate): training error (grey), validation mean (solid red), validation median (dashed red). b) Ratios against epoch: validation to training (grey) and mean to median of the validation error (blue).

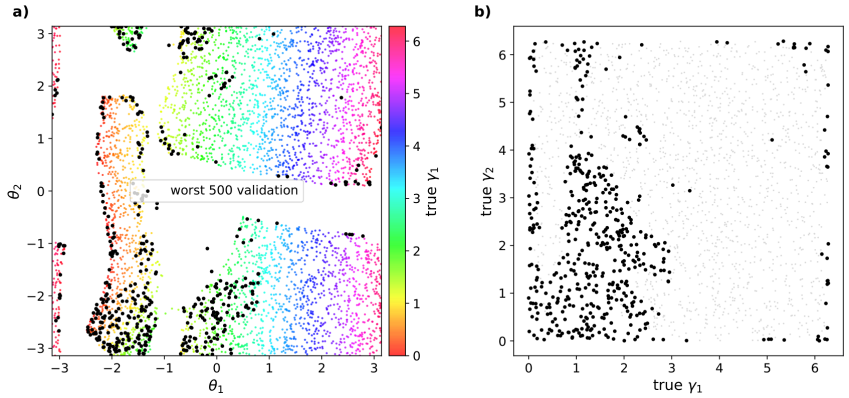


Figure 8.9: Localization of the reconstruction error, of the toroidal variational autoencoder with the flat dataset with edge crossing. a) Latent coordinates (θ_1, θ_2) of the validation set, coloured by the ground truth γ_1; the 500 points with the largest reconstruction error are marked in black. b) The same points in ground-truth coordinates (γ_1, γ_2), with the worst 500 in black and the remainder in grey.

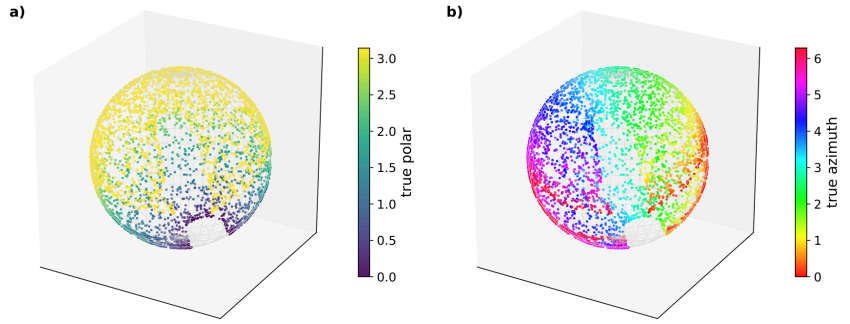


Figure 8.10: Spherical latent representation of a flat dataset with edge crossing. a) and b) represent the two ground truth angles of variation, 3 dimensional plot.

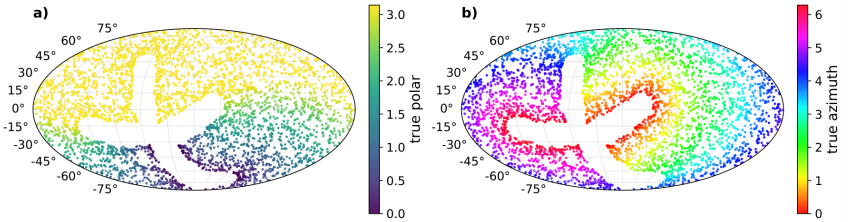


Figure 8.11: Spherical latent representation of a flat dataset with edge crossing. a) and b) represent the two ground truth angles of variation, 2 dimensional plot.

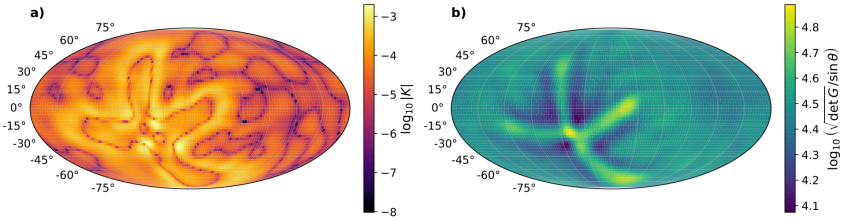


Figure 8.12: a) absolute curvature and b) magnification (both logarithmically scaled) of the Fisher metric of the decoder of a flat dataset with edge crossing in a spherical latent space.

8.3 Ressources and References

Claude.ai (Anthropic) was used for troubleshooting and improving the efficiency of the application's code. Therefore, the originality of parts of the code cannot be guaranteed. The implementation of the spherical dataset, depicted in Fig. 5.4, was a solution proposed by Claude.ai. Concepts and ideas were developed independently except stated otherwhise (see references for further information). Research, referencing and spell- and grammar-checking (changes in text were done manually by the author) were aided by Claude.ai and Perplexity.ai. The outlook in particular relied on research done in collaboration with Claude.ai. Additionally, Claude.ai was used to verify the correctness of mathematical equations and derivations (all derivations and mathematical work relevant to this thesis

were carried out by the author). The actual writing and all theoretical work of this thesis were done *solely by the author* using the named references. No scientific contribution of this work rests on the use of Claude.ai, but solely on the author's scientific judgement and the foundational work provided by the named references.

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All mistakes that remain are mine alone.

Declaration

I hereby declare that I have written this thesis independently and have not used any sources or aids other than those stated.

Ich versichere, dass ich diese Arbeit selbstständig verfasst und keine anderen als die angegebenen Quellen und Hilfsmittel benutzt habe.

Heidelberg, September 3, 2026

Signature